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Electrostatics

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 248 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Two point charges $q_1 = 3\mu C$ and $q_2 = -4\mu C$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ respectively. Force on charge q_2 is ____ N. (Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI Units)

Open solution

(A) $(12\hat{i} + 24\hat{j} + 24\hat{k}) \times 10^{-3}$

(B) $(4\hat{i} + 8\hat{j} + 8\hat{k}) \times 10^{-3}$

(C) $(3\hat{i} + 6\hat{j} + 6\hat{k}) \times 10^{-3}$

(D) $(-4\hat{i} - 8\hat{j} - 8\hat{k}) \times 10^{-3}$

Q2. JEE Main 2026 (Online) 6th April Evening Shift

The electric potential as a function of x, y is given by $V = 5\left(x^2 - y^2\right)V$. The electric field at a point $(2, 3)\mathrm{m}$ is ____ $\mathrm{V/m}$.

Open solution

(A) $(-20\hat{i} + 30\hat{j})$

(B) $(20\hat{i} - 30\hat{j})$

(C) $(20\hat{i} + 45\hat{j})$

(D) $(-4\hat{i} + 6\hat{j})$

Q3. JEE Main 2026 (Online) 6th April Morning Shift

A thin half ring of radius 35 cm is uniformly charged with a total charge of Q coulomb. If the magnitude of the electric field at centre of the half ring is $100\ \mathrm{V/m}$, then the value of Q is ____ nC . $\left(\epsilon_0 = 8.85 \times 10^{-12}\mathrm{C^2/Nm^2}\right.$ and $\left.\pi = 3.14\right)$

Open solution

(A) 2.14

(B) 2.44

(C) 3.25

(D) 0.7

Q4. JEE Main 2026 (Online) 4th April Evening Shift

A rigid dipole undergoes a simple harmonic motion about its centre in the presence of an electric field $\vec{E}_1 = E_0\hat{x}$. If another electric field $\vec{E}_2 = 2E_0(\hat{y} + \hat{z})$ is introduced to the system, what will be the percentage change in the frequency of the oscillation (approximate)?

Open solution

(A) 73%

(B) 63%

(C) 83%

(D) 53%

Q5. JEE Main 2026 (Online) 4th April Evening Shift

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Assertion A : In electrostatics, a conductor does not store any net charge inside.

Reason R : Inside the capacitor (with no dielectric medium), the free charge carriers, if placed between the plates of capacitor, experience force and drift.

Choose the correct answer from the options given below

Open solution

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

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Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Two metal plates **(A, B)** are kept horizontally with separation of $\left(\frac{12}{\pi}\right)$ **cm**, with plate A on the top.

An atomizer jet sprays oil (density **1.5 g/cm^3**) droplets of radius **1 mm** horizontally. All oil droplets carry a charge **5 nC**. The potentials $\mathbf{V_A}$ and $\mathbf{V_B}$ are required on plates A and B respectively in order to ensure the droplets do not descend. The values of $\mathbf{V_A}$ and $\mathbf{V_B}$ are _____.

(Neglect the air resistance to the droplets and take **g = 10 m/s^2**)

(A) 100 V and 580 V (B) 580 V and 100 V (C) 60 V and 400 V (D) 0 V and -200 V

Q7. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Two point charges $8\ \mu C$ and $-2\ \mu C$ are located at $x = 2$ cm and $x = 4$ cm, respectively on the x-axis. The ratio of electric flux due to these charges through two spheres of radii 3 cm and 5 cm with their centers at the origin is _____.

(A) 4 : 1 (B) 3 : 4 (C) 4 : 3 (D) 4 : 5

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

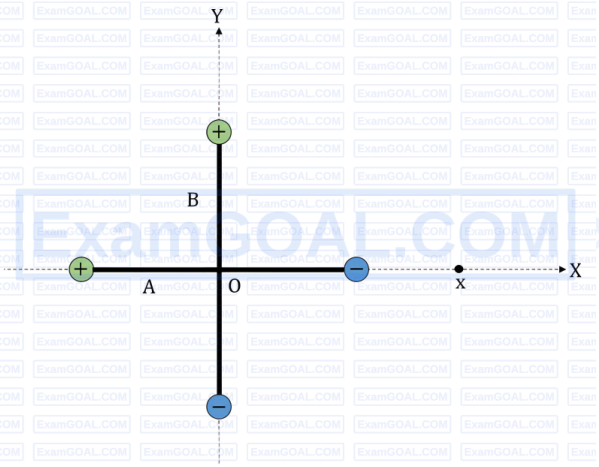
Two charged conducting spheres S_1 and S_2 of radii 8 cm and 18 cm are connected to each other by a wire. After equilibrium is established, the ratio of electric fields on S_1 and S_2 spheres are E_{S1} and E_{S2} respectively. The value of $\frac{E_{S1}}{E_{S2}}$ is _____.

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{4}{9}$ (D) $\frac{9}{4}$

Q9. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Two short electric dipoles **A** and **B** having dipole moment p_1 and p_2 respectively are placed with their axes mutually perpendicular as shown in the figure. The resultant electric field at a point x is making an angle of 60° with the line joining points O and x. The ratio of the dipole moments p_2/p_1 is _____.



(A) $\frac{\sqrt{3}}{2}$ (B) $2\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

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Q10. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Identify the correct statements :

A. Electrostatic field lines form closed loops.

B. The electric field lines point radially outward when charge is greater than zero.

C. The Gauss - Law is valid only for inverse-square force.

D. The work done in moving a charged particle in a static electric field around a closed path is zero.

E. The motion of a particle under Coulomb's force must take place in a plane.

Choose the **correct** answer from the options given below :

(A) A, C, E Only (B) A, B, D, E Only (C) B, C, D, E Only (D) A, B, C, D Only

Q11. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Two point charges of 1 nC and 2 nC are placed at the two corners of equilateral triangle of side 3 cm . The work done in bringing a charge of 3 nC from infinity to the third corner of the triangle is ____ μJ.

$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \mathrm{N}\cdot\ \mathrm{m}^2/\mathrm{C}^2$

(A) 3.3 (B) 27 (C) 2.7 (D) 5.4

Q12. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

The electrostatic potential in a charged spherical region of radius r varies as $V = ar^3 + b$, where a and b are constants. The total charge in the sphere of unit radius is $\alpha \times \pi a\epsilon_0$. The value of α is ____ .

(permittivity of vacuum is ϵ_0)

(A) -12 (B) -8 (C) -9 (D) -6

Q13. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

There are three concentric conducting spherical shells A, B and C of radii a, b and c respectively ($c > b > a$) and they are charged with charge q_1, q_2 and q_3 respectively. The potentials of the spheres A, B and C respectively, are :

(A) $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{b}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{c}\right)$

(B) $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{a}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{b}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{c}\right)$

(C) $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{a}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right)$

(D) $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{c}\right)$

Q14. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Three charges $+2q, +3q$ and $-4q$ are situated at $(0, -3a), (2a, 0)$ and $(-2a, 0)$ respectively in the xy plane. The resultant dipole moment about origin is ____ .

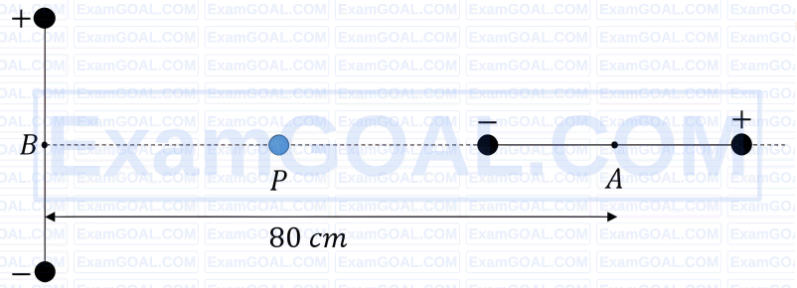
(A) $2qa(7\hat{i} - 3\hat{j})$ (B) $2qa(3\hat{j} - 7\hat{i})$ (C) $2qa(3\hat{i} - 7\hat{j})$ (D) $2qa(3\hat{j} - \hat{i})$

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Q15. JEE Main 2026 (Online) 23rd January Evening Shift

Two short dipoles (A, B), A having charges $\pm 2\mu C$ and length 1 cm and B having charges $\pm 4\mu C$ and length 1 cm are placed with their centres 80 cm apart as shown in the figure. The electric field at a point P, equi-distant from the centres of both dipoles is ____ N/C.

Open solution



(A) $\frac{9}{16}\sqrt{2} \times 10^5$

(B) $\frac{9}{16}\sqrt{2} \times 10^4$

(C) $9\sqrt{2} \times 10^4$

(D) $4.5\sqrt{2} \times 10^4$

Q16. JEE Main 2026 (Online) 23rd January Evening Shift

Two charges $7\mu C$ and $-2\mu C$ are placed at $(-9, 0, 0)\text{cm}$ and $(9, 0, 0)\text{cm}$ respectively in an external field $E = \frac{A}{r^2}\hat{r}$, where $A = 9 \times 10^5\ \text{N/C.m}^2$. Considering the potential at infinity is 0 , the electrostatic energy of the configuration is ____ J.

Open solution

(A) 49.3

(B) 1.4

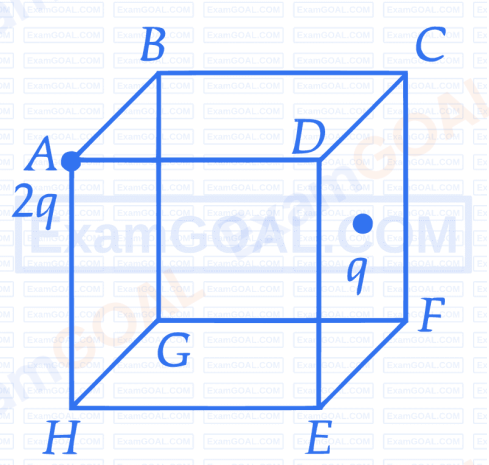
(C) 24.3

(D) -90.7

Q17. JEE Main 2026 (Online) 23rd January Morning Shift

Two point charges $2q$ and q are placed at vertex A and centre of face $CDEF$ of the cube as shown in figure. The electric flux passing through the cube is:

Open solution



(A) $\frac{3q}{2\epsilon_0}$

(B) $\frac{3q}{4\epsilon_0}$

(C) $\frac{q}{\epsilon_0}$

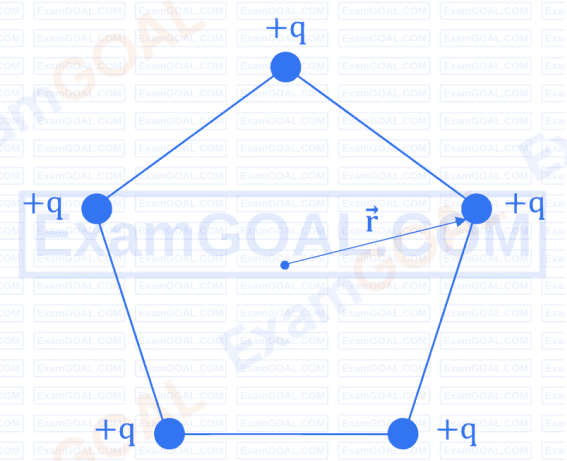
(D) $\frac{3q}{\epsilon_0}$

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Q18. JEE Main 2026 (Online) 22nd January Evening Shift

Five positive charges each having charge q are placed at the vertices of a pentagon as shown in the figure. The electric potential (V) and the electric field $(\vec{E})$ at the center O of the pentagon due to these five positive charges are :

Open solution



(A) $V = \frac{5q}{4\pi\varepsilon_o r}$ and $\vec{E} = \frac{5\sqrt{3}q}{8\pi\varepsilon_o r^2}\hat{r}$

(B) $V = \frac{5q}{4\pi\varepsilon_o r}$ and $\vec{E} = 0$

(C) $V = 0$ and $\vec{E} = 0$

(D) $V = \frac{5q}{4\pi\varepsilon_0 r}$ and $\vec{E} = \frac{5q}{4\pi\varepsilon_o r^2}\hat{r}$

Q19. JEE Main 2026 (Online) 22nd January Evening Shift

Three small identical bubbles of water having same charge on each coalesce to form a bigger bubble. Then the ratio of the potentials on one initial bubble and that on the resultant bigger bubble is :

Open solution

(A) $1 : 3^{2/3}$ (B) $3^{2/3} : 1$ (C) $1 : 3^{1/3}$ (D) $1 : 2^{2/3}$

Q20. JEE Main 2026 (Online) 22nd January Morning Shift

Electric field in a region is given by $\vec{E} = Ax\hat{i} + By\hat{j}$, where $A = 10\ \mathrm{V/m^2}$ and $B = 5\ \mathrm{V/m^2}$. If the electric potential at a point $(10, 20)$ is 500 V , then the electric potential at origin is ____ V.

Open solution

(A) 1000 (B) 0 (C) 2000 (D) 500

Q21. JEE Main 2026 (Online) 22nd January Morning Shift

A simple pendulum has a bob with mass m and charge q. The pendulum string has negligible mass. When a uniform and horizontal electric field $\vec{E}$ is applied, the tension in the string changes. The final tension in the string, when pendulum attains an equilibrium position is ____ .

Open solution

(A) $mg - qE$ (B) $\sqrt{m^2g^2 + q^2E^2}$ (C) $m\,\mathrm{g} + qE$ (D) $\sqrt{m^2g^2 - q^2E^2}$

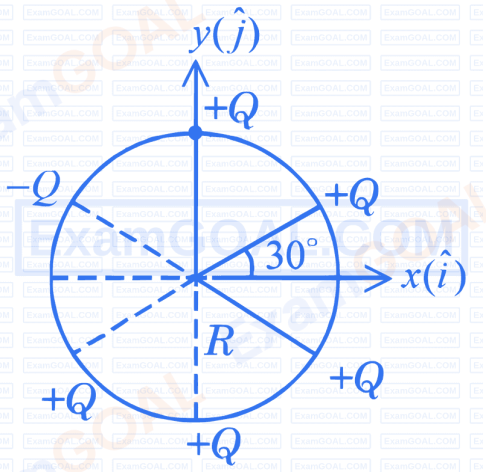
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Q22. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Six point charges are kept 60° apart from each other on the circumference of a circle of radius R as shown in figure. The net electric field at the center of the circle is ____ .

(ϵ_0 is permittivity of free space)



(A) $-\left(\frac{5Q}{8\pi\epsilon_0 R^2}\right)(\hat{i}-3\hat{j})$

(B) $\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i}-\hat{j})$

(C) $-\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i}-\hat{j})$

(D) $-\frac{5Q}{8\pi\epsilon_0 R^2}(\hat{i}+\sqrt{3}\hat{j})$

Q23. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Consider two identical metallic spheres of radius R each having charge Q and mass m. Their centers have an initial separation of $4R$. Both the spheres are given an initial speed of u towards each other. The minimum value of u, so that they can just touch each other is:

(Take $k=\frac{1}{4\pi\epsilon_0}$ and assume $kQ^2 > Gm^2$ where G is the Gravitational constant)

(A) $\sqrt{\frac{kQ^2}{4mR}\left(1+\frac{Gm^2}{kQ^2}\right)}$

(B) $\sqrt{\frac{kQ^2}{2mR}\left(1-\frac{Gm^2}{2kQ^2}\right)}$

(C) $\sqrt{\frac{kQ^2}{2mR}\left(1-\frac{Gm^2}{kQ^2}\right)}$

(D) $\sqrt{\frac{kQ^2}{4mR}\left(1-\frac{Gm^2}{kQ^2}\right)}$

Q24. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

A point charge of 10^{-8}C is placed at origin. The work done in moving a point charge $2\mu\text{C}$ from point $A(4,4,2)\text{m}$ to $B(2,2,1)\text{m}$ is ____ J. $\left(\frac{1}{4\pi\epsilon_0}=9\times10^9 \text{ in SI units}\right)$

(A) 30×10^{-6} (B) 0 (C) 15×10^{-6} (D) 45×10^{-6}

Q25. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Two metal spheres of radius **R** and **3R** have same surface charge density **σ**. If they are brought in contact and then separated, the surface charge density on smaller and bigger sphere becomes σ_1 and σ_2, respectively. The ratio $\frac{\sigma_1}{\sigma_2}$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{9}$ (C) 9 (D) 3

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Q26. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : Work done in moving a test charge between two points inside a uniformly charged spherical shell is zero, no matter which path is chosen.

Reason R : Electrostatic potential inside a uniformly charged spherical shell is constant and is same as that on the surface of the shell.

In the light of the above statements, choose the correct answer from the options given below.

(A) A is true but R is false

(B) Both A and R are true and R is the correct explanation of A

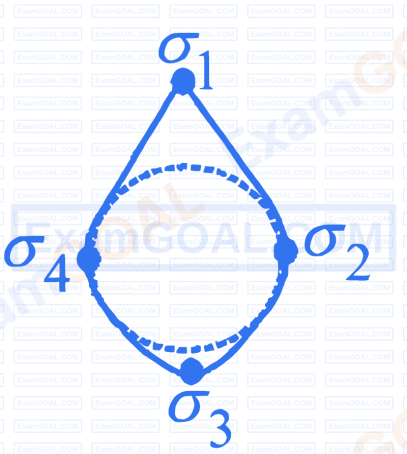
(C) Both A and R are true but R is NOT the correct explanation of A

(D) A is false but R is true

Q27. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Electric charge is transferred to an irregular metallic disk as shown in the figure. If σ_1, σ_2, σ_3 and σ_4 are charge densities at given points then, choose the correct answer from the options given below:



A. $\sigma_1 > \sigma_3; \sigma_2 = \sigma_4$

B. $\sigma_1 > \sigma_2; \sigma_3 > \sigma_4$

C. $\sigma_1 > \sigma_3 > \sigma_2 = \sigma_4$

D. $\sigma_1 < \sigma_3 < \sigma_2 = \sigma_4$

E. $\sigma_1 = \sigma_2 = \sigma_3 = \sigma_4$

(A) B and C Only (B) A and C Only (C) D and E Only (D) A, B and C Only

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Q28. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

An infinitely long wire has uniform linear charge density $\lambda = 2$ nC/m. The net flux through a Gaussian cube of side length $\sqrt{3}$ cm, if the wire passes through any two corners of the cube, that are maximally displaced from each other, would be x Nm^2C^{-1}, where x is:

[Neglect any edge effects and use $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI units]

(A) 6.48π (B) 0.72π (C) 1.44π (D) 2.16π

Q29. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

A dipole with two electric charges of 2 μC magnitude each, with separation distance 0.5 μm, is placed between the plates of a capacitor such that its axis is parallel to an electric field established between the plates when a potential difference of 5 V is applied. Separation between the plates is 0.5 mm. If the dipole is rotated by 30° from the axis, it tends to realign in the direction due to a torque. The value of torque is:

(A) 2.5×10^{-9} Nm (B) 2.5×10^{-12} Nm (C) 5×10^{-3} Nm (D) 5×10^{-9} Nm

Q30. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : The outer body of an aircraft is made of metal which protects persons sitting inside from lightning strikes.

Reason (R) : The electric field inside the cavity enclosed by a conductor is zero.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(B) (A) is correct but (R) is not correct

(C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

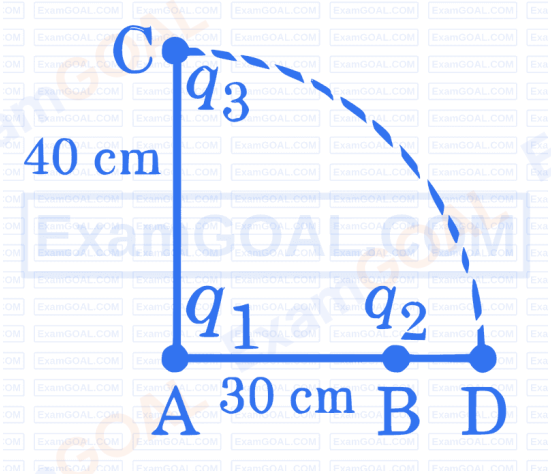
(D) (A) is not correct but (R) is correct

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Q31. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Two charges q_1 and q_2 are separated by a distance of 30 cm . A third charge q_3 initially at ' C ' as shown in the figure, is moved along the circular path of radius 40 cm from C to D . If the difference in potential energy due to movement of q_3 from C to D is given by $\frac{q_3 K}{4\pi\epsilon_0}$, the value of K is :

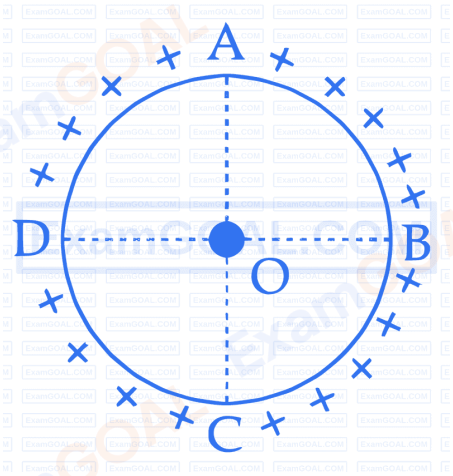


(A) $6q_2$ (B) $6q_1$ (C) $8q_1$ (D) $8q_2$

Q32. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

A metallic ring is uniformly charged as shown in figure. AC and BD are two mutually perpendicular diameters. Electric field due to arc AB at ' O ' is ' E ' in magnitude. What would be the magnitude of electric field at ' O ' due to arc ABC ?



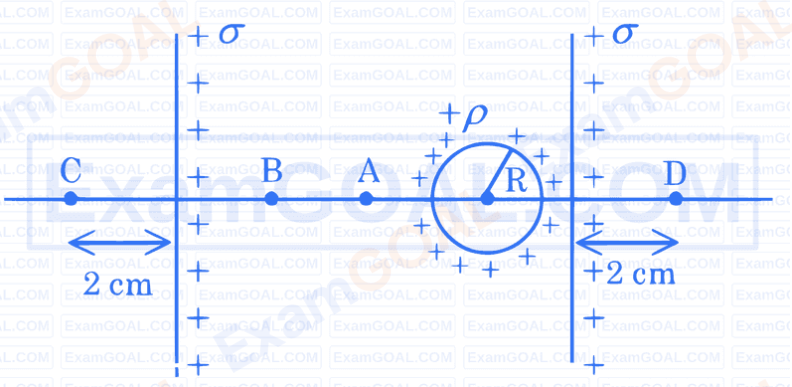
(A) 2E (B) Zero (C) E/2 (D) $\sqrt{2}$E

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Q33. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Two infinite identical charged sheets and a charged spherical body of charge density ' ρ ' are arranged as shown in figure. Then the correct relation between the electrical fields at $\mathrm{A}, \mathrm{B}, \mathrm{C}$ and D points is:



(A) $\left|\vec{E}_A\right| = \left|\vec{E}_B\right|; \vec{E}_C > \vec{E}_D$

(B) $\vec{E}_A = \vec{E}_B; \vec{E}_C = \vec{E}_D$

(C) $\vec{E}_C \neq \vec{E}_D; \vec{E}_A > \vec{E}_B$

(D) $\vec{E}_A > \vec{E}_B; \vec{E}_C = \vec{E}_D$

Q34. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Two small spherical balls of mass 10 g each with charges $-2\mu\mathrm{C}$ and $2\mu\mathrm{C}$, are attached to two ends of very light rigid rod of length 20 cm . The arrangement is now placed near an infinite nonconducting charge sheet with uniform charge density of $100\mu\mathrm{C/m^2}$ such that length of rod makes an angle of 30° with electric field generated by charge sheet. Net torque acting on the rod is: (Take $\varepsilon_o : 8.85 \times 10^{-12}\mathrm{C^2/Nm^2}$)

(A) 1.12 Nm

(B) 2.24 Nm

(C) 11.2 Nm

(D) 112 Nm

Q35. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

The electrostatic potential on the surface of uniformly charged spherical shell of radius $\mathrm{R} = 10\ \mathrm{cm}$ is 120 V . The potential at the centre of shell, at a distance $\mathrm{r} = 5\ \mathrm{cm}$ from centre, and at a distance $\mathrm{r} = 15$ cm from the centre of the shell respectively, are:

(A) $0\ \mathrm{V}, 120\ \mathrm{V}, 40\ \mathrm{V}$

(B) $120\ \mathrm{V}, 120\ \mathrm{V}, 80\ \mathrm{V}$

(C) $40\ \mathrm{V}, 40\ \mathrm{V}, 80\ \mathrm{V}$

(D) $0\ \mathrm{V}, 0\ \mathrm{V}, 80\ \mathrm{V}$

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Q36. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Net dipole moment of a polar linear isotropic dielectric substance is not zero even in the absence of an external electric field.

Reason (R) : In absence of an external electric field, the different permanent dipoles of a polar dielectric substance are oriented in random directions.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

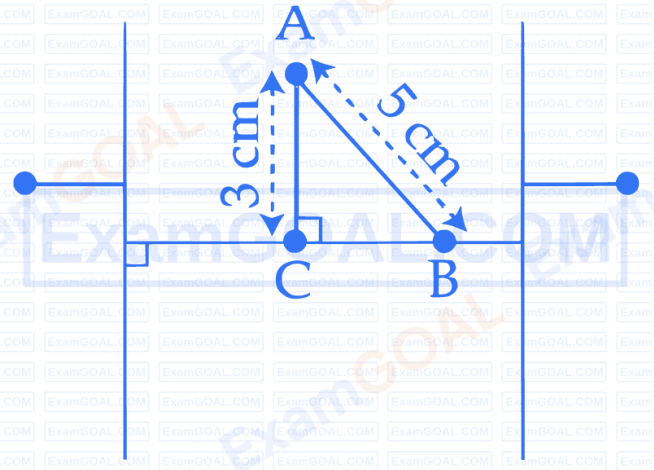
(C) (A) is not correct but (R) is correct

(D) (A) is correct but (R) is not correct

Q37. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Two large plane parallel conducting plates are kept 10 cm apart as shown in figure. The potential difference between them is V . The potential difference between the points A and B (shown in the figure) is :



(A) 1 V (B) $\frac{1}{4}$ V (C) $\frac{3}{4}$ V (D) $\frac{2}{5}$ V

Q38. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Consider a circular loop that is uniformly charged and has a radius $\mathrm{a}\sqrt{2}$. Find the position along the positive z-axis of the cartesian coordinate system where the electric field is maximum if the ring was assumed to be placed in xy plane at the origin :

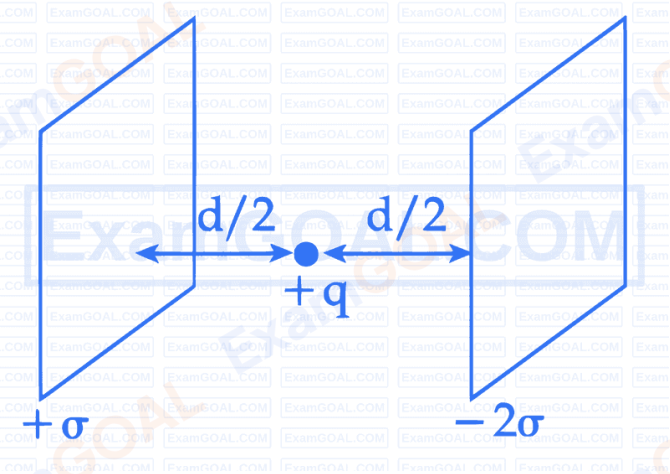
(A) a (B) $a/2$ (C) 0 (D) $a/\sqrt{2}$

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Q39. JEE Main 2025 (Online) 2nd April Morning Shift

Consider two infinitely large plane parallel conducting plates as shown below. The plates are uniformly charged with a surface charge density $+\sigma$ and -2σ. The force experienced by a point charge +q placed at the mid point between two plates will be:

Open solution



(A) $\frac{3\sigma q}{4\epsilon_0}$

(B) $\frac{3\sigma q}{2\epsilon_0}$

(C) $\frac{\sigma q}{4\epsilon_0}$

(D) $\frac{\sigma q}{2\epsilon_0}$

Q40. JEE Main 2025 (Online) 2nd April Morning Shift

A point charge $+q$ is placed at the origin. A second point charge $+9q$ is placed at $(d, 0, 0)$ in Cartesian coordinate system. The point in between them where the electric field vanishes is:

Open solution

(A) $(3d/4, 0, 0)$

(B) $(d/4, 0, 0)$

(C) $(4d/3, 0, 0)$

(D) $(d/3, 0, 0)$

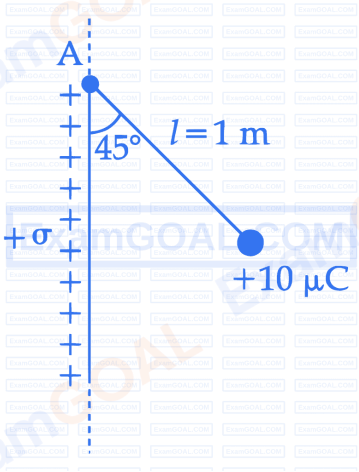
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Q41. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

A small bob of mass 100 mg and charge $+10\mu C$ is connected to an insulating string of length 1 m . It is brought near to an infinitely long non-conducting sheet of charge density ' σ ' as shown in figure. If string subtends an angle of 45° with the sheet at equilibrium the charge density of sheet will be.

(Given, $\epsilon_0 = 8.85 \times 10^{-12} \frac{F}{m}$ and acceleration due to gravity, $g = 10 \frac{m}{s^2}$)



(A) $1.77\,nC/m^2$

(B) $0.885\,nC/m^2$

(C) $885\,nC/m^2$

(D) $17.7\,nC/m^2$

Q42. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

A point charge causes an electric flux of $-2 \times 10^4 Nm^2C^{-1}$ to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is :

(Given $\epsilon_0 = 8.85 \times 10^{-12} C^2\,N^{-1}\,m^{-2}$)

(A) $17.7 \times 10^{-8} C$

(B) $-17.7 \times 10^{-8} C$

(C) $15.7 \times 10^{-8} C$

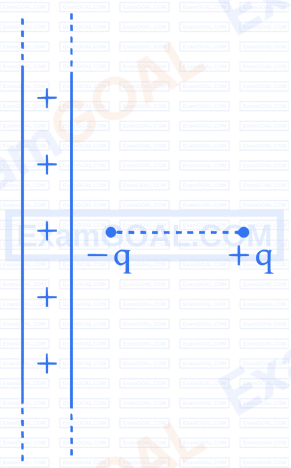
(D) $-15.7 \times 10^{-8} C$

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Q43. JEE Main 2025 (Online) 29th January Evening Shift

An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_o. Choose the correct option from the following.

Open solution



(A) Torque on dipole is zero and net force is directed away from the sheet.

(B) Potential energy of dipole is minimum and torque is zero.

(C) Potential energy and torque both are maximum.

(D) Torque on dipole is zero and net force acts towards the sheet.

Q44. JEE Main 2025 (Online) 29th January Morning Shift

An electric dipole of mass m, charge q, and length l is placed in a uniform electric field $\vec{E} = E_0\hat{i}$. When the dipole is rotated slightly from its equilibrium position and released, the time period of its oscillations will be :

Open solution

(A) $\frac{1}{2\pi}\sqrt{\frac{ml}{2qE_0}}$

(B) $2\pi\sqrt{\frac{ml}{2qE_0}}$

(C) $2\pi\sqrt{\frac{ml}{qE_0}}$

(D) $\frac{1}{2\pi}\sqrt{\frac{2\,ml}{qE_0}}$

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Q45. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

Match List - I with List - II.

List - I	List - II
(A) Electric field inside (distance r > 0 from center) of a uniformly charged spherical shell with surface charge density σ, and radius R.	(I) σ/ε_0
(B) Electric field at distance r>0 from a uniformly charged infinite plane sheet with surface charge density σ.	(II) $\sigma/2\varepsilon_0$
(C) Electric field outside (distance r>0 from center) of a uniformly charged spherical shell with surface charge density σ, and radius R.	(III) 0
(D) Electric field between 2 oppositely charged infinite plane parallel sheets with uniform surface charge density σ.	(IV) $\frac{\sigma}{\epsilon_0 r^2}$

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(B) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(C) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(D) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

Q46. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Three infinitely long wires with linear charge density λ are placed along the x-axis, y-axis and z-axis respectively. Which of the following denotes an equipotential surface?

(A) $\left(x^2+y^2\right)\left(y^2+z^2\right)\left(z^2+x^2\right)=$ constant

(B) $xyz=$ constant

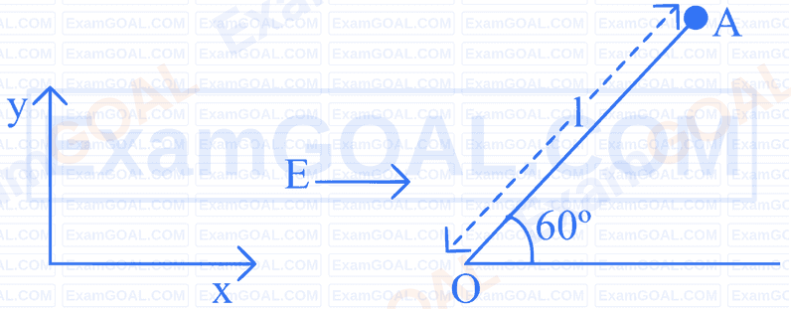
(C) $xy+yz+zx=$ constant

(D) $(x+y)(y+z)(z+x)=$ constant

Q47. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

A particle of mass ' m ' and charge ' q ' is fastened to one end ' A ' of a massless string having equilibrium length l, whose other end is fixed at point ' O '. The whole system is placed on a frictionless horizontal plane and is initially at rest. If uniform electric field is switched on along the direction as shown in figure, then the speed of the particle when it crosses the x-axis is



(A) $\sqrt{\frac{qEl}{2m}}$

(B) $\sqrt{\frac{qEl}{4m}}$

(C) $\sqrt{\frac{qEl}{m}}$

(D) $\sqrt{\frac{2qEl}{m}}$

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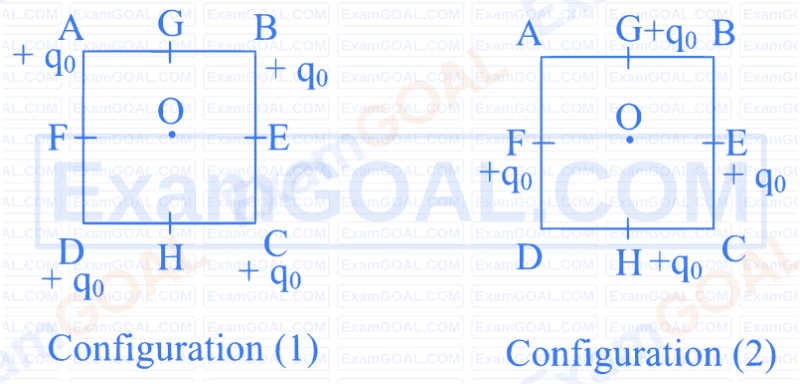
Q48. JEE Main 2025 (Online) 24th January Evening Shift

A small uncharged conducting sphere is placed in contact with an identical sphere but having $4 \times 10^{-8}\mathrm{C}$ charge and then removed to a distance such that the force of repulsion between them is $9 \times 10^{-3}\ \mathrm{N}$. The distance between them is (Take $\frac{1}{4\pi\epsilon_o}$ as 9×10^9 in SI units)

(A) 1 cm (B) 2 cm (C) 4 cm (D) 3 cm

Open solution

Q49. JEE Main 2025 (Online) 24th January Evening Shift



Open solution

In the first configuration (1) as shown in the figure, four identical charges (q_0) are kept at the corners A, B, C and D of square of side length ' a '. In the second configuration (2), the same charges are shifted to mid points G, E, H and F, of the square. If $K = \frac{1}{4\pi\epsilon_0}$, the difference between the potential energies of configuration (2) and (1) is given by :

(A) $\frac{Kq_0^2}{a}(4 - 2\sqrt{2})$ (B) $\frac{Kq_0^2}{a}(4\sqrt{2} - 2)$

(C) $\frac{\mathrm{Kq_0^2}}{\mathrm{a}}(3 - \sqrt{2})$ (D) $\frac{\mathrm{Kq_0^2}}{\mathrm{a}}(3\sqrt{2} - 2)$

Q50. JEE Main 2025 (Online) 23rd January Evening Shift

Two point charges $-4\mu c$ and $4\mu c$, constituting an electric dipole, are placed at $(-9, 0, 0)\mathrm{cm}$ and $(9, 0, 0)\mathrm{cm}$ in a uniform electric field of strength $10^4\mathrm{NC}^{-1}$. The work done on the dipole in rotating it from the equilibrium through 180° is :

(A) 18.4 mJ (B) 12.4 mJ (C) 16.4 mJ (D) 14.4 mJ

Open solution

Q51. JEE Main 2025 (Online) 23rd January Evening Shift

Two charges $7\mu c$ and $-4\mu c$ are placed at $(-7\ \mathrm{cm}, 0, 0)$ and $(7\ \mathrm{cm}, 0, 0)$ respectively. Given, $\epsilon_0 = 8.85 \times 10^{-12}\mathrm{C}^2\ \mathrm{N}^{-1}\ \mathrm{m}^{-2}$, the electrostatic potential energy of the charge configuration is :

(A) -2.0 J (B) -1.5 J (C) -1.2 J (D) -1.8 J

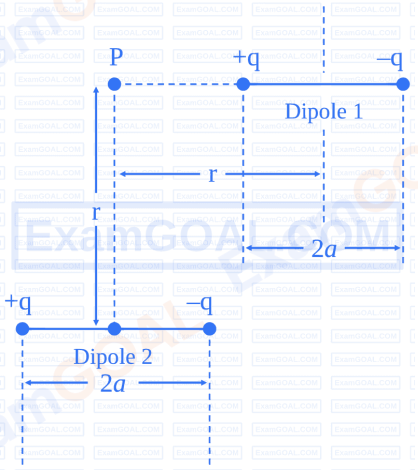
Open solution

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Q52. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

A point particle of charge Q is located at P along the axis of an electric dipole 1 at a distance r as shown in the figure. The point P is also on the equatorial plane of a second electric dipole 2 at a distance r. The dipoles are made of opposite charge q separated by a distance $2a$. For the charge particle at P not to experience any net force, which of the following correctly describes the situation?

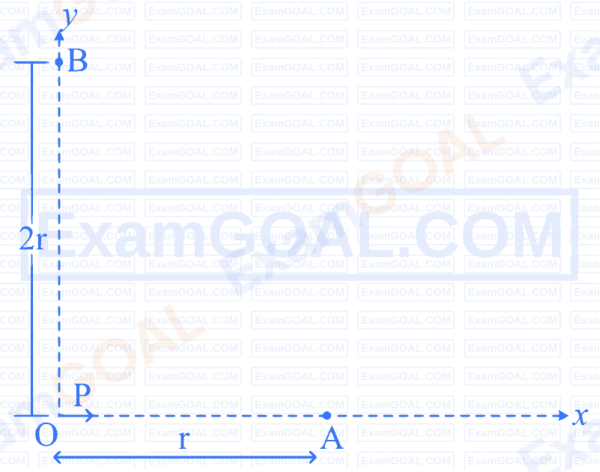


(A) $\frac{a}{r} \sim 0.5$ (B) $\frac{a}{r} \sim 3$ (C) $\frac{a}{r} \sim 10$ (D) $\frac{a}{r} \sim 20$

Q53. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

For a short dipole placed at origin O , the dipole moment P is along x-axis, as shown in the figure. If the electric potential and electric field at A are V_0 and E_0, respectively, then the correct combination of the electric potential and electric field, respectively, at point B on the y-axis is given by



(A) $\frac{V_0}{2}$ and $\frac{E_0}{16}$ (B) zero and $\frac{E_0}{8}$

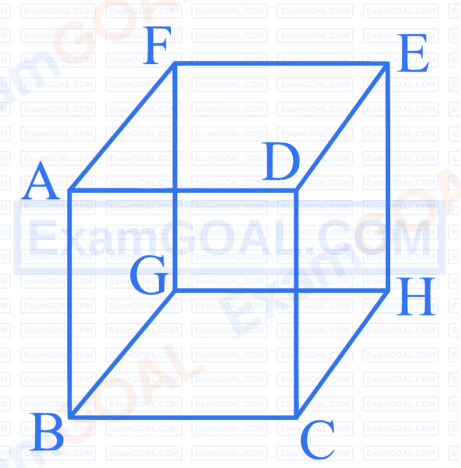
(C) V_0 and $\frac{E_0}{4}$ (D) zero and $\frac{E_0}{16}$

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Q54. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

A line charge of length $\frac{a}{2}$ is kept at the center of an edge BC of a cube ABCDEFGH having edge length ' a ' as shown in the figure. If the density of line charge is λC per unit length, then the total electric flux through all the faces of the cube will be _____ . (Take, ϵ_0 as the free space permittivity)



(A) $\frac{\lambda a}{2\epsilon_0}$

(B) $\frac{\lambda a}{8\epsilon_0}$

(C) $\frac{\lambda a}{4\epsilon_0}$

(D) $\frac{\lambda a}{16\epsilon_0}$

Q55. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

An electron is made to enter symmetrically between two parallel and equally but oppositely charged metal plates, each of 10 cm length. The electron emerges out of the electric field region with a horizontal component of velocity $10^6\ \mathrm{m/s}$. If the magnitude of the electric field between the plates is $9.1\ \mathrm{V/cm}$, then the vertical component of velocity of electron is (mass of electron $= 9.1 \times 10^{-31}\ \mathrm{kg}$ and charge of electron $= 1.6 \times 10^{-19}\mathrm{C}$)

(A) $1 \times 10^6\ \mathrm{m/s}$

(B) $16 \times 10^6\ \mathrm{m/s}$

(C) $16 \times 10^4\ \mathrm{m/s}$

(D) 0

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Q56. JEE Main 2024 (Online) 9th April Evening Shift

Five charges $+q, +5q, -2q, +3q$ and $-4q$ are situated as shown in the figure. The electric flux due to this configuration through the surface S is :



Open solution

(A) $\frac{q}{\epsilon_0}$ (B) $\frac{3q}{\epsilon_0}$ (C) $\frac{5q}{\epsilon_0}$ (D) $\frac{4q}{\epsilon_0}$

Q57. JEE Main 2024 (Online) 8th April Morning Shift

Two charged conducting spheres of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is:

(A) ab (B) $\frac{b}{a}$ (C) $\frac{a}{b}$ (D) $\sqrt{ab}$

Open solution

Q58. JEE Main 2024 (Online) 6th April Evening Shift

Two identical conducting spheres P and S with charge Q on each, repel each other with a force $16\ \mathrm{N}$. A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S is :

(A) 1 N (B) 6 N (C) 12 N (D) 4 N

Open solution

Q59. JEE Main 2024 (Online) 6th April Morning Shift

σ is the uniform surface charge density of a thin spherical shell of radius R. The electric field at any point on the surface of the spherical shell is :

(A) $\sigma/\epsilon_o R$ (B) σ/ϵ_o (C) $\sigma/2\epsilon_o$ (D) $\sigma/4\epsilon_o$

Open solution

Q60. JEE Main 2024 (Online) 5th April Evening Shift

The vehicles carrying inflammable fluids usually have metallic chains touching the ground:

(A) To protect tyres from catching dirt from ground

(B) It is a custom

(C) To alert other vehicles

(D) To conduct excess charge due to air friction to ground and prevent sparking

Open solution

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Q61. JEE Main 2024 (Online) 5th April Morning Shift

In hydrogen like system the ratio of coulombian force and gravitational force between an electron and a proton is in the order of :

(A) 10^{19} (B) 10^{39} (C) 10^{29} (D) 10^{36}

Open solution

Q62. JEE Main 2024 (Online) 4th April Evening Shift

A charge q is placed at the center of one of the surfaces of a cube. The flux linked with the cube is:

(A) $\frac{q}{2\epsilon_0}$ (B) Zero (C) $\frac{q}{4\epsilon_0}$ (D) $\frac{q}{8\epsilon_0}$

Open solution

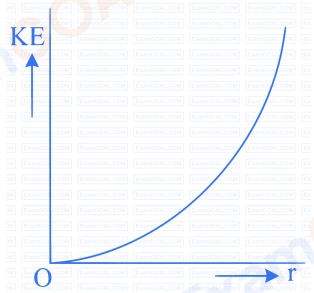
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Q63. JEE Main 2024 (Online) 4th April Morning Shift

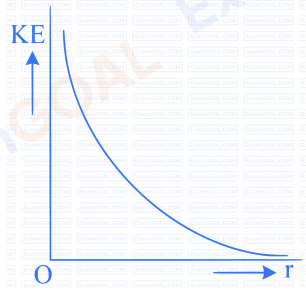
Open solution

An infinitely long positively charged straight thread has a linear charge density $\lambda\ \mathrm{Cm}^{-1}$. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is :

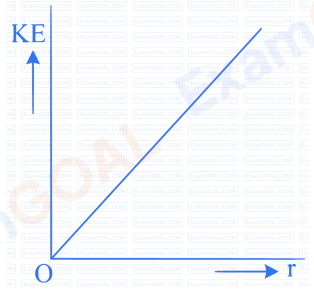
(A)



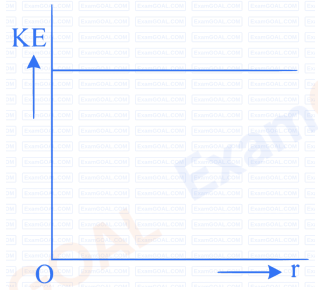
(B)



(C)



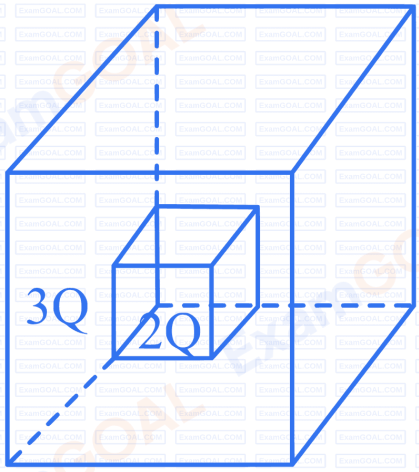
(D)



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Q64. JEE Main 2024 (Online) 1st February Evening Shift

C_1 and C_2 are two hollow concentric cubes enclosing charges $2Q$ and $3Q$ respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is :



Open solution

(A) $3:2$ (B) $5:2$ (C) $2:5$ (D) $2:3$

Q65. JEE Main 2024 (Online) 31st January Evening Shift

Force between two point charges q_1 and q_2 placed in vacuum at 'r' cm apart is F. Force between them when placed in a medium having dielectric constant $K = 5$ at '$r/5$' cm apart will be:

(A) $5F$ (B) $25F$ (C) $F/5$ (D) $F/25$

Open solution

Q66. JEE Main 2024 (Online) 31st January Morning Shift

Two charges q and $3q$ are separated by a distance 'r' in air. At a distance x from charge q, the resultant electric field is zero. The value of x is :

(A) $\frac{r}{3(1+\sqrt{3})}$ (B) $\frac{(1+\sqrt{3})}{r}$ (C) $\frac{r}{(1+\sqrt{3})}$ (D) $r(1+\sqrt{3})$

Open solution

Q67. JEE Main 2024 (Online) 30th January Evening Shift

A particle of charge '$-q$' and mass 'm' moves in a circle of radius 'r' around an infinitely long line charge of linear charge density '$+\lambda$'. Then time period will be given as :

(Consider k as Coulomb's constant)

(A) $T^2 = \frac{4\pi^2 m}{2k\lambda q} r^3$ (B) $T = \frac{1}{2\pi r}\sqrt{\frac{m}{2k\lambda q}}$

(C) $T = \frac{1}{2\pi}\sqrt{\frac{2k\lambda q}{m}}$ (D) $T = 2\pi r\sqrt{\frac{m}{2k\lambda q}}$

Open solution

Q68. JEE Main 2024 (Online) 30th January Morning Shift

The electrostatic potential due to an electric dipole at a distance 'r' varies as :

(A) $\frac{1}{r^3}$ (B) $\frac{1}{\mathrm{r}}$ (C) $\frac{1}{r^2}$ (D) r

Open solution

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Q69. JEE Main 2024 (Online) 29th January Evening Shift

An electric field is given by $(6\hat{i}+5\hat{j}+3\hat{k})\mathrm{N/C}$. The electric flux through a surface area $30\hat{i}\,\mathrm{m}^2$ lying in YZ-plane (in SI unit) is :

(A) 60 (B) 90 (C) 180 (D) 150

Open solution

Q70. JEE Main 2024 (Online) 29th January Morning Shift

Two charges of $5Q$ and $-2Q$ are situated at the points $(3a,0)$ and $(-5a,0)$ respectively. The electric flux through a sphere of radius '$4a$' having center at origin is :

(A) $\frac{2Q}{\varepsilon_0}$ (B) $\frac{7Q}{\varepsilon_0}$ (C) $\frac{3Q}{\varepsilon_0}$ (D) $\frac{5Q}{\varepsilon_0}$

Open solution

Q71. JEE Main 2024 (Online) 27th January Evening Shift

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Work done by electric field on moving a positive charge on an equipotential surface is always zero.

Reason (R) : Electric lines of forces are always perpendicular to equipotential surfaces.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) (A) is correct but (R) is not correct

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) (A) is not correct but (R) is correct

Open solution

Q72. JEE Main 2024 (Online) 27th January Morning Shift

An electric charge $10^{-6}\mu\mathrm{C}$ is placed at origin $(0,0)$ m of $\mathrm{X}-\mathrm{Y}$ co-ordinate system. Two points P and Q are situated at $(\sqrt{3},\sqrt{3})\mathrm{m}$ and $(\sqrt{6},0)\mathrm{m}$ respectively. The potential difference between the points P and Q will be :

(A) $\sqrt{3}\,\mathrm{V}$ (B) $\sqrt{6}\,\mathrm{V}$ (C) $0\,\mathrm{V}$ (D) $3\,\mathrm{V}$

Open solution

Q73. JEE Main 2023 (Online) 15th April Morning Shift

The electric field due to a short electric dipole at a large distance (r) from center of dipole on the equatorial plane varies with distance as :

(A) $\frac{1}{r^2}$ (B) $\frac{1}{r}$ (C) r (D) $\frac{1}{r^3}$

Open solution

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Q74. JEE Main 2023 (Online) 13th April Evening Shift

A $10\ \mu C$ charge is divided into two parts and placed at $1\ cm$ distance so that the repulsive force between them is maximum. The charges of the two parts are:

(A) $9\ \mu C, 1\ \mu C$
(B) $5\ \mu C, 5\ \mu C$
(C) $8\ \mu C, 2\ \mu C$
(D) $7\ \mu C, 3\ \mu C$

Open solution

Q75. JEE Main 2023 (Online) 13th April Morning Shift

Two charges each of magnitude $0.01\ C$ and separated by a distance of $0.4\ mm$ constitute an electric dipole. If the dipole is placed in an uniform electric field '$\vec{E}$' of 10 dyne/C making 30° angle with $\vec{E}$, the magnitude of torque acting on dipole is:

(A) $4.0 \times 10^{-10}\ Nm$
(B) $1.5 \times 10^{-9}\ Nm$
(C) $1.0 \times 10^{-8}\ Nm$
(D) $2.0 \times 10^{-10}\ Nm$

Open solution

Q76. JEE Main 2023 (Online) 12th April Morning Shift

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : If an electric dipole of dipole moment $30 \times 10^{-5}\ C\ m$ is enclosed by a closed surface, the net flux coming out of the surface will be zero.

Reason R : Electric dipole consists of two equal and opposite charges.

In the light of the above statements, choose the correct answer from the options given below.

(A) A is true but R is false
(B) A is false but R is true
(C) Both A and R are true but R is NOT the correct explanation of A
(D) Both A and R are true and R is the correct explanation of A

Open solution

Q77. JEE Main 2023 (Online) 11th April Evening Shift

A capacitor of capacitance C is charged to a potential V. The flux of the electric field through a closed surface enclosing the positive plate of the capacitor is :

(A) Zero
(B) $\frac{CV}{\varepsilon_0}$
(C) $\frac{CV}{2\varepsilon_0}$
(D) $\frac{2CV}{\varepsilon_0}$

Open solution

Q78. JEE Main 2023 (Online) 8th April Evening Shift

Electric potential at a point 'P' due to a point charge of $5 \times 10^{-9} C$ is $50\ V$. The distance of 'P' from the point charge is:

(Assume, $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^{+9}\ Nm^2C^{-2}$)

(A) 0.9 cm
(B) 90 cm
(C) 3 cm
(D) 9 cm

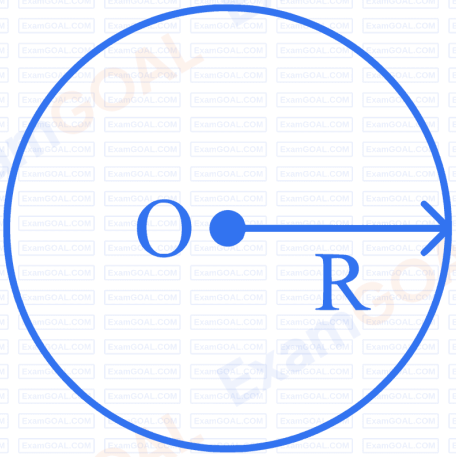
Open solution

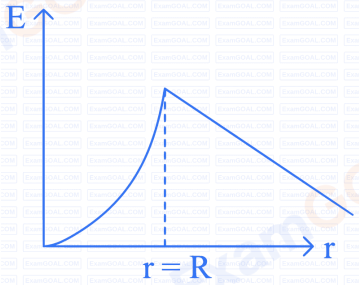
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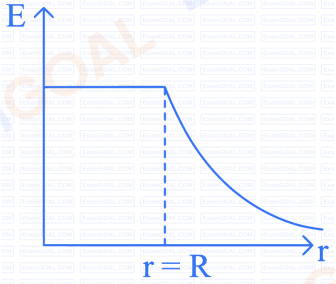
Q79. JEE Main 2023 (Online) 8th April Morning Shift

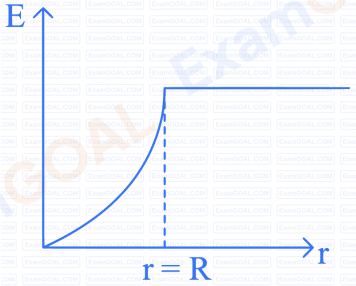
Open solution

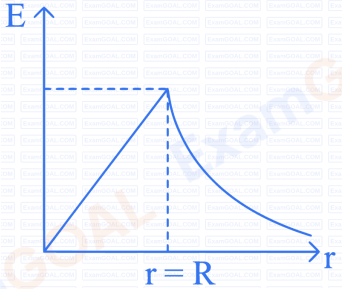
Graphical variation of electric field due to a uniformly charged insulating solid sphere of radius R, with distance r from the centre O is represented by:



(A)


(B)


(C)


(D)


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Q80. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

A dipole comprises of two charged particles of identical magnitude q and opposite in nature. The mass 'm' of the positive charged particle is half of the mass of the negative charged particle. The two charges are separated by a distance 'l'. If the dipole is placed in a uniform electric field '$\bar{E}$'; in such a way that dipole axis makes a very small angle with the electric field, '$\bar{E}$'. The angular frequency of the oscillations of the dipole when released is given by:

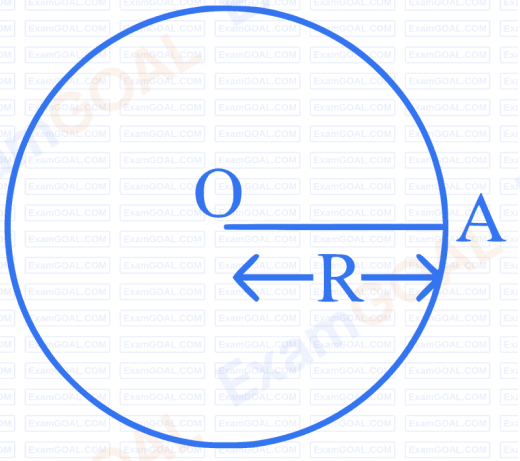
(A) $\sqrt{\frac{3qE}{2ml}}$ (B) $\sqrt{\frac{4qE}{ml}}$ (C) $\sqrt{\frac{8qE}{3ml}}$ (D) $\sqrt{\frac{8qE}{ml}}$

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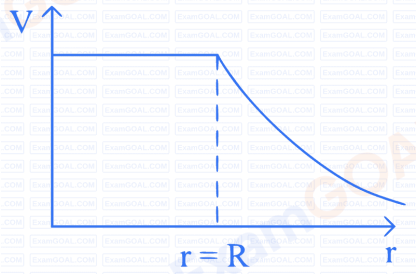
Q81. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

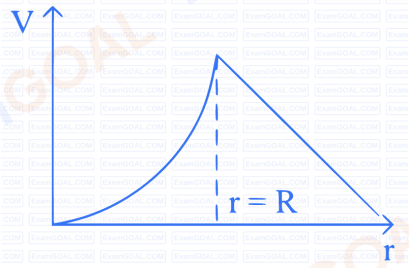
For a uniformly charged thin spherical shell, the electric potential (V) radially away from the centre (O) of shell can be graphically represented as -



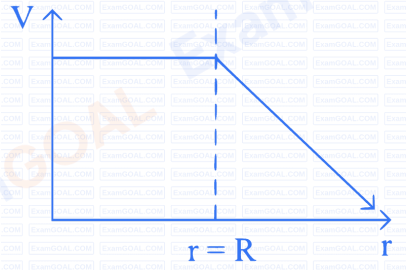
(A)



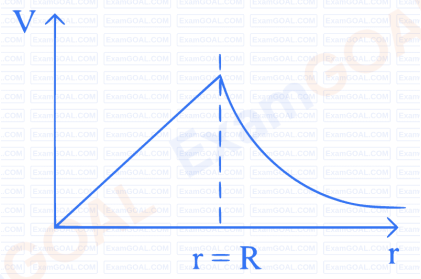
(B)



(C)



(D)

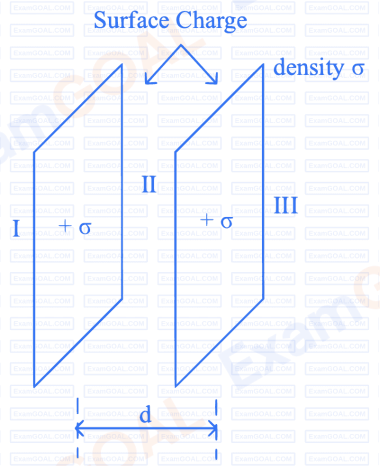


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Q82. JEE Main 2023 (Online) 1st February Morning Shift

Let σ be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric fields in three different regions E_I, E_{II} and E_{III} are:

Open solution



(A) $\vec{E}_I = 0, \vec{E}_{II} = \frac{\sigma}{\epsilon_0}\hat{n}, E_{III} = 0$

(B) $\vec{E}_I = \frac{\sigma}{2\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{2\epsilon_0}\hat{n}$

(C) $\vec{E}_I = -\frac{\sigma}{\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{\epsilon_0}\hat{n}$

(D) $\vec{E}_I = \frac{2\sigma}{\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{2\sigma}{\epsilon_0}\hat{n}$

Q83. JEE Main 2023 (Online) 31st January Evening Shift

Considering a group of positive charges, which of the following statements is correct ?

Open solution

(A) Net potential of the system cannot be zero at a point but net electric field can be zero at that point

(B) Net potential of the system at a point can be zero but net electric field can't be zero at that point.

(C) Both the net potential and the net electric field cannot be zero at a point.

(D) Both the net potential and the net field can be zero at a point.

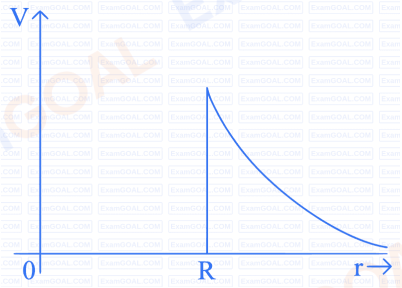
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Q84. JEE Main 2023 (Online) 31st January Morning Shift

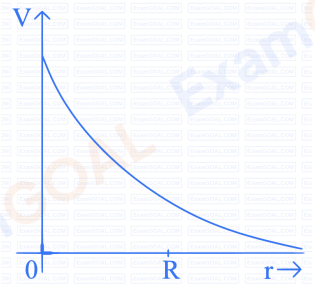
Which of the following correctly represents the variation of electric potential (V) of a charged spherical conductor of radius (R) with radial distance (r) from the center?

Open solution

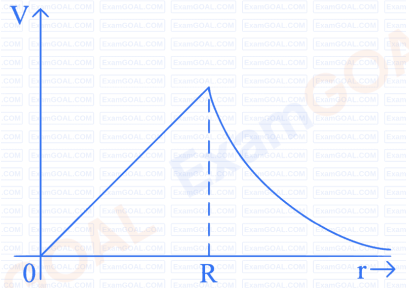
(A)



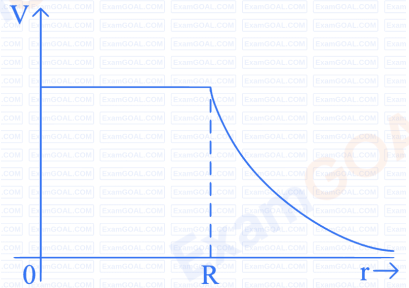
(B)



(C)



(D)



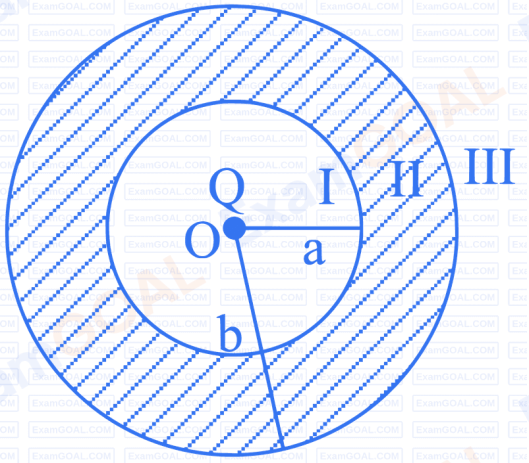
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Q85. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b. The electric field due to charge Q in three different regions I, II and III is given by:

$(\mathrm{I}: r < a, \mathrm{II}: a < r < b, \mathrm{III}: r > b)$



(A) $E_I = 0, E_{II} = 0, E_{III} \neq 0$

(B) $E_I \neq 0, E_{II} = 0, E_{III} = 0$

(C) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$

(D) $E_I = 0, E_{II} = 0, E_{III} = 0$

Q86. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Electric field in a certain region is given by $\vec{\mathrm{E}} = \left(\frac{\mathrm{A}}{x^2}\hat{i} + \frac{\mathrm{B}}{y^3}\hat{j}\right)$. The SI unit of A and B are :

(A) $\mathrm{Nm^2C; Nm^3C}$

(B) $\mathrm{Nm^3C^{-1}; Nm^2C^{-1}}$

(C) $\mathrm{Nm^3C; Nm^2C}$

(D) $\mathrm{Nm^2C^{-1}; Nm^3C^{-1}}$

Q87. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Two isolated metallic solid spheres of radii R and 2R are charged such that both have same charge density σ. The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ'. The ratio $\frac{\sigma'}{\sigma}$ is :

(A) $\frac{5}{3}$

(B) $\frac{5}{6}$

(C) $\frac{9}{4}$

(D) $\frac{4}{3}$

Q88. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

A point charge 2×10^{-2} C is moved from P to S in a uniform electric field of $30\ \mathrm{NC^{-1}}$ directed along positive x-axis. If coordinates of P and S are (1, 2, 0) m and (0, 0, 0) m respectively, the work done by electric field will be

(A) 600 mJ

(B) -1200 mJ

(C) 1200 mJ

(D) -600 mJ

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Q89. JEE Main 2023 (Online) 29th January Morning Shift

In a cuboid of dimension $2\,\mathrm{L} \times 2\,\mathrm{L} \times \mathrm{L}$, a charge q is placed at the center of the surface 'S' having area of $4\,\mathrm{L}^2$. The flux through the opposite surface to 'S' is given by

(A) $\frac{q}{2\epsilon_0}$ (B) $\frac{q}{3\epsilon_0}$ (C) $\frac{q}{12\epsilon_0}$ (D) $\frac{q}{6\in_0}$

Open solution

Q90. JEE Main 2023 (Online) 25th January Evening Shift

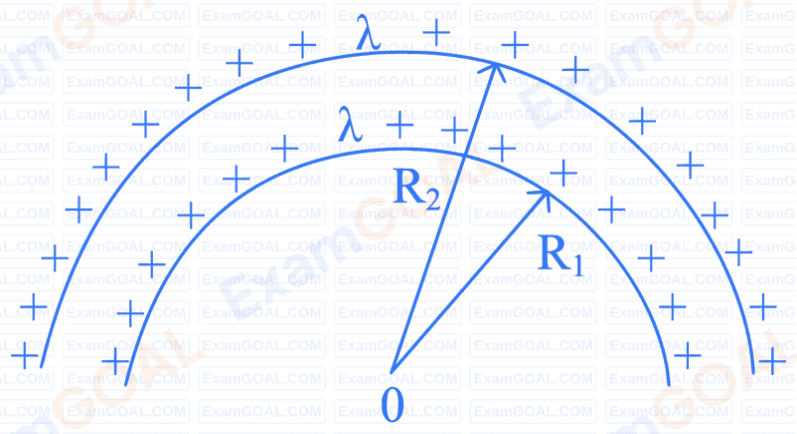
A point charge of 10 μC is placed at the origin. At what location on the X-axis should a point charge of 40 μC be placed so that the net electric field is zero at $x = 2$cm on the X-axis?

(A) $x = 6$ cm (B) $x = 8$ cm (C) $x = 4$ cm (D) $x = -4$ cm

Open solution

Q91. JEE Main 2023 (Online) 24th January Evening Shift

The electric potential at the centre of two concentric half rings of radii R_1 and R_2, having same linear charge density λ is :



(A) $\frac{\lambda}{2\in_0}$ (B) $\frac{\lambda}{\in_0}$ (C) $\frac{2\lambda}{\in_0}$ (D) $\frac{\lambda}{4\in_0}$

Open solution

Q92. JEE Main 2023 (Online) 24th January Morning Shift

If two charges q_1 and q_2 are separated with distance 'd' and placed in a medium of dielectric constant K. What will be the equivalent distance between charges in air for the same electrostatic force?

(A) $d\sqrt{k}$ (B) $1.5d\sqrt{k}$ (C) $k\sqrt{d}$ (D) $2d\sqrt{k}$

Open solution

Q93. JEE Main 2022 (Online) 30th July Evening Shift

A charge q is uniformly distributed along the length L of a rod. It is then bent in the shape of a semicircle. The magnitude of electric field at the centre of semicircle will be :

(A) $\frac{q}{2\epsilon_0 L^2}$ (B) $\frac{q}{2\pi\epsilon_0 L^2}$ (C) $\frac{q}{2\pi\epsilon_0 L}$ (D) $\frac{q}{4\pi^2\epsilon_0 L}$

Open solution

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Q94. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

Two identical metallic spheres A and B when placed at certain distance in air repel each other with a force of F. Another identical uncharged sphere C is first placed in contact with A and then in contact with B and finally placed at midpoint between spheres A and B. The force experienced by sphere C will be:

(A) 3F/2 (B) 3F/4 (C) F (D) 2F

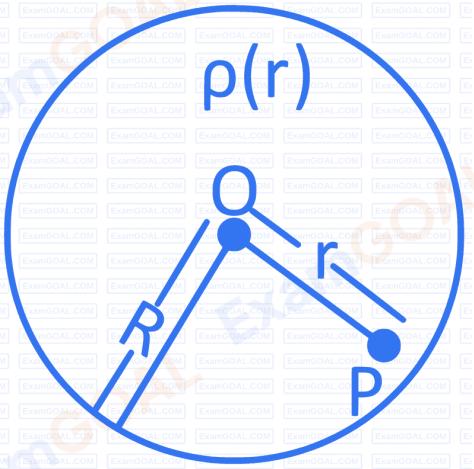
Q95. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

A spherically symmetric charge distribution is considered with charge density varying as

$$\rho(r) = \begin{cases} \rho_0\left(\frac{3}{4} - \frac{r}{R}\right) & \text{for } r \leq R \\ \text{zero} & \text{for } r > R \end{cases}$$

Where, $r(r < R)$ is the distance from the centre O (as shown in figure). The electric field at point P will be:



(A) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{3}{4} - \frac{r}{R}\right)$

(B) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{3}{4} - \frac{r}{R}\right)$

(C) $\frac{\rho_0 r}{4\varepsilon_0}\left(1 - \frac{r}{R}\right)$

(D) $\frac{\rho_0 r}{5\varepsilon_0}\left(1 - \frac{r}{R}\right)$

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Q96. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

Given below are two statements.

Statement I : Electric potential is constant within and at the surface of each conductor.

Statement II : Electric field just outside a charged conductor is perpendicular to the surface of the conductor at every point.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

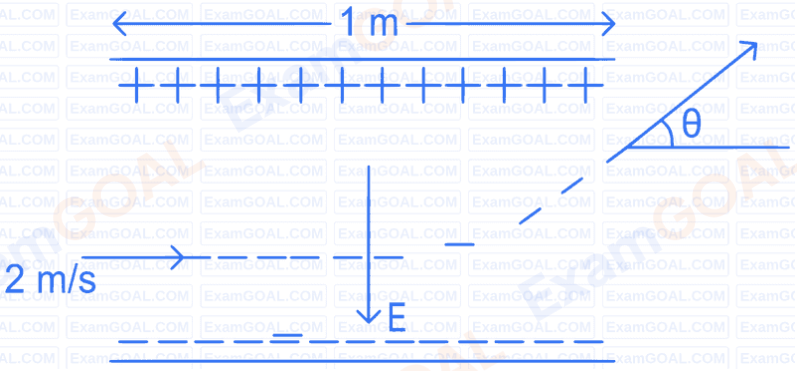
(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q97. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

A uniform electric field $\mathrm{E} = (8\ \mathrm{m/e})\ \mathrm{V/m}$ is created between two parallel plates of length $1\ \mathrm{m}$ as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of $2\ \mathrm{m/s}$. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be _____.



(A) $\tan^{-1}(4)$

(B) $\tan^{-1}(2)$

(C) $\tan^{-1}\left(\frac{1}{3}\right)$

(D) $\tan^{-1}(3)$

Q98. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

A charge of $4\ \mu\mathrm{C}$ is to be divided into two. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be :

(A) $1\ \mu\mathrm{C}$ and $3\ \mu\mathrm{C}$

(B) $2\ \mu\mathrm{C}$ and $2\ \mu\mathrm{C}$

(C) 0 and $4\ \mu\mathrm{C}$

(D) $1.5\ \mu\mathrm{C}$ and $2.5\ \mu\mathrm{C}$

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Q99. JEE Main 2022 (Online) 27th July Morning Shift

Two identical positive charges Q each are fixed at a distance of '2a' apart from each other. Another point charge q_0 with mass 'm' is placed at midpoint between two fixed charges. For a small displacement along the line joining the fixed charges, the charge q_0 executes SHM. The time period of oscillation of charge q_0 will be :

Open solution

(A) $\sqrt{\frac{4\pi^3\varepsilon_0 ma^3}{q_0 Q}}$ (B) $\sqrt{\frac{q_0 Q}{4\pi^3\varepsilon_0 ma^3}}$

(C) $\sqrt{\frac{2\pi^2\varepsilon_0 ma^3}{q_0 Q}}$ (D) $\sqrt{\frac{8\pi^3\varepsilon_0 ma^3}{q_0 Q}}$

Q100. JEE Main 2022 (Online) 26th July Evening Shift

Two uniformly charged spherical conductors A and B of radii $5\ \text{mm}$ and $10\ \text{mm}$ are separated by a distance of $2\ \text{cm}$. If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitudes of the electric fields at the surface of the sphere A and B will be :

Open solution

(A) 1 : 2 (B) 2 : 1 (C) 1 : 1 (D) 1 : 4

Q101. JEE Main 2022 (Online) 29th June Evening Shift

Two point charges Q each are placed at a distance d apart. A third point charge q is placed at a distance x from mid-point on the perpendicular bisector. The value of x at which charge q will experience the maximum Coulomb's force is :

Open solution

(A) x = d (B) $x = \frac{d}{2}$ (C) $x = \frac{d}{\sqrt{2}}$ (D) $x = \frac{d}{2\sqrt{2}}$

Q102. JEE Main 2022 (Online) 29th June Evening Shift

If the electric potential at any point (x, y, z) m in space is given by V = $3x^2$ volt. The electric field at the point (1, 0, 3) m will be :

Open solution

(A) 3 Vm^{-1}, directed along positive x-axis. (B) 3 Vm^{-1}, directed along negative x-axis.

(C) 6 Vm^{-1}, directed along positive x-axis. (D) 6 Vm^{-1}, directed along negative x-axis.

Q103. JEE Main 2022 (Online) 29th June Morning Shift

A positive charge particle of 100 mg is thrown in opposite direction to a uniform electric field of strength 1×10^5 NC^{-1}. If the charge on the particle is 40 μC and the initial velocity is 200 ms^{-1}, how much distance it will travel before coming to the rest momentarily :

Open solution

(A) 1 m (B) 5 m (C) 10 m (D) 0.5 m

Q104. JEE Main 2022 (Online) 28th June Evening Shift

Two point charges A and B of magnitude $+8 \times 10^{-6}$ C and -8×10^{-6} C respectively are placed at a distance d apart. The electric field at the middle point O between the charges is 6.4×10^4 NC^{-1}. The distance 'd' between the point charges A and B is :

Open solution

(A) 2.0 m (B) 3.0 m (C) 1.0 m (D) 4.0 m

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Q105. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Given below are two statements :

Statement I : A point charge is brought in an electric field. The value of electric field at a point near to the charge may increase if the charge is positive.

Statement II : An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero.

Choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

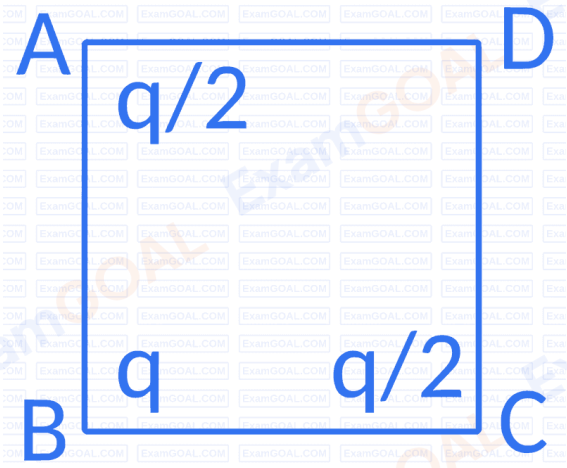
(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q106. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

The three charges q/2, q and q/2 are placed at the corners A, B and C of a square of side 'a' as shown in figure. The magnitude of electric field (E) at the corner D of the square, is :



(A) $\frac{q}{4\pi\in_0 a^2}\left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)$

(B) $\frac{q}{4\pi\in_0 a^2}\left(1+\frac{1}{\sqrt{2}}\right)$

(C) $\frac{q}{4\pi\in_0 a^2}\left(1-\frac{1}{\sqrt{2}}\right)$

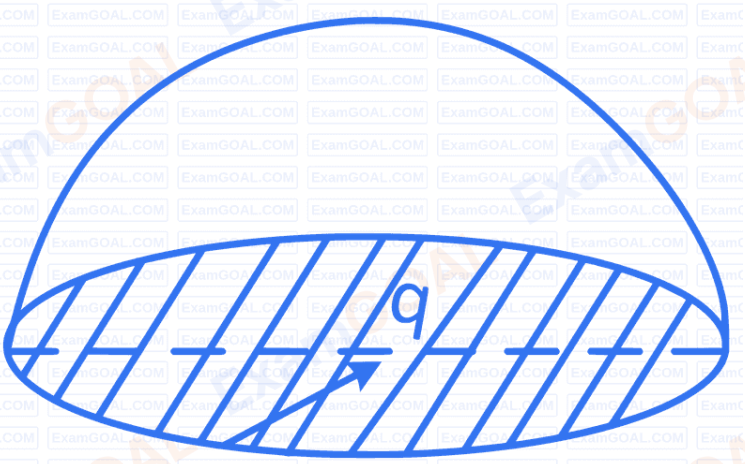
(D) $\frac{q}{4\pi\in_0 a^2}\left(\frac{1}{\sqrt{2}}-\frac{1}{2}\right)$

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Q107. JEE Main 2022 (Online) 27th June Evening Shift

If a charge q is placed at the centre of a closed hemispherical non-conducting surface, the total flux passing through the flat surface would be :

Open solution



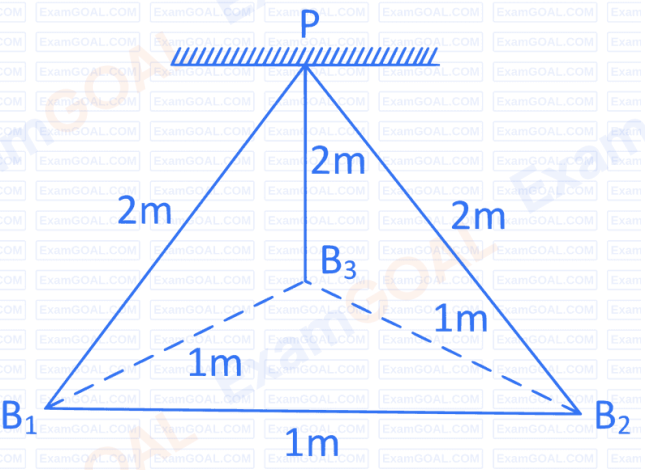
(A) $\frac{q}{\epsilon_0}$ (B) $\frac{q}{2\epsilon_0}$ (C) $\frac{q}{4\epsilon_0}$ (D) $\frac{q}{2\pi\epsilon_0}$

Q108. JEE Main 2022 (Online) 27th June Evening Shift

Three identical charged balls each of charge 2 C are suspended from a common point P by silk threads of 2 m each (as shown in figure). They form an equilateral triangle of side 1m.

The ratio of net force on a charged ball to the force between any two charged balls will be :

Open solution



(A) 1 : 1 (B) 1 : 4 (C) $\sqrt{3}$: 2 (D) $\sqrt{3}$: 1

Q109. JEE Main 2022 (Online) 26th June Evening Shift

Sixty four conducting drops each of radius 0.02 m and each carrying a charge of 5 μC are combined to form a bigger drop. The ratio of surface charge density of bigger drop to the smaller drop will be :

Open solution

(A) 1 : 4 (B) 4 : 1 (C) 1 : 8 (D) 8 : 1

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Q110. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

Given below two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Non-polar materials do not have any permanent dipole moment.

Reason (R) : When a non-polar material is placed in an electric field, the centre of the positive charge distribution of it's individual atom or molecule coincides with the centre of the negative charge distribution.

In the light of above statements, choose the most appropriate answer from the options given below.

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(B) Both (A) and (R) are correct and (R) is not the correct explanation of (A).

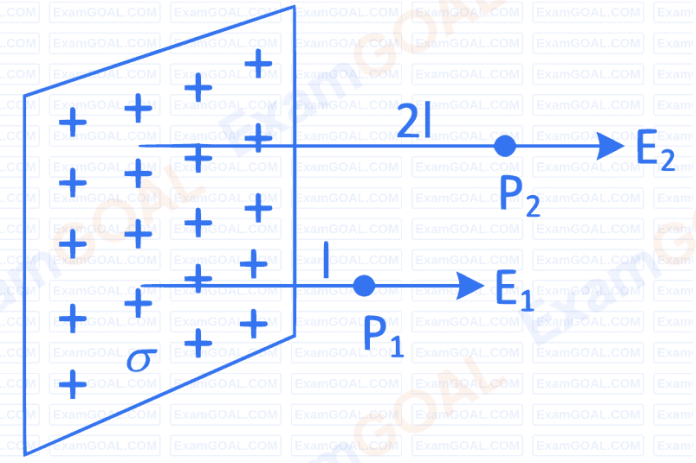
(C) (A) is correct but (R) is not correct.

(D) (A) is not correct but (R) is correct.

Q111. JEE Main 2022 (Online) 25th June Morning Shift

Open solution

In the figure, a very large plane sheet of positive charge is shown. P_1 and P_2 are two points at distance l and 2l from the charge distribution. If σ is the surface charge density, then the magnitude of electric fields E_1 and E_2 at P_1 and P_2 respectively are :



(A) $E_1 = \sigma/\varepsilon_0,\ E_2 = \sigma/2\varepsilon_0$

(B) $E_1 = 2\sigma/\varepsilon_0,\ E_2 = \sigma/\varepsilon_0$

(C) $E_1 = E_2 = \sigma/2\varepsilon_0$

(D) $E_1 = E_2 = \sigma/\varepsilon_0$

Q112. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

Two identical charged particles each having a mass 10 g and charge 2.0×10^{-7}C are placed on a horizontal table with a separation of L between them such that they stay in limiting equilibrium. If the coefficient of friction between each particle and the table is 0.25, find the value of L. [Use g = 10 ms^{-2}]

(A) 12 cm **(B)** 10 cm **(C)** 8 cm **(D)** 5 cm

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Q113. JEE Main 2022 (Online) 24th June Evening Shift

A long cylindrical volume contains a uniformly distributed charge of density ρ. The radius of cylindrical volume is R. A charge particle (q) revolves around the cylinder in a circular path. The kinetic energy of the particle is :

Open solution

(A) $\frac{\rho q R^2}{4\varepsilon_0}$ (B) $\frac{\rho q R^2}{2\varepsilon_0}$

(C) $\frac{q\rho}{4\varepsilon_0 R^2}$ (D) $\frac{4\varepsilon_0 R^2}{q\rho}$

Q114. JEE Main 2022 (Online) 24th June Morning Shift

A vertical electric field of magnitude 4.9×10^5 N/C just prevents a water droplet of a mass 0.1 g from falling. The value of charge on the droplet will be :

Open solution

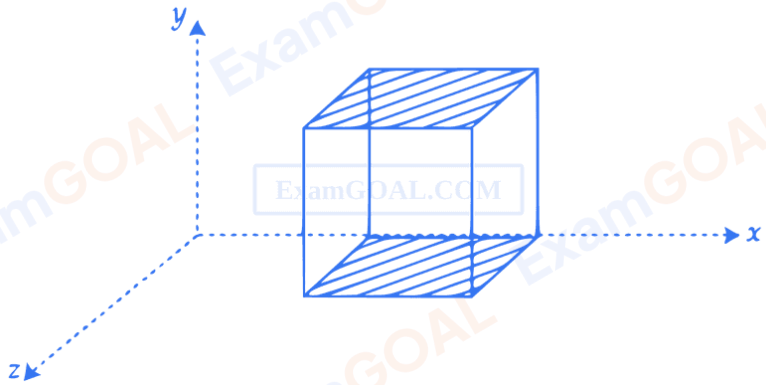
(Given : g = 9.8 m/s^2)

(A) 1.6×10^{-9} C (B) 2.0×10^{-9} C (C) 3.2×10^{-9} C (D) 0.5×10^{-9} C

Q115. JEE Main 2021 (Online) 1st September Evening Shift

A cube is placed inside an electric field, $\vec{E} = 150y^2\hat{j}$. The side of the cube is 0.5 m and is placed in the field as shown in the given figure. The charge inside the cube is :

Open solution



(A) 3.8×10^{-11} C (B) 8.3×10^{-11} C (C) 3.8×10^{-12} C (D) 8.3×10^{-12} C

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Q116. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

Choose the incorrect statement :

(1) The electric lines of force entering into a Gaussian surface provide negative flux.

(2) A charge 'q' is placed at the centre of a cube. The flux through all the faces will be the same.

(3) In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero.

(4) When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

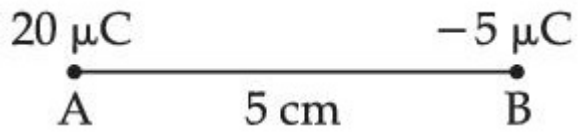
Choose the most appropriate answer from the options given below

(A) (3) and (4) only **(B)** (2) and (4) only **(C)** (4) only **(D)** (1) and (3) only

Q117. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

Two particles A and B having charges $20\mu C$ and $-5\mu C$ respectively are held fixed with a separation of 5 cm. At what position a third charged particle should be placed so that it does not experience a net electric force?



(A) At 5 cm from 20 μC on the left side of system

(B) At 5 cm from -5 μC on the right side

(C) At 1.25 cm from -5 μC between two charges

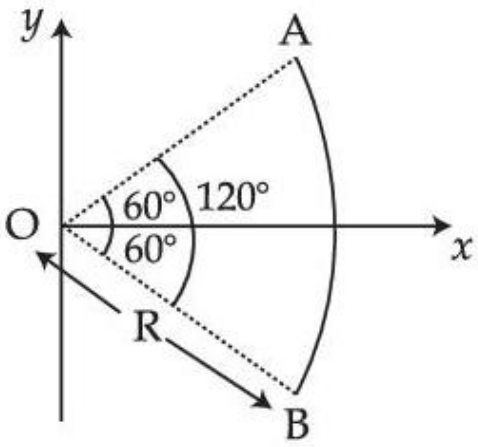
(D) At midpoint between two charges

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Q118. JEE Main 2021 (Online) 27th August Evening Shift

Figure shows a rod AB, which is bent in a 120° circular arc of radius R. A charge (−Q) is uniformly distributed over rod AB. What is the electric field $\vec{E}$ at the centre of curvature O ?

Open solution



(A) $\frac{3\sqrt{3}Q}{8\pi\varepsilon_0 R^2}(\hat{i})$

(B) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(\hat{i})$

(C) $\frac{3\sqrt{3}Q}{16\pi^2\varepsilon_0 R^2}(\hat{i})$

(D) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(-\hat{i})$

Q119. JEE Main 2021 (Online) 27th August Morning Shift

A uniformly charged disc of radius R having surface charge density σ is placed in the xy plane with its center at the origin. Find the electric field intensity along the z-axis at a distance Z from origin :-

Open solution

(A) $E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{Z}{(Z^2+R^2)^{1/2}}\right)$

(B) $E = \frac{\sigma}{2\varepsilon_0}\left(1 + \frac{Z}{(Z^2+R^2)^{1/2}}\right)$

(C) $E = \frac{2\varepsilon_0}{\sigma}\left(\frac{1}{(Z^2+R^2)^{1/2}} + Z\right)$

(D) $E = \frac{\sigma}{2\varepsilon_0}\left(\frac{1}{(Z^2+R^2)} + \frac{1}{Z^2}\right)$

Q120. JEE Main 2021 (Online) 26th August Evening Shift

The two thin coaxial rings, each of radius 'a' and having charges +Q and −Q respectively are separated by a distance of 's'. The potential difference between the centres of the two rings is :

Open solution

(A) $\frac{Q}{2\pi\varepsilon_0}\left[\frac{1}{a} + \frac{1}{\sqrt{s^2+a^2}}\right]$

(B) $\frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a} + \frac{1}{\sqrt{s^2+a^2}}\right]$

(C) $\frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a} - \frac{1}{\sqrt{s^2+a^2}}\right]$

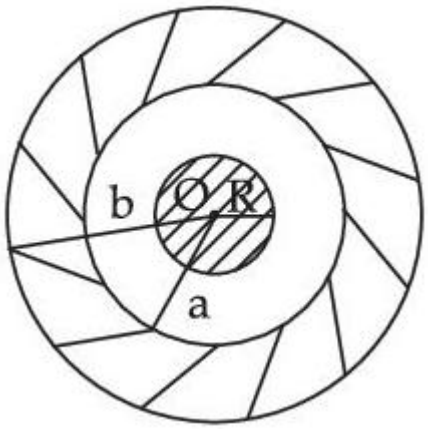
(D) $\frac{Q}{2\pi\varepsilon_0}\left[\frac{1}{a} - \frac{1}{\sqrt{s^2+a^2}}\right]$

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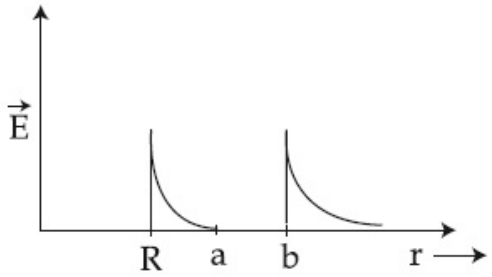
Q121. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

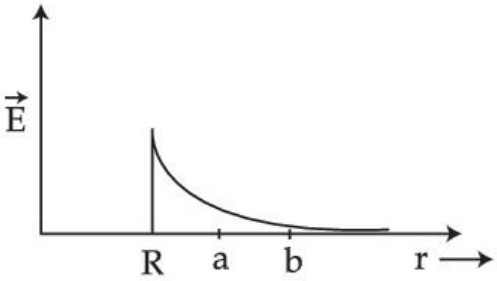
A solid metal sphere of radius R having charge q is enclosed inside the concentric spherical shell of inner radius a and outer radius b as shown in the figure. The approximate variation of electric field $\vec{E}$ as a function of distance r from centre O is given by



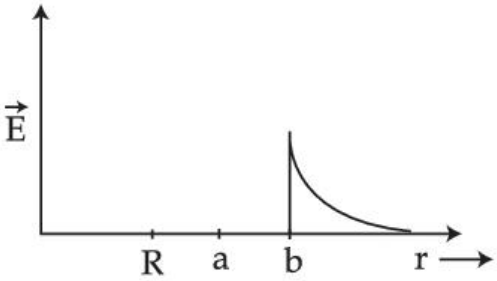
(A)



(B)

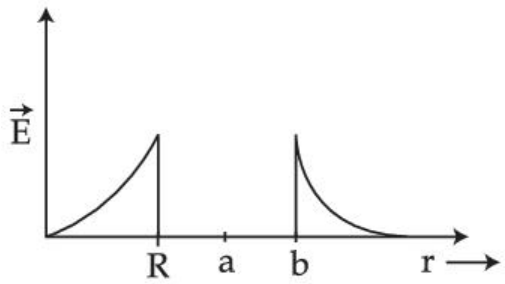


(C)



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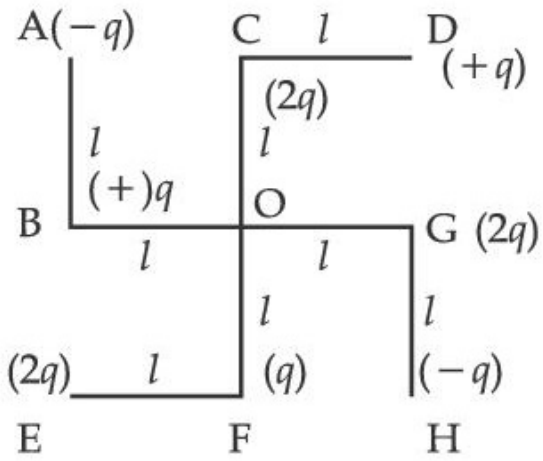
(D)



Q122. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

What will be the magnitude of electric field at point O as shown in the figure? Each side of the figure is l and perpendicular to each other.



(A) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{q}{(2l^2)}\left(2\sqrt{2}-1\right)$

(C) $\frac{q}{4\pi\varepsilon_0(2l)^2}$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{2l^2}\left(\sqrt{2}\right)$

Q123. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

Two identical tennis balls each having mass 'm' and charge 'q' are suspended from a fixed point by threads of length 'l'. What is the equilibrium separation when each thread makes a small angle 'θ' with the vertical?

(A) $x=\left(\frac{q^2l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{2}}$

(B) $x=\left(\frac{q^2l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(C) $x=\left(\frac{q^2l^2}{2\pi\varepsilon_0 m^2g}\right)^{\frac{1}{3}}$

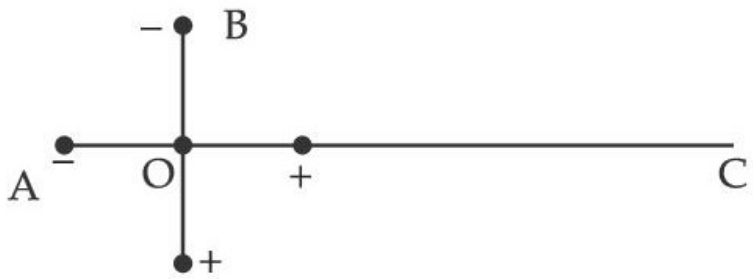
(D) $x=\left(\frac{q^2l^2}{2\pi\varepsilon_0 m^2g^2}\right)^{\frac{1}{3}}$

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Q124. JEE Main 2021 (Online) 25th July Evening Shift

Two ideal electric dipoles A and B, having their dipole moment p_1 and p_2 respectively are placed on a plane with their centres at O as shown in the figure. At point C on the axis of dipole A, the resultant electric field is making an angle of 37° with the axis. The ratio of the dipole moment of A and B, $\frac{p_1}{p_2}$ is : (take $\sin 37^\circ = \frac{3}{5}$)

Open solution



(A) $\frac{3}{8}$ (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{4}{3}$

Q125. JEE Main 2021 (Online) 22th July Evening Shift

An electric dipole is placed on x-axis in proximity to a line charge of linear charge density 3.0×10^{-6} C/m. Line charge is placed on z-axis and positive and negative charge of dipole is at a distance of 10 mm and 12 mm from the origin respectively. If total force of 4N is exerted on the dipole, find out the amount of positive or negative charge of the dipole.

Open solution

(A) 0.485 mC (B) 815.1 nC (C) 8.8 μC (D) 4.44 μC

Q126. JEE Main 2021 (Online) 20th July Morning Shift

A certain charge Q is divided into two parts q and (Q − q). How should the charges Q and q be divided so that q and (Q − q) placed at a certain distance apart experience maximum electrostatic repulsion?

Open solution

(A) Q = 2q (B) Q = 4q (C) Q = 3q (D) Q = $\frac{q}{2}$

Q127. JEE Main 2021 (Online) 18th March Morning Shift

An oil drop of radius 2 mm with a density 3g cm^{-3} is held stationary under a constant electric field 3.55×10^5 V m^{-1} in the Millikan's oil drop experiment. What is the number of excess electrons that the oil drop will possess? (consider g = 9.81 m/s^2)

Open solution

(A) 48.8×10^{11} (B) 1.73×10^{10} (C) 17.3×10^{10} (D) 1.73×10^{12}

Q128. JEE Main 2021 (Online) 16th March Evening Shift

Find out the surface charge density at the intersection of point x = 3 m plane and x-axis, in the region of uniform line charge of 8 nC/m lying along the z-axis in free space.

Open solution

(A) 0.424 nC m^{-2} (B) 4.0 nC m^{-2} (C) 47.88 C/m (D) 0.07 nC m^{-2}

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Q129. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

Given below are two statements:

Statement I : An electric dipole is placed at the center of a hollow sphere. The flux of the electric field through the sphere is zero but the electric field is not zero anywhere in the sphere.

Statement II : If R is the radius of a solid metallic sphere and Q be the total charge on it. The electric field at any point on the spherical surface of radius r (< R) is zero but the electric flux passing through this closed spherical surface of radius r is not zero.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

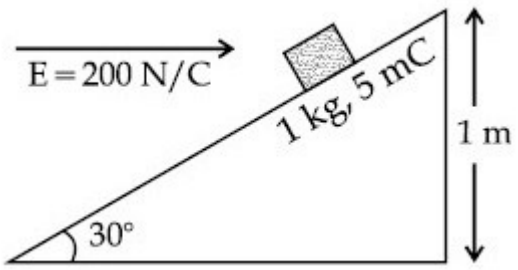
(D) Both Statement I and Statement II are false

Q130. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

An inclined plane making an angle of 30° with the horizontal is placed in a uniform horizontal electric field $200\frac{N}{C}$ as shown in the figure. A body of mass 1 kg and charge 5 mC is allowed to slide down from rest at a height of 1 m. If the coefficient of friction is 0.2, find the time taken by the body to reach the bottom.

[g = 9.8 m/s^2; $\sin 30^\circ = \frac{1}{2}$; $\cos 30^\circ = \frac{\sqrt{3}}{2}$]



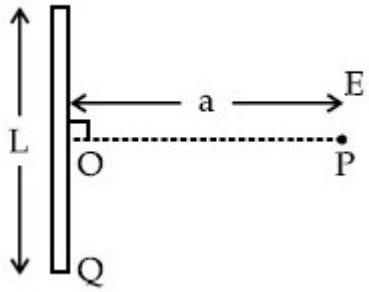
(A) 0.46 s (B) 0.92 s (C) 1.3 s (D) 2.3 s

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Q131. JEE Main 2021 (Online) 26th February Morning Shift

Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the wire is a = $\frac{\sqrt{3}}{2}L$.

Open solution



(A) $\frac{Q}{4\pi\varepsilon_0 L^2}$
(B) $\frac{Q}{3\pi\varepsilon_0 L^2}$
(C) $\frac{Q}{2\sqrt{3}\pi\varepsilon_0 L^2}$
(D) $\frac{\sqrt{3}Q}{4\pi\varepsilon_0 L^2}$

Q132. JEE Main 2021 (Online) 25th February Evening Shift

An electron with kinetic energy K_1 enters between parallel plates of a capacitor at an angle 'α' with the plates. It leaves the plates at angle 'β' with kinetic energy K_2. Then the ratio of kinetic energies $K_1 : K_2$ will be :

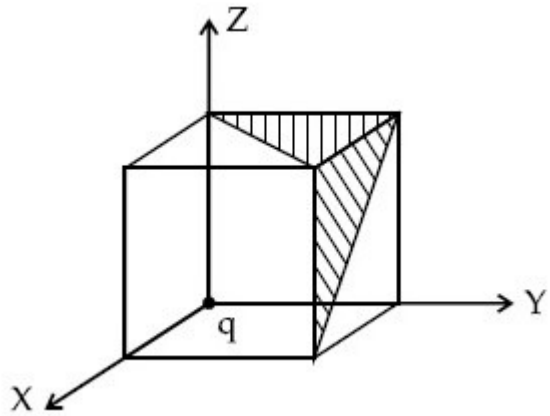
Open solution

(A) $\frac{\cos^2\beta}{\cos^2\alpha}$
(B) $\frac{\cos\beta}{\cos\alpha}$
(C) $\frac{\sin^2\beta}{\cos^2\alpha}$
(D) $\frac{\cos\beta}{\sin\alpha}$

Q133. JEE Main 2021 (Online) 25th February Evening Shift

A charge 'q' is placed at one corner of a cube as shown in figure. The flux of electrostatic field $\vec{E}$ through the shaded area is :

Open solution



(A) $\frac{q}{24\varepsilon_0}$
(B) $\frac{q}{48\varepsilon_0}$
(C) $\frac{q}{4\varepsilon_0}$
(D) $\frac{q}{8\varepsilon_0}$

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Q134. JEE Main 2021 (Online) 24th February Evening Shift

Two electrons each are fixed at a distance '2d'. A third charge proton placed at the midpoint is displaced slightly by a distance x (x << d) perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency : (m = mass of charged particle)

Open solution

(A) $\left(\frac{2q^2}{\pi\varepsilon_0 md^3}\right)^{\frac{1}{2}}$

(B) $\left(\frac{q^2}{2\pi\varepsilon_0 md^3}\right)^{\frac{1}{2}}$

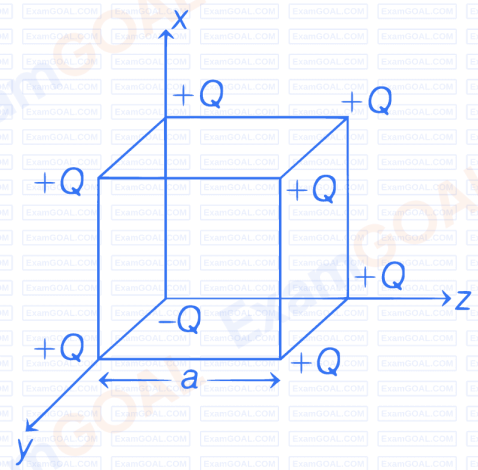
(C) $\left(\frac{2\pi\varepsilon_0 md^3}{q^2}\right)^{\frac{1}{2}}$

(D) $\left(\frac{\pi\varepsilon_0 md^3}{2q^2}\right)^{\frac{1}{2}}$

Q135. JEE Main 2021 (Online) 24th February Morning Shift

A cube of side 'a' has point charges +Q located at each of its vertices except at the origin where the charge is $-$Q. The electric field at the centre of cube is :

Open solution



(A) $\frac{2Q}{3\sqrt{3}\pi\varepsilon_0 a^2}\left(\hat{x}+\hat{y}+\hat{z}\right)$

(B) $\frac{-Q}{3\sqrt{3}\pi\varepsilon_0 a^2}\left(\hat{x}+\hat{y}+\hat{z}\right)$

(C) $\frac{Q}{3\sqrt{3}\pi\varepsilon_0 a^2}\left(\hat{x}+\hat{y}+\hat{z}\right)$

(D) $\frac{-2Q}{3\sqrt{3}\pi\varepsilon_0 a^2}\left(\hat{x}+\hat{y}+\hat{z}\right)$

Q136. JEE Main 2020 (Online) 6th September Evening Slot

Consider the force F on a charge 'q' due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F, if 'q' is placed at distance r from the centre of the shell?

Open solution

(A) $\frac{1}{4\pi\varepsilon_0}\frac{qQ}{R^2} > F > 0$ for r < R

(B) $F = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}$ for r > R

(C) $F = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}$ for all r

(D) $F = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{R^2}$ for r < R

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Q137. JEE Main 2020 (Online) 6th September Evening Slot

Two identical electric point dipoles have dipole moments $\overrightarrow{p_1} = p\hat{i}$ and $\overrightarrow{p_2} = -p\hat{i}$ and are held on the x axis at distance 'a' from each other. When released, they move along the x-axis with the direction of their dipole moments remaining unchanged. If the mass of each dipole is 'm', their speed when they are infinitely far apart is :

Open solution

(A) $\frac{p}{a}\sqrt{\frac{3}{2\pi\in_0 ma}}$

(B) $\frac{p}{a}\sqrt{\frac{1}{\pi\in_0 ma}}$

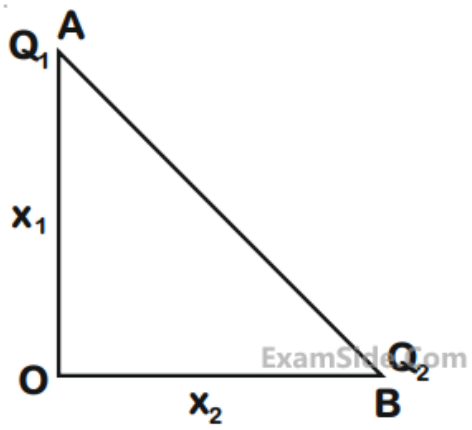
(C) $\frac{p}{a}\sqrt{\frac{1}{2\pi\in_0 ma}}$

(D) $\frac{p}{a}\sqrt{\frac{2}{\pi\in_0 ma}}$

Q138. JEE Main 2020 (Online) 6th September Morning Slot

Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then $\frac{Q_1}{Q_2}$ is proportional to :

Open solution



(A) $\frac{x_1^3}{x_2^3}$

(B) $\frac{x_2^2}{x_1^2}$

(C) $\frac{x_1}{x_2}$

(D) $\frac{x_2}{x_1}$

Q139. JEE Main 2020 (Online) 5th September Evening Slot

Ten charges are placed on the circumference of a circle of radius R with constant angular separation between successive charges. Alternate charges 1, 3, 5, 7, 9 have charge (+q) each, while 2, 4, 6, 8, 10 have charge (−q) each. The potential V and the electric field E at the centre of the circle are respectively.
(Take V = 0 at infinity)

Open solution

(A) V = 0; E = 0

(B) $V = \frac{10q}{4\pi\varepsilon_0 R}$; $E = \frac{10q}{4\pi\varepsilon_0 R^2}$

(C) $V = \frac{10q}{4\pi\varepsilon_0 R}$; E = 0

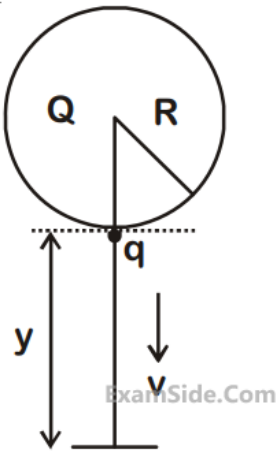
(D) V = 0; $E = \frac{10q}{4\pi\varepsilon_0 R^2}$

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Q140. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

A solid sphere of radius R carries a charge Q + q distributed uniformly over its volume. A very small point like piece of it of mass m gets detached from the bottom of the sphere and falls down vertically under gravity. This piece carries charge q. If it acquires a speed v when it has fallen through a vertical height y (see figure), then :
(assume the remaining portion to be spherical).



(A) $v^2 = y\left[\frac{qQ}{4\pi\varepsilon_0 R(R+y)m} + g\right]$

(B) $v^2 = 2y\left[\frac{qQR}{4\pi\varepsilon_0 (R+y)^3 m} + g\right]$

(C) $v^2 = 2y\left[\frac{qQ}{4\pi\varepsilon_0 R(R+y)m} + g\right]$

(D) $v^2 = y\left[\frac{qQ}{4\pi\varepsilon_0 R^2 ym} + g\right]$

Q141. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x-direction, where a and E_0 are constants. Initially the particle was at rest at x = 0. Other than the initial position the kinetic energy of the particle becomes zero when the distance of the particle from the origin is :

(A) a (B) $\sqrt{\frac{2}{a}}$ (C) $\sqrt{\frac{3}{a}}$ (D) $\sqrt{\frac{1}{a}}$

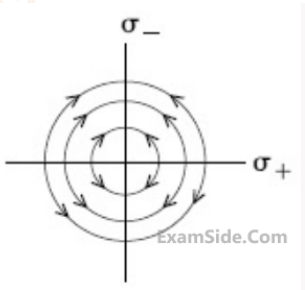
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Q142. JEE Main 2020 (Online) 4th September Morning Slot

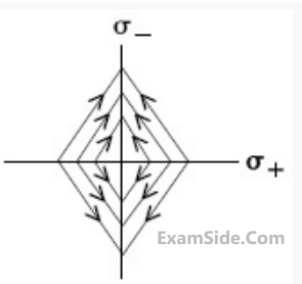
Open solution

Two charged thin infinite plane sheets of uniform surface charge density σ_+ and σ_-, where $|\sigma_+| > |\sigma_-|$, intersect at right angle. Which of the following best represents the electric field lines for this system :

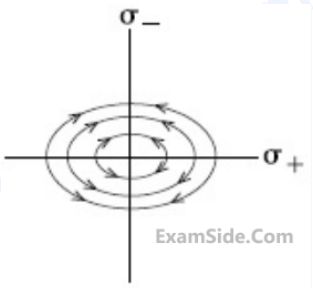
(A)



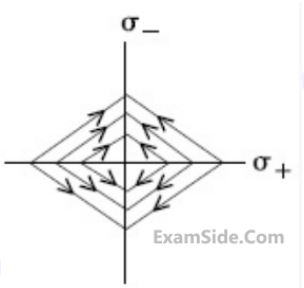
(B)



(C)



(D)

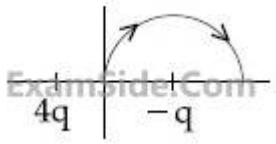


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Q143. JEE Main 2020 (Online) 4th September Morning Slot

A two point charges 4q and -q are fixed on the x-axis at x = $-\frac{d}{2}$ and x = $\frac{d}{2}$ respectively. If a third point charge 'q' is taken from the origin to x = d along the semicircle as shown in the figure, the energy of the charge will :

Open solution



(A) increase by $\frac{3q^2}{4\pi\varepsilon_0 d}$

(B) increase by $\frac{2q^2}{3\pi\varepsilon_0 d}$

(C) decrease by $\frac{q^2}{4\pi\varepsilon_0 d}$

(D) decrease by $\frac{4q^2}{3\pi\varepsilon_0 d}$

Q144. JEE Main 2020 (Online) 3rd September Evening Slot

Concentric metallic hollow spheres of radii R and 4R hold charges Q_1 and Q_2 respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference V(R) − V(4R) is :

Open solution

(A) $\frac{3Q_2}{4\pi\varepsilon_0 R}$

(B) $\frac{3Q_1}{4\pi\varepsilon_0 R}$

(C) $\frac{3Q_1}{16\pi\varepsilon_0 R}$

(D) $\frac{Q_2}{4\pi\varepsilon_0 R}$

Q145. JEE Main 2020 (Online) 3rd September Evening Slot

Hydrogen ion and singly ionized helium atom are accelerated, from rest, through the same potential difference. The ratio of final speeds of hydrogen and helium ions is close to :

Open solution

(A) 2 : 1 (B) 1 : 2 (C) 5 : 7 (D) 10 : 7

Q146. JEE Main 2020 (Online) 3rd September Morning Slot

Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}R$ and $\frac{1}{3}R$ have 12 μC and −3 μC charges, respectively, and are at a large distance from each other. They are now connected by a conducting wire. A long time after this is done the charges on S_1 and S_2 are respectively :

Open solution

(A) 4.5 μC on both

(B) +4.5 μC and −4.5 μC

(C) 6 μC and 3 μC

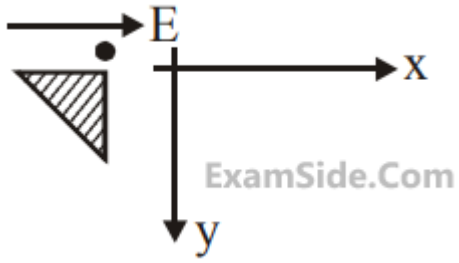
(D) 3 μC and 6 μC

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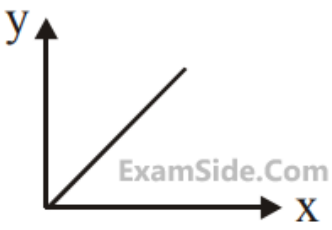
Q147. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

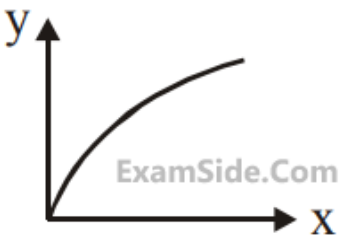
A small point mass carrying some positive charge on it, is released from the edge of a table. There is a uniform electric field in this region in the horizontal direction. Which of the following options then correctly describe the trajectory of the mass?
(Curves are drawn schematically and are not to scale).



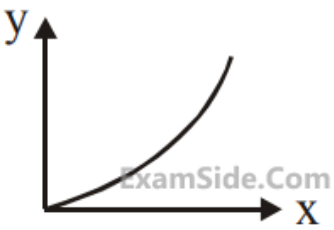
(A)



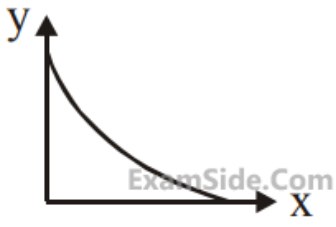
(B)



(C)



(D)

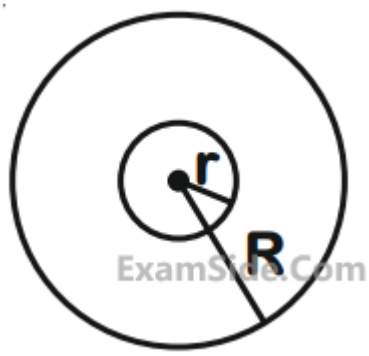


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Q148. JEE Main 2020 (Online) 2nd September Evening Slot

A charge Q is distributed over two concentric conducting thin spherical shells radii r and R (R > r). If the surface charge densities on the two shells are equal, the electric potential at the common centre is :

Open solution



(A) $\frac{1}{4\pi\varepsilon_0}\frac{(R+r)}{(R^2+r^2)}$ Q

(B) $\frac{1}{4\pi\varepsilon_0}\frac{(R+r)}{2(R^2+r^2)}Q$

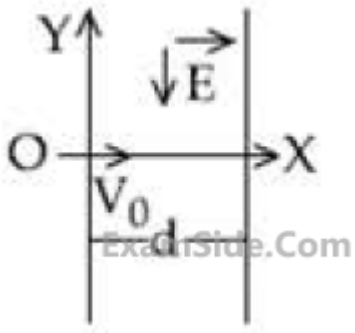
(C) $\frac{1}{4\pi\varepsilon_0}\frac{(R+2r)Q}{2(R^2+r^2)}$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{(2R+r)}{(R^2+r^2)}Q$

Q149. JEE Main 2020 (Online) 2nd September Morning Slot

A charged particle (mass m and charge q) moves along X-axis with velocity V_0. When it passes through the origin it enters a region having uniform electric field $\vec{E} = -E\hat{j}$ which extends upto x = d.
Equation of path of electron in the region x > d is

Open solution



(A) y = $\frac{qEd}{mV_0^2}(x-d)$

(B) y = $\frac{qEd}{mV_0^2}\left(\frac{d}{2}-x\right)$

(C) y = $\frac{qEd}{mV_0^2}x$

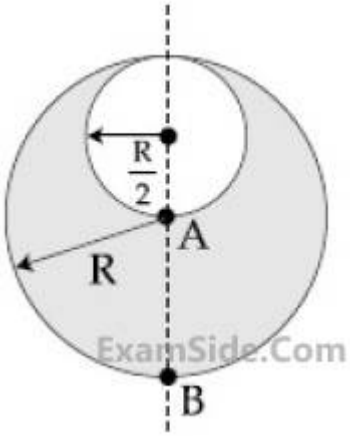
(D) y = $\frac{qEd^2}{mV_0^2}x$

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Q150. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

Consider a sphere of radius R which carries a uniform charge density ρ. If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\frac{\left|\overrightarrow{E_A}\right|}{\left|\overrightarrow{E_B}\right|}$ of magnitude of electric field $\overrightarrow{E_A}$ and $\overrightarrow{E_B}$, respectively, at points A and B due to the remaining portion is :



(A) $\frac{17}{54}$ (B) $\frac{18}{54}$ (C) $\frac{18}{34}$ (D) $\frac{21}{34}$

Q151. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

An electric dipole of moment
$\overrightarrow{p}=\left(-\hat{i}-3\hat{j}+2\hat{k}\right)\times 10^{-29}$ C.m is
at the origin (0, 0, 0). The electric field due to this dipole at
$\overrightarrow{r}=+\hat{i}+3\hat{j}+5\hat{k}$ (note that $\overrightarrow{r}.\overrightarrow{p}=0$) is parallel to :

(A) $\left(+\hat{i}+3\hat{j}-2\hat{k}\right)$ (B) $\left(+\hat{i}-3\hat{j}-2\hat{k}\right)$
(C) $\left(-\hat{i}+3\hat{j}-2\hat{k}\right)$ (D) $\left(-\hat{i}-3\hat{j}+2\hat{k}\right)$

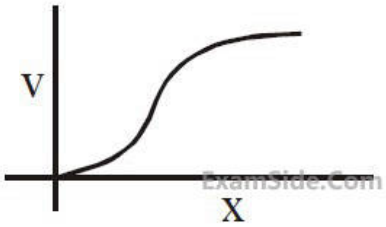
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Q152. JEE Main 2020 (Online) 8th January Evening Slot

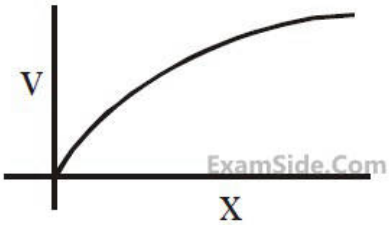
A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale)

Open solution

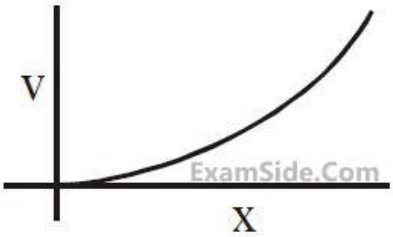
(A)



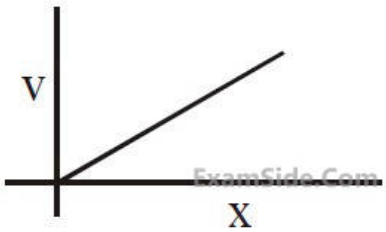
(B)



(C)



(D)



Q153. JEE Main 2020 (Online) 8th January Evening Slot

Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2, respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that $E_1/E_2 = R_1/R_2$. Then the ratio V_1 (on S_1) / V_2 (on S_2) of the electrostatic potentials on each sphere is :

Open solution

(A) $(R_1/R_2)^2$ (B) (R_2/R_1) (C) $(R_1/R_2)^3$ (D) R_1/R_2

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Q154. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

In finding the electric field using Gauss Law the formula $\left|\vec{E}\right| = \frac{q_{enc}}{\varepsilon_0 |A|}$ is applicable. In the formula ε_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. The equation can be used in which of the following situation?

(A) Only when $\left|\vec{E}\right|$ = constant on the surface.

(B) For any choice of Gaussian surface.

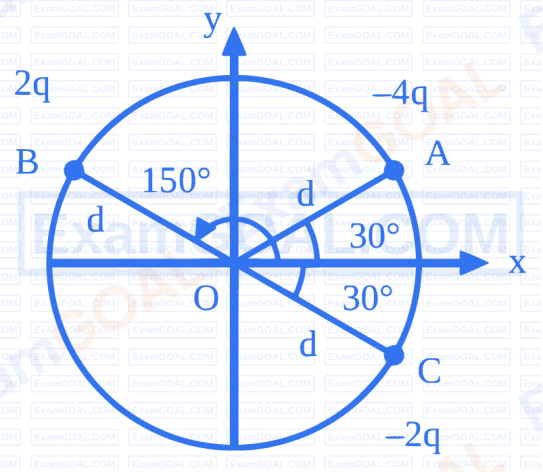
(C) Only when the Gaussian surface is an equipotential surface.

(D) Only when the Gaussian surface is an equipotential surface and $\left|\vec{E}\right|$ is constant on the surface.

Q155. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

Three charged particles A, B and C with charges −4q, 2q and −2q are present on the circumference of a circle of radius d. The charged particles A, C and centre O of the circle form an equilateral triangle as shown in figure. Electric field at O along x-direction is :



(A) $\frac{3\sqrt{3}q}{4\pi\varepsilon_0 d^2}$

(B) $\frac{\sqrt{3}q}{4\pi\varepsilon_0 d^2}$

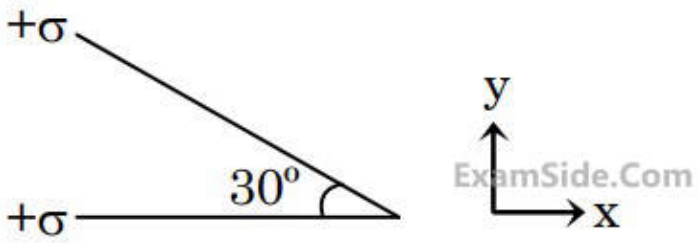
(C) $\frac{\sqrt{3}q}{\pi\varepsilon_0 d^2}$

(D) $\frac{2\sqrt{3}q}{\pi\varepsilon_0 d^2}$

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Q156. JEE Main 2020 (Online) 7th January Morning Slot

Two infinite planes each with uniform surface charge density σ are kept in such a way that the angle between them is 30°. The electric field in the region shown between them is given by :



(A) $\frac{\sigma}{\in_0}\left[\left(1+\frac{\sqrt{3}}{2}\right)\hat{y}+\frac{\hat{x}}{2}\right]$

(B) $\frac{\sigma}{2\in_0}\left[\left(1+\sqrt{3}\right)\hat{y}+\frac{\hat{x}}{2}\right]$

(C) $\frac{\sigma}{2\in_0}\left[\left(1+\sqrt{3}\right)\hat{y}-\frac{\hat{x}}{2}\right]$

(D) $\frac{\sigma}{2\in_0}\left[\left(1-\frac{\sqrt{3}}{2}\right)\hat{y}-\frac{\hat{x}}{2}\right]$

Open solution

Q157. JEE Main 2019 (Online) 12th April Evening Slot

Let a total charge 2Q be distributed in a sphere of radius R, with the charge density given by ρ(r) = kr, where r is the distance from the centre. Two charges A and B, of –Q each, are placed on diametrically opposite points, at equal distance, a from the centre. If A and B do not experience any force, then :

(A) $a = 8^{-1/4}R$ (B) $a = 2^{-1/4}R$ (C) $a = \frac{3R}{2^{1/4}}$ (D) $a = \frac{R}{\sqrt{3}}$

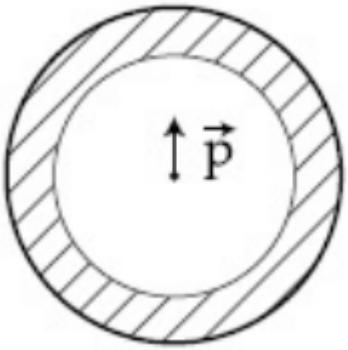
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Q158. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

Shown in the figure is a shell made of a conductor. It has inner radius a and outer radius b, and carries charge Q. At its centre is a dipole $\vec{P}$ as shown. In this case :



(A) surface charge density on the inner surface is uniform and equal to $\frac{(Q/2)}{4\pi a^2}$

(B) surface charge density on the inner surface of the shell is zero everywhere

(C) surface charge density on the outer surface depends on $\left|\vec{P}\right|$

(D) electric field outside the shell is the same as that of a point charge at the centre of the shell

Q159. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

A point dipole $\vec{p} = -p_0\hat{x}$ is kept at the origin. The potential and electric field due to this dipole on the y-axis at a distance d are, respectively: (Take V= 0 at infinity)

(A) $\frac{\left|\vec{p}\right|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$

(B) $0, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$

(C) $\frac{\left|\vec{p}\right|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$

(D) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$

Q160. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

In free space, a particle A of charge 1μC is held fixed at a point P. Another particle B of the same charge and mass 4μg is kept at a distance of 1 mm from P. If B is released, then its velocity at a distance of 9 mm from P is : $\left[Take\ \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 Nm^2C^{-2}\right]$

(A) 1.0 m/s

(B) 6.32×10^4 m/s

(C) 2.0×10^3 m/s

(D) 1.5×10^2 m/s

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Q161. JEE Main 2019 (Online) 10th April Morning Slot

A uniformly charged ring of radius 3a and total charge q is placed in xy-plane centred at origin. A point charge q is moving towards the ring along the z-axis and has speed u at z = 4a. The minimum value of u such that it crosses the origin is :

Open solution

(A) $\sqrt{\frac{2}{m}}\left(\frac{2}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)^{1/2}$

(B) $\sqrt{\frac{2}{m}}\left(\frac{1}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)^{1/2}$

(C) $\sqrt{\frac{2}{m}}\left(\frac{1}{5}\frac{q^2}{4\pi\varepsilon_0 a}\right)^{1/2}$

(D) $\sqrt{\frac{2}{m}}\left(\frac{4}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)^{1/2}$

Q162. JEE Main 2019 (Online) 9th April Evening Slot

Four point charges −q, +q, +q and −q are placed on y-axis at y = −2d, y = −d, y = +d and y = +2d, respectively. The magnitude of the electric field E at a point on the x-axis at x = D, with D >> d, will behave as :-

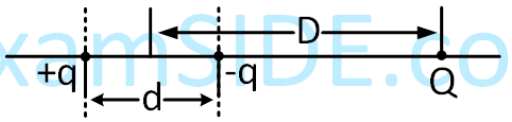
Open solution

(A) $E \propto \frac{1}{D^3}$ (B) $E \propto \frac{1}{D}$ (C) $E \propto \frac{1}{D^4}$ (D) $E \propto \frac{1}{D^2}$

Q163. JEE Main 2019 (Online) 9th April Morning Slot

A system of three charges are placed as shown in the figure :

Open solution



ExamSIDE.com

If D >> d, the potential energy of the system is best given by :

(A) $\frac{1}{4\pi\varepsilon_0}\left[\frac{q^2}{d}+\frac{qQd}{D^2}\right]$

(B) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}-\frac{qQd}{2D^2}\right]$

(C) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}-\frac{qQd}{D^2}\right]$

(D) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}+\frac{2qQd}{D^2}\right]$

Q164. JEE Main 2019 (Online) 8th April Evening Slot

An electric dipole is formed by two equal and opposite charges q with separation d. The charges have same mass m. It is kept in a uniform electric field E. If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is :-

Open solution

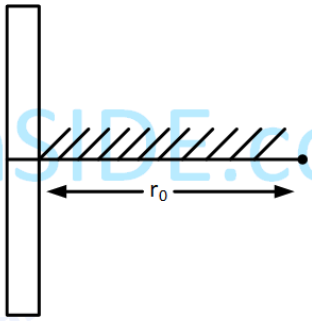
(A) $\sqrt{\frac{qE}{md}}$ (B) $\sqrt{\frac{qE}{2md}}$ (C) $\sqrt{\frac{qE}{-2md}}$ (D) $\sqrt{\frac{2qE}{md}}$

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Q165. JEE Main 2019 (Online) 8th April Evening Slot

A positive point charge is released from rest at a distance r_0 from a positive line charge with uniform density. The speed (v) of the point charge, as a function of instantaneous distance r from line charge, is proportional to :-

Open solution



ExamSIDE.com

(A) $v \propto \left(\frac{r}{r_0}\right)$

(B) $v \propto \ln\left(\frac{r}{r_0}\right)$

(C) $v \propto e^{+r/r_0}$

(D) $v \propto \sqrt{\ln\left(\frac{r}{r_0}\right)}$

Q166. JEE Main 2019 (Online) 8th April Evening Slot

The electric field in a region is given by $\vec{E} = (Ax + B)\hat{i}$, where E is in NC^{-1} and x is in metres. The values of constants are A = 20 SI unit and B = 10 SI unit. If the potential at x = 1 is V_1 and that at x = −5 is V_2, then $V_1 - V_2$ is :-

Open solution

(A) −520 V (B) 180 V (C) −48 V (D) 320 V

Q167. JEE Main 2019 (Online) 8th April Morning Slot

The bob of a simple pendulum has mass 2g and a charge of 5.0 μC. It is at rest in a uniform horizontal electric field of intensity 2000 V/m. At equilibrium, the angle that the pendulum makes with the vertical is : (take g = 10 m/s^2)

Open solution

(A) $\tan^{-1}(5.0)$ (B) $\tan^{-1}(2.0)$ (C) $\tan^{-1}(0.5)$ (D) $\tan^{-1}(0.2)$

Q168. JEE Main 2019 (Online) 8th April Morning Slot

A solid conducting sphere, having a charge Q, is surrounded by an uncharged conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V. If the shell is now given a charge of −4 Q, the new potential difference between the same two surfaces is :

Open solution

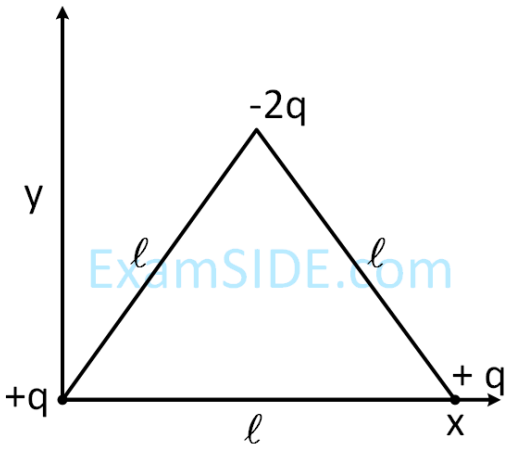
(A) V (B) 2V (C) −2V (D) 4V

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Q169. JEE Main 2019 (Online) 12th January Morning Slot

Determine the electric dipole moment of the system of the three charges, placed on the vertices of an equilateral triangle, as shown in the figure :

Open solution



ExamSIDE.com

(A) $2q\ell\hat{j}$

(B) $(q\ell)\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

(C) $\sqrt{3}\,q\ell\frac{\hat{j}-\hat{i}}{\sqrt{2}}$

(D) $-\sqrt{3}\,q\ell\hat{j}$

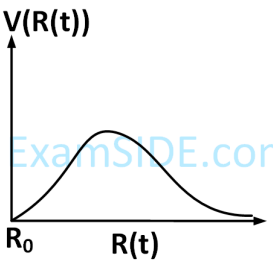
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Q170. JEE Main 2019 (Online) 12th January Morning Slot

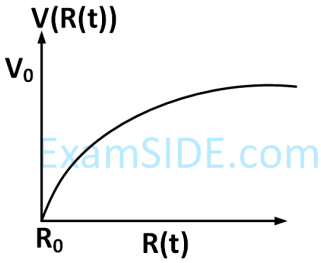
Open solution

There is a uniform spherically symmetric surface charge density at a distance R_0 from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed V (R(t)) of the distribution as a function of its instantaneous radius R (t) is :

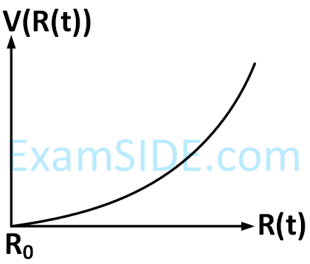
(A)



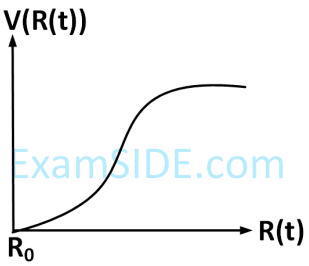
(B)



(C)



(D)



Q171. JEE Main 2019 (Online) 11th January Evening Slot

Open solution

An electric field of 1000 V/m is applied to an electric dipole at angle of 45°. The value of electric dipole moment is 10^{-29} C.m. What is the potential energy of the electric dipole?

(A) -7×10^{-27} J

(B) -9×10^{-20} J

(C) -10×10^{-29} J

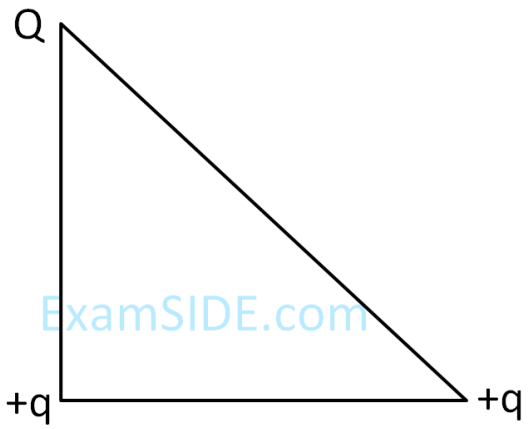
(D) -20×10^{-18} J

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Q172. JEE Main 2019 (Online) 11th January Morning Slot

Three charges Q, + q and + q are placed at the vertices of a right-angle isosceles triangle as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is :

Open solution



ExamSIDE.com

(A) $\frac{-q}{1+\sqrt{2}}$

(B) + q

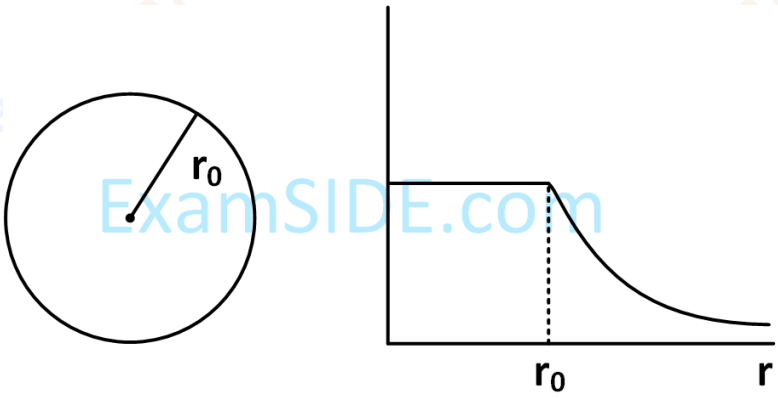
(C) $-$ 2q

(D) $\frac{-\sqrt{2}q}{\sqrt{2}+1}$

Q173. JEE Main 2019 (Online) 11th January Morning Slot

The given graph shows variation (with distance r from centre) of :

Open solution



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(A) Electric field of a uniformly charged sphere

(B) Electric field of a uniformly charged spherical shell

(C) Potential of a uniformly charged sphere

(D) Potential of a uniformly charged spherical shell

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Q174. JEE Main 2019 (Online) 10th January Evening Slot

Four equal point charges Q each are placed in the xy plane at (0, 2), (4, 2), (4, −2) and (0, −2). The work required to put a fifth charge Q at the origin of the coordinate system will be -

Open solution

(A) $\frac{Q^2}{4\pi\varepsilon_0}$

(B) $\frac{Q^2}{2\sqrt{2}\pi\varepsilon_0}$

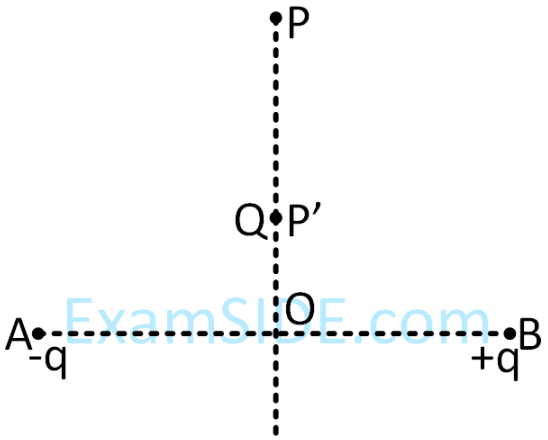
(C) $\frac{Q^2}{4\pi\varepsilon_0}\left(1+\frac{1}{\sqrt{3}}\right)$

(D) $\frac{Q^2}{4\pi\varepsilon_0}\left(1+\frac{1}{\sqrt{5}}\right)$

Q175. JEE Main 2019 (Online) 10th January Evening Slot

Charges −q and +q located at A and B, respectively, constitude an electric dipole. Distance AB = 2a, O is the mid point of the dipole and OP is perpendicular to AB. A charge Q is placed at P where OP = y and y >> 2a. The charge Q experiences an electrostatic force F. If Q is now moved along the equatorial line to P' such that OP' = $\left(\frac{y}{3}\right)$, the force on Q will be close to - $\left(\frac{y}{3} >> 2a\right)$

Open solution



(A) 9F

(B) 3F

(C) F/3

(D) 27F

Q176. JEE Main 2019 (Online) 10th January Morning Slot

A charge Q is distributed over three concentric spherical shells of radii a, b, c (a < b < c) such that their surface charge densities are equal to one another. The total potential at a point at distance r from their common centre, where r < a, would be -

Open solution

(A) $\frac{Q(a^2+b^2+c^2)}{4\pi\varepsilon_0(a^3+b^3+c^3)}$

(B) $\frac{Q}{4\pi\varepsilon_0(a+b+c)}$

(C) $\frac{Q}{12\pi\varepsilon_0}\frac{ab+bc+ca}{abc}$

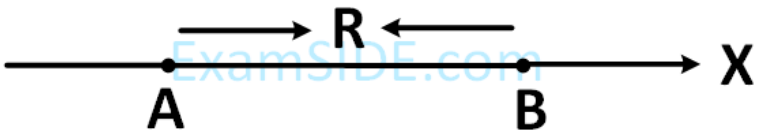
(D) $\frac{Q(a+b+c)}{4\pi\varepsilon_0(a^2+b^2+c^2)}$

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Q177. JEE Main 2019 (Online) 10th January Morning Slot

Two electric dipoles, A, B with respective dipole moments $\vec{d_A} = -4qai$ and $\vec{d_B} = -2qai$ are placed on the x-axis with a separation R, as shown in the figure. The distance from A at which both of them produce the same potential is -

Open solution



(A) $\frac{\sqrt{2}R}{\sqrt{2}+1}$

(B) $\frac{R}{\sqrt{2}+1}$

(C) $\frac{\sqrt{2}R}{\sqrt{2}-1}$

(D) $\frac{R}{\sqrt{2}-1}$

Q178. JEE Main 2019 (Online) 9th January Evening Slot

Two point charges $q_1\left(\sqrt{10}\mu C\right)$ and $q_2(-25\ \mu C)$ are placed on the x-axis at x = 1 m and x = 4 m respectively. The electric field (in V/m) at a point y = 3 m on y-axis is,
[take $\frac{1}{4\pi\epsilon_0}$ = 9 × 10^9 Nm^2C^{-2}]

Open solution

(A) $\left(63\hat{i} - 27\hat{j}\right) \times 10^2$

(B) $\left(-63\hat{i} + 27\hat{j}\right) \times 10^2$

(C) $\left(81\hat{i} - 81\hat{j}\right) \times 10^2$

(D) $\left(-81\hat{i} + 81\hat{j}\right) \times 10^2$

Q179. JEE Main 2019 (Online) 9th January Evening Slot

Charge is distributed within a sphere of radius R with a volume charge density $\rho(r) = \frac{A}{r^2}e^{-2r/a}$, where A and a are constants. If Q is the total charge of this charge distribution, the radius R is :

Open solution

(A) a log $\left(1 - \frac{Q}{2\pi aA}\right)$

(B) $\frac{a}{2}$ log $\left(\frac{1}{1-\frac{Q}{2\pi aA}}\right)$

(C) a log $\left(\frac{1}{1-\frac{Q}{2\pi aA}}\right)$

(D) $\frac{a}{2}$ log $\left(1 - \frac{Q}{2\pi aA}\right)$

Q180. JEE Main 2019 (Online) 9th January Morning Slot

Three charges + Q, q, + Q are placed respectively, at distance, 0, d/2 and d from the origin, on the x-axis. If the net force experienced by + Q, placed at x = 0, is zero, then value of q is :

Open solution

(A) $-\frac{Q}{4}$

(B) + $\frac{Q}{2}$

(C) + $\frac{Q}{4}$

(D) $-\frac{Q}{2}$

Q181. JEE Main 2019 (Online) 9th January Morning Slot

For a uniformly charged ring of radius R, the electric field on its axis has the largest magnitude at a distance h from its center. Then value of h is :

Open solution

(A) $\frac{R}{\sqrt{5}}$

(B) $\frac{R}{\sqrt{2}}$

(C) R

(D) R$\sqrt{2}$

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Q182. JEE Main 2018 (Online) 16th April Morning Slot

Two identical conducting spheres A and B, carry equal charge. They are separated by a distance much larger than their diameters, and the force between them is F. A third identical conducting sphere, C, is uncharged. Sphere C is first touched to A, then to B, and then removed. As a result, the force between A and B would be equal to :

Open solution

(A) F (B) $\frac{3F}{4}$ (C) $\frac{3F}{8}$ (D) $\frac{F}{2}$

Q183. JEE Main 2018 (Offline)

Three concentric metal shells A, B and C of respective radii a, b and c (a < b < c) have surface charge densities $+\sigma$, $-\sigma$ and $+\sigma$ respectively. The potential of shell B is :

Open solution

(A) $\frac{\sigma}{\in_0}\left[\frac{b^2-c^2}{c}+a\right]$ (B) $\frac{\sigma}{\in_0}\left[\frac{a^2-b^2}{a}+c\right]$

(C) $\frac{\sigma}{\in_0}\left[\frac{a^2-b^2}{b}+c\right]$ (D) $\frac{\sigma}{\in_0}\left[\frac{b^2-c^2}{b}+a\right]$

Q184. JEE Main 2018 (Online) 15th April Evening Slot

A solid ball of radius R has a charge density ρ given by $\rho = \rho_0\left(1-\frac{r}{R}\right)$ for $0 \le r \le R$. The electric field outside the ball is :

Open solution

(A) $\frac{\rho_o R^3}{\in_o r^2}$ (B) $\frac{\rho_o R^3}{12\in_o r^2}$

(C) $\frac{4\rho_o R^3}{3\in_o r^2}$ (D) $\frac{3\rho_o R^3}{4\in_o r^2}$

Q185. JEE Main 2018 (Online) 15th April Morning Slot

A body of mass M and charge q is connected to spring of spring constant k. It is oscillating along x-direction about its equilibrium position, taken to be at $x = 0$, with an amplitude A. An electric field E is applied along the x-direction. Which of the following statements is **correct** ?

Open solution

(A) The new equilibrium position is at a distance $\frac{qE}{2k}$ from $x = 0$.

(B) The total energy of the system is $\frac{1}{2}m\omega^2A^2 + \frac{1}{2}\frac{q^2E^2}{k}$.

(C) The total energy of the system is $\frac{1}{2}m\omega^2A^2 - \frac{1}{2}\frac{q^2E^2}{k}$.

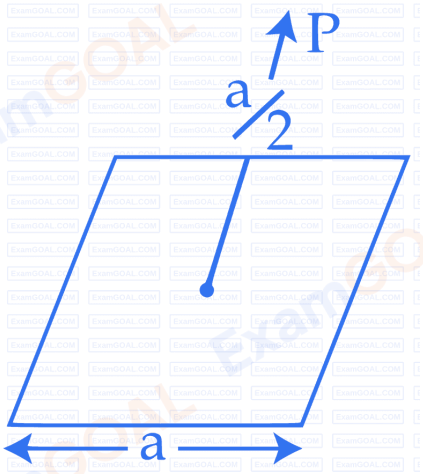
(D) The new equilibrium position is at a distance $\frac{2qE}{k}$ from $x = 0$.

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Q186. JEE Main 2018 (Online) 15th April Morning Slot

A charge Q is placed at a distance $a/2$ above the center of the square surface of edge a as shown in the figure.

Open solution



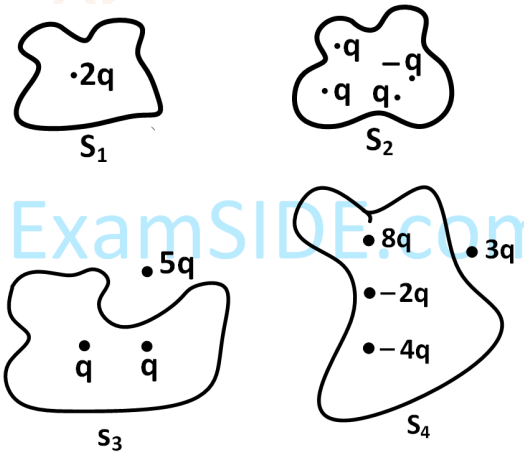
The electric flux through the square surface is

(A) $\frac{Q}{\epsilon_0}$ (B) $\frac{Q}{2\epsilon_0}$ (C) $\frac{Q}{3\epsilon_0}$ (D) $\frac{Q}{6\epsilon_0}$

Q187. JEE Main 2017 (Online) 9th April Morning Slot

Four closed surfaces and corresponding charge distributions are shown below.

Open solution



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Let the respective electric fluxes through the surfaces be Φ_1, Φ_2, Φ_3 and Φ_4. Then :

(A) $\Phi_1 < \Phi_2 = \Phi_3 > \Phi_4$ (B) $\Phi_1 > \Phi_2 > \Phi_3 > \Phi_4$

(C) $\Phi_1 = \Phi_2 = \Phi_3 = \Phi_4$ (D) $\Phi_1 > \Phi_3$; $\Phi_2 < \Phi_4$

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Q188. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

There is a uniform electrostatic field in a region. The potential at various points on a small sphere centred at P, in the region, is found to vary between the limits 589.0 V to 589.8 V. What is the potential at a point on the sphere whose radius vector makes an angle of 60° with the direction of the field ?

(A) 589.5 V (B) 589.2 V (C) 589.4 V (D) 589.6 V

Q189. JEE Main 2017 (Offline)

Open solution

An electric dipole has a fixed dipole moment $\vec{p}$, which makes angle θ with respect to x-axis. When subjected to an electric field $\vec{E_1} = E\hat{i}$, it experiences a torque $\vec{T_1} = \tau\hat{k}$. When subjected to another electric field $\vec{E_2} = \sqrt{3}E_1\hat{j}$ it experiences a torque $\vec{T_2} = -\vec{T_1}$. The angle θ is:

(A) 90° (B) 45° (C) 30° (D) 60°

Q190. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

Within a spherical charge distribution of charge density ρ(r), N equipotential surfaces of potential V_0, $V_0 + \Delta V$, $V_0 + 2\Delta V$, ... $V_0 + N\Delta V$ ($\Delta V > 0$), are drawn and have increasing radii r_0, r_1, r_2,.....r_N, respectively. If the difference in the radii of the surfaces is constant for all values of V_0 and ΔV then :

(A) ρ (r) $\propto$ r (B) ρ (r) = constant

(C) ρ (r) $\propto \frac{1}{r}$ (D) ρ (r) $\propto \frac{1}{r^2}$

Q191. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

The potential (in volts) of a charge distribution is given by.

V(z) = $30 - 5z^2$ for $|z| \leq 1$ m.
V(z) = $35 - 10\,|z|$ for $|z| \geq 1$ m.

V(z) does not depend on x and y. If this potential is generated by a constant charge per unit volume ρ_0 (in units of ε_0) which is spread over a certain region, then choose the correct statement.

(A) ρ_0 = 10 ε_0 for $|z| \leq 1$ m and $\rho_0 = 0$ elsewhere

(B) ρ_0 = 20 ε_0 in the entire region

(C) ρ_0 = 40 ε_0 in the entire region

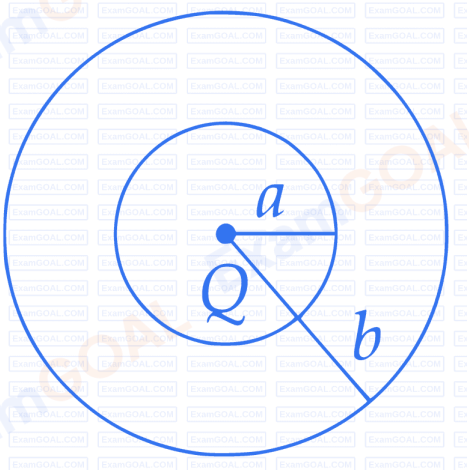
(D) ρ_0 = 20 ε_0 for $|z| \leq 1$ m and $\rho_0 = 0$ elsewhere

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Q192. JEE Main 2016 (Offline)

Open solution

The region between two concentric spheres of radii $'a'$ and $'b'$, respectively (see figure), has volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the center. At the center of the spheres is a point charge Q. The value of A such that the electric field in the region between the spheres will be constant, is :



(A) $\frac{2Q}{\pi(a^2-b^2)}$

(B) $\frac{2Q}{\pi a^2}$

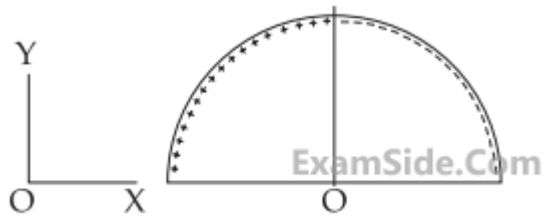
(C) $\frac{Q}{2\pi a^2}$

(D) $\frac{Q}{2\pi (b^2-a^2)}$

Q193. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

A wire, of length L(=20 cm), is bent into a semi-circular arc. If the two equal halves, of the arc, were each to be uniformly charged with charges $\pm$Q, [$|Q|$ = $10^3\ \varepsilon_0$ Coulomb where ε_0 is the permittivity (in SI units) of free space] the net electric field at the center O of the semi-circular arc would be :



(A) $(50 \times 10^3\ \text{N/C})\ \hat{j}$

(B) $(25 \times 10^3\ \text{N/C})\ \hat{i}$

(C) $(25 \times 10^3\ \text{N/C})\ \hat{j}$

(D) $(50 \times 10^3\ \text{N/C})\ \hat{i}$

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Q194. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

An electric field

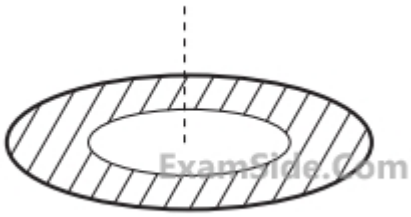
$\overrightarrow{E} = \left(25\hat{i} + 30\hat{j}\right) NC^{-1}$ exists in a region of space. If the potential at the origin is taken to be zero then the potential at x = 2 m, y = 2 m is :

(A) $-$ 130 V (B) $-$ 120 V (C) $-$ 140 V (D) $-$ 110 V

Q195. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

A thin disc of radius b = 2a has a concentric hole of radius 'a' in it (see figure). It carries uniform surface charge 'σ' on it. If the electric field on its axis at height 'h' (h < < a) from its centre is given as 'Ch' then value of 'C' is :



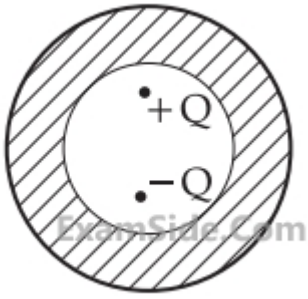
(A) $\frac{\sigma}{a\in_0}$ (B) $\frac{\sigma}{2\,a\in_0}$ (C) $\frac{\sigma}{4\,a\in_0}$ (D) $\frac{\sigma}{8\,a\in_0}$

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Q196. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

Shown in the figure are two point charges + Q and − Q inside the cavity of a spherical shell. The charges are kept near the surface of the cavity on opposite sides of the centre of the shell. If σ_1 is the surface charge on the inner surface and Q_1 net charge on it and σ_2 the surface charge on the outer surface and Q_2 net charge on it then :



(A) $\sigma_1 \neq 0,\ Q_1 \neq 0$

$\sigma_2 \neq 0,\ Q_2 \neq 0$

(B) $\sigma_1 \neq 0,\ Q_1 = 0$

$\sigma_2 \neq 0,\ Q_2 = 0$

(C) $\sigma_1 \neq 0,\ Q_1 = 0$

$\sigma_2 = 0,\ Q_2 = 0$

(D) $\sigma_1 = 0,\ Q_1 = 0$

$\sigma_2 = 0,\ Q_2 = 0$

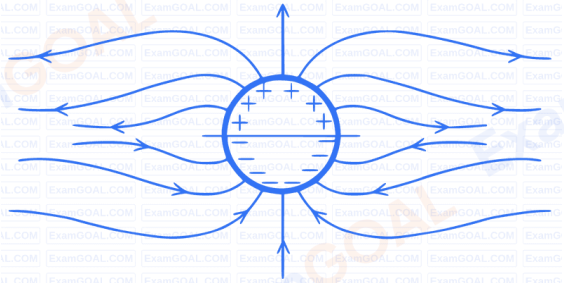
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Q197. JEE Main 2015 (Offline)

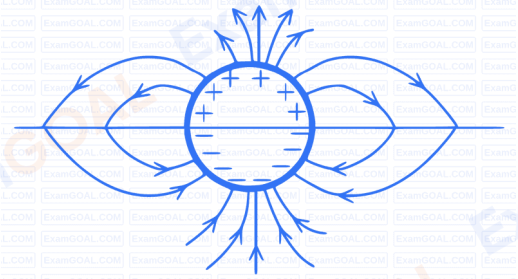
Open solution

A long cylindrical shell carries positive surface charge σ in the upper half and negative surface charge - σ in the lower half. The electric field lines around the cylinder will look like figure given in : (*figures are schematic and not drawn to scale*)

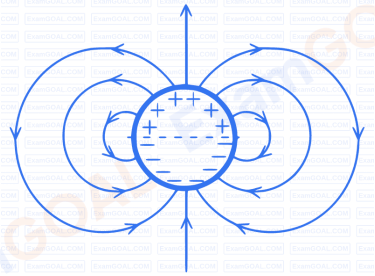
(A)



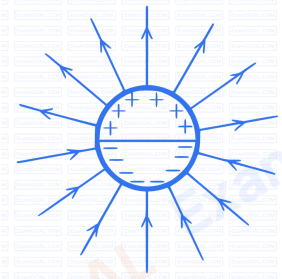
(B)



(C)



(D)



Q198. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

The electric field in a region of space is given by, $\vec{E} = E_0\hat{i} + 2E_0\hat{j}$ where E_0 = 100 N/C. The flux of this field through a circular surface of a radius 0.02 m parallel to the Y-Z plane is nearly :

(A) 0.125 Nm^2/C (B) 0.02 Nm^2/C (C) 0.005 Nm^2/C (D) 3.14 Nm^2/C

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Q199. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

A spherically symmetric charge distribution is characterised by a charge density having the following variation :

$\rho(r) = \rho_0 \left(1 - \frac{r}{R}\right)$ for r < R

$\rho(r) = 0$ for r $\geq$ R

Where r is the distance from the center of the charge distribution and ρ_0 is a constant. The electric field at an internal point (r < R) is :

(A) $\frac{\rho_0}{4\in_0}\left(\frac{r}{3} - \frac{r^2}{4R}\right)$

(B) $\frac{\rho_0}{\in_0}\left(\frac{r}{3} - \frac{r^2}{4R}\right)$

(C) $\frac{\rho_0}{3\in_0}\left(\frac{r}{3} - \frac{r^2}{4R}\right)$

(D) $\frac{\rho_0}{12\in_0}\left(\frac{r}{3} - \frac{r^2}{4R}\right)$

Q200. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

A cone of base radius R and height h is located in a uniform electric field $\vec{E}$ parallel to its base. The electric flux entering the cone is :

(A) $\frac{1}{2}$ E h R (B) E h R (C) 2 E h R (D) 4 E h R

Q201. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

The magnitude of the average electric field normally present in the atmosphere just above the surface of the Earth is about 150 N/C, directed inward towards the center of the Earth. This gives the total net surface charge carried by the Earth to be :

[Given $\in_0 = 8.85 \times 10^{-12}$ C^2/N-m^2, $R_E = 6.37 \times 10^6$ m]

(A) + 670 kC (B) $-$ 670 kC (C) $-$ 680 kC (D) + 680 kC

Q202. JEE Main 2014 (Offline)

Open solution

Assume that an electric field $\vec{E} = 30x^2\hat{i}$ exists in space. Then the potential difference $V_A - V_O$, where V_O is the potential at the origin and V_A the potential at $x = 2\ m$ is :

(A) $120\ J/C$ (B) $-120\ J/C$ (C) $-80\ J/C$ (D) $80\ J/C$

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Q203. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: No work is required to be done to move a test charge between any two points on an equipotential surface.
Statement 2: Electric lines of force at the equipotential surfaces are mutually perpendicular to each other.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

(B) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is false, Statement 2 is true.

Q204. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

The surface charge density of a thin charged disc of radius R is σ. The value of the electric field at the centre of the disc is $\frac{\sigma}{2\epsilon_0}$. With respect to the field at the centre, the electric field along the axis at a distance R from the centre of the disc:

(A) reduces by 70.7% (B) reduces by 29.3% (C) reduces by 9.7% (D) reduces by 14.6%

Q205. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

Consider a finite insulated, uncharged conductor placed near a finite positively charged conductor. The uncharged body must have a potential :

(A) less than the charged conductor and more than at infinity.

(B) more than the charged conductor and less than at infinity.

(C) more than the charged conductor and more than at infinity.

(D) less than the charged conductor and less than at infinity.

Q206. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

A liquid drop having 6 excess electrons is kept stationary under a uniform electric field of 25.5 kVm^{-1}. The density of liquid is 1.26×10^3 kg m^{-3}. The radius of the drop is (neglect buoyancy).

(A) 4.3×10^{-7} m (B) 7.8×10^{-7} m (C) 0.078×10^{-7} m (D) 3.4×10^{-7} m

Q207. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

A point charge of magnitude $+1\mu C$ is fixed at (0, 0, 0). An isolated uncharged spherical conductor, is fixed with its center at (4, 0, 0). The potential and the induced electric field at the centre of the sphere is :

(A) 1.8×10^5 V and -5.625×10^6 V/m (B) 0 V and 0 V/m

(C) 2.25×10^5 V and -5.625×10^6 V/m (D) 2.25×10^5 V and 0 V/m

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Q208. JEE Main 2013 (Online) 22th April Morning Slot

Two small equal point charges of magnitude q are suspended from a common point on the ceiling by insulating mass less strings of equal lengths. They come to equilibrium with each string making angle θ from the vertical. If the mass of each charge is m, then the electrostatic potential at the centre of line joining them will be $\left(\frac{1}{4\pi\epsilon_0} = k\right)$.

Open solution

(A) $2\sqrt{kmg\tan\theta}$ (B) $\sqrt{kmg\tan\theta}$ (C) $4\sqrt{kmg\tan\theta}$ (D) $4\sqrt{kmg/\tan\theta}$

Q209. JEE Main 2013 (Online) 9th April Morning Slot

Two point dipoles of dipole moment $\vec{p_1}$ and $\vec{p_2}$ are at a distance x from each other and $\vec{p_1} \| \vec{p_2}$. The force between the dipoles is :

Open solution

(A) $\frac{1}{4\pi\epsilon_0}\frac{4p_1p_2}{x^4}$ (B) $\frac{1}{4\pi\epsilon_0}\frac{3p_1p_2}{x^3}$

(C) $\frac{1}{4\pi\epsilon_0}\frac{6p_1p_2}{x^4}$ (D) $\frac{1}{4\pi\epsilon_0}\frac{8p_1p_2}{x^4}$

Q210. JEE Main 2013 (Online) 9th April Morning Slot

Two balls of same mass and carrying equal charge are hung from a fixed support by threads of length l. At electrostatic equilibrium, assuming that angles made by each thread is small, the separation, x between the balls is proportional to :

Open solution

(A) l (B) l^2 (C) $l^{2/3}$ (D) $l^{1/3}$

Q211. JEE Main 2013 (Online) 9th April Morning Slot

A uniform electric field $\vec{E}$ exists between the plates of a charged condenser. A charged particle enters the space between the plates and perpendicular to $\vec{E}$. The path of the particle between the plates is a:

Open solution

(A) straight line (B) hyperbola (C) parabola (D) circle

Q212. JEE Main 2013 (Offline)

Two charges, each equals to q, are kept at $x = -a$ and $x = a$ on the x-axis. A particle of mass m and charge $q_0 = \frac{q}{2}$ is placed at the origin. If charge q_0 is given a small displacement $(y << a)$ along the y-axis, the net force acting on the particle is proportional to

Open solution

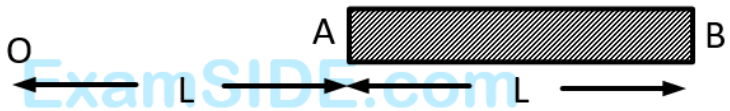
(A) y (B) $-y$ (C) $\frac{1}{y}$ (D) $-\frac{1}{y}$

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Q213. JEE Main 2013 (Offline)

A charge Q is uniformly distributed over a long rod AB of length L as shown in the figure. The electric potential at the point O lying at distance L from the end A is

Open solution



(A) $\frac{Q}{8\pi\varepsilon_0 L}$

(B) $\frac{3Q}{4\pi\varepsilon_0 L}$

(C) $\frac{Q}{4\pi\varepsilon_0 L \ln 2}$

(D) $\frac{Q \ln 2}{4\pi\varepsilon_0 L}$

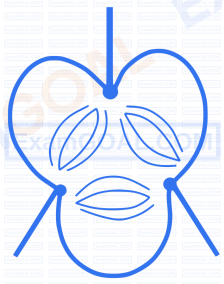
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Q214. AIEEE 2012 (Online) 26th May

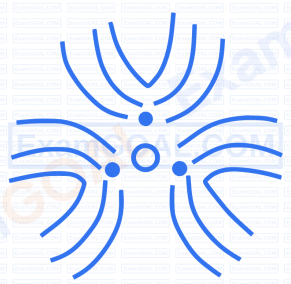
Three positive charges of equal value q are placed at vertices of an equilateral triangle. The resulting lines of force should be sketched as in

Open solution

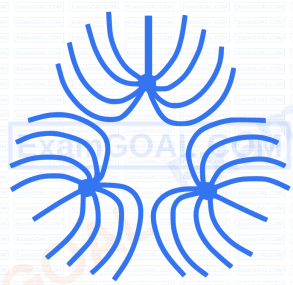
(A)



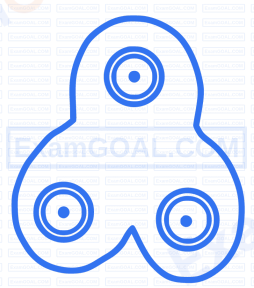
(B)



(C)



(D)



Q215. AIEEE 2012 (Online) 19th May

A charge of total amount Q is distributed over two concentric hollow spheres of radii r and $R\,(R>r)$ such that the surface charge densities on the two spheres are equal. The electric potential at the common centre is

Open solution

(A) $\frac{1}{4\pi\varepsilon_0}\frac{(R-r)Q}{(R^2+r^2)}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{(R+r)Q}{2(R^2+r^2)}$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{(R+r)Q}{(R^2+r^2)}$

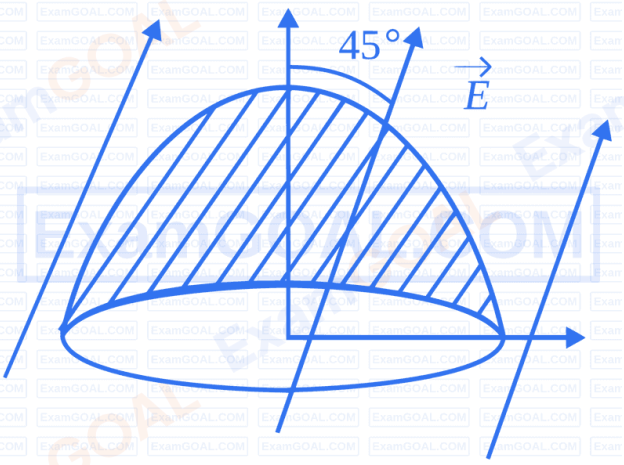
(D) $\frac{1}{4\pi\varepsilon_0}\frac{(R-r)Q}{2(R^2+r^2)}$

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Q216. AIEEE 2012 (Online) 19th May

The flat base of a hemisphere of radius a with no charge inside it lies in a horizontal plane. A uniform electric field $\vec{E}$ is applied at an angle $\frac{\pi}{4}$ with the vertical direction. The electric flux through the curved surface of the hemisphere is

Open solution



(A) $\pi a^2 E$

(B) $\frac{\pi a^2 E}{\sqrt{2}}$

(C) $\frac{\pi a^2 E}{2\sqrt{2}}$

(D) $\frac{(\pi+2)\pi a^2 E}{(2\sqrt{2})^2}$

Q217. AIEEE 2012 (Online) 7th May

The electric potential $V(x)$ in a region around the origin is given by $V(x) = 4x^2$ volts. The electric charge enclosed in a cube of $1\ \mathrm{m}$ side with its centre at the origin is (in coulomb)

Open solution

(A) $8\varepsilon_0$

(B) $-4\varepsilon_0$

(C) 0

(D) $-8\varepsilon_0$

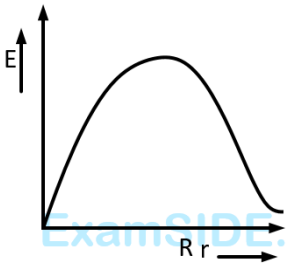
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Q218. AIEEE 2012

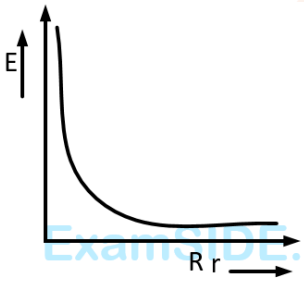
Open solution

In a uniformly charged sphere of total charge Q and radius R, the electric field E is plotted as a function of distance from the center. The graph which would correspond to the above will be:

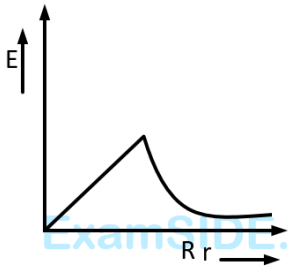
(A)



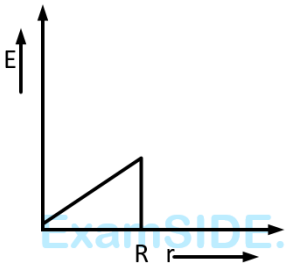
(B)



(C)



(D)



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Q219. AIEEE 2012

Open solution

This question has statement- 1 and statement- 2. Of the four choices given after the statements, choose the one that best describes the two statements.
An insulating solid sphere of radius R has a uniformly positive charge density ρ. As a result of this uniform charge distribution there is a finite value of electric potential at the center of the sphere, at the surface of the sphere and also at a point outside the sphere. The electric potential at infinity is zero.
Statement- 1 : When a charge q is taken from the centre to the surface of the sphere its potential energy changes by $\frac{q\rho}{3\varepsilon_0}$
Statement- 2 : The electric field at a distance $r\,(r < R)$ from the center of the sphere is $\frac{\rho r}{3\varepsilon_0}$.

(A) Statement- 1 is true, Statement- 2 is true; Statement- 2 is not the correct explanation of Statement- 1.

(B) Statement 1 is true, Statement 2 is false.

(C) Statement 1 is false, Statement 2 is true.

(D) Statement- 1 is true, Statement- 2 is true; Statement- 2 is the correct explanation of Statement- 1.

Q220. AIEEE 2011

Open solution

Two identical charged spheres suspended from a common point by two massless strings of length l are initially a distance $d\,(d << l)$ apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result charges approach each other with a velocity v. Then as a function of distance x between them,

(A) $v \propto x^{-1}$

(B) $v \propto x^{1/2}$

(C) $v \propto x$

(D) $v \propto x^{-1/2}$

Q221. AIEEE 2011

Open solution

The electrostatic potential inside a charged spherical ball is given by $\phi = ar^2 + b$ where r is the distance from the center and a, b are constants. Then the charge density inside the ball is:

(A) $-6a\varepsilon_0 r$ (B) $-24\pi a\varepsilon_0$ (C) $-6a\varepsilon_0$ (D) $-24\pi\varepsilon_0 r$

Q222. AIEEE 2010

Open solution

Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30° with each other. When suspended in a liquid of density $0.8g\ cm^{-3}$, the angle remains the same. If density of the material of the sphere is $1.6\ g\ cm^{-3}$, the dielectric constant of the liquid is

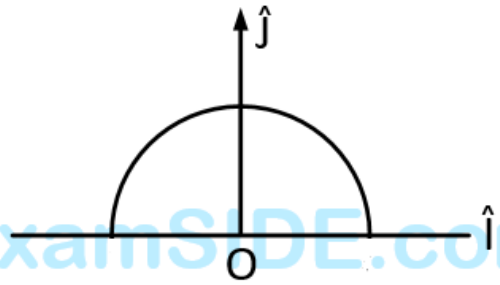
(A) 4 (B) 3 (C) 2 (D) 1

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Q223. AIEEE 2010

A thin semi-circular ring of radius r has a positive charges q distributed uniformly over it. The net field $\overrightarrow{E}$ at the center O is

Open solution



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(A) $\frac{q}{4\pi^2\varepsilon_0 r^2}\, j$
(B) $-\frac{q}{4\pi^2\varepsilon_0 r^2}\, j$
(C) $-\frac{q}{2\pi^2\varepsilon_0 r^2}\, j$
(D) $\frac{q}{2\pi^2\varepsilon_0 r^2}\, j$

Q224. AIEEE 2010

Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0\left(\frac{5}{4} - \frac{r}{R}\right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance $r\,(r < R)$ from the origin is given by

Open solution

(A) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{3} - \frac{r}{R}\right)$
(B) $\frac{4\pi\rho_0 r}{3\varepsilon_0}\left(\frac{5}{3} - \frac{r}{R}\right)$
(C) $\frac{4\rho_0 r}{4\varepsilon_0}\left(\frac{5}{4} - \frac{r}{R}\right)$
(D) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4} - \frac{r}{R}\right)$

Q225. AIEEE 2009

Two points P and Q are maintained at the potentials of $10\ V$ and $-4\ V$, respectively. The work done in moving 100 electrons from P to Q is :

Open solution

(A) $9.60 \times 10^{-17} J$
(B) $-2.24 \times 10^{-16} J$
(C) $2.24 \times 10^{-16} J$
(D) $-9.60 \times 10^{-17} J$

Q226. AIEEE 2009

A charge Q is placed at each of the opposite corners of a square. A charge q is placed at each of the other two corners. If the net electrical force on Q is zero, then Q/q equals:

Open solution

(A) -1
(B) 1
(C) $-\frac{1}{\sqrt{2}}$
(D) $-2\sqrt{2}$

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Q227. AIEEE 2009

Open solution

(This question contains Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.)
Statement-1 : For a charged particle moving from point P to point Q, the net work done by an electrostatic field on the particle is independent of the path connecting point P to point Q.
Statement-2 : The net work done by a conservative force on an object moving along a closed loop is zero.

(A) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1.

(B) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1.

(C) Statement- 1 is false, Statement- 2 is true.

(D) Statement- 1 true, Statement- 2 is false

Q228. AIEEE 2009

Open solution

Let $P(r) = \frac{Q}{\pi R^4} r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point $'p'$ inside the sphere at distance r_1 from the center of the sphere, the magnitude of electric field is :

(A) $\frac{Q}{4\pi \in_0 r_1^2}$

(B) $\frac{Qr_1^2}{4\pi \in_0 R^4}$

(C) $\frac{Qr_1^2}{3\pi \in_0 R^4}$

(D) 0

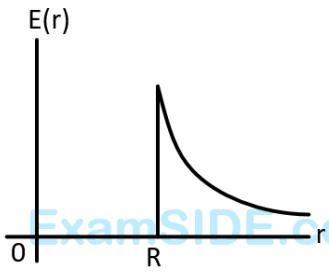
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Q229. AIEEE 2008

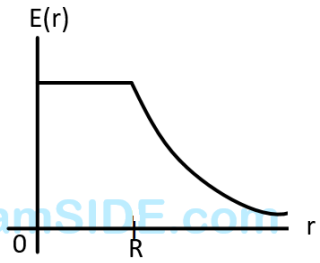
A thin spherical shell of radius R has charge Q spread uniformly over its surface. Which of the following graphs most closely represents the electric field $E(r)$ produced by the shell in the range $0 \leq r < \infty$, where r is the distance from the center of the shell?

Open solution

(A)

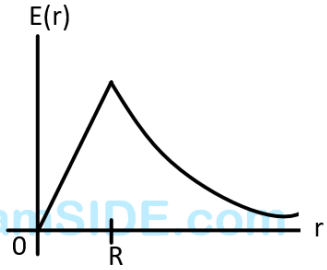


(B)

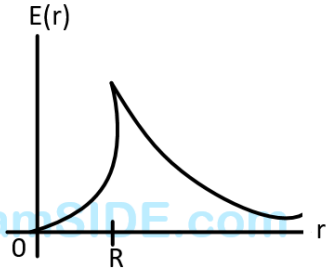


c

(C)



(D)



Q230. AIEEE 2007

An electric charge $10^{-3}\ \mu C$ is placed at the origin $(0, 0)$ of $X - Y$ co-ordinate system. Two points A and B are situated at $\left(\sqrt{2}, \sqrt{2}\right)$ and $(2, 0)$ respectively. The potential difference between the points A and B will be

Open solution

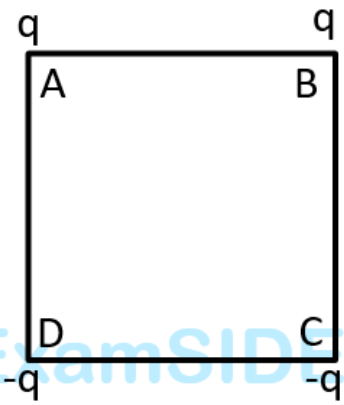
(A) 4.5 volts (B) 9 volts (C) zero (D) 2 volts

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Q231. AIEEE 2007

Open solution

Charges are placed on the vertices of a square as shown. Let $\vec{E}$ be the electric field and V the potential at the center. If the charges on A and B are interchanged with those on D and C respectively, then



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(A) $\vec{E}$ changes, V remains unchanged

(B) $\vec{E}$ remains unchanged, V changes

(C) both $\vec{E}$ and V change

(D) $\vec{E}$ and V remain unchanged

Q232. AIEEE 2007

Open solution

The potential at a point x (measured in μm) due to some charges situated on the x-axis is given by $V(x) = 20/\left(x^2 - 4\right)$ volt
The electric field E at $x = 4\,\mu m$ is given by

(A) $(10/9)$ volt / μm and in the $+ve\ x$ direction

(B) $(5/3)$ volt/ μm and in the $-ve\ x$ direction

(C) $(5/3)$ volt/μm and in the $+ve\ x$ direction

(D) $(10/9)$ volt/ μm and in the $-ve\ x$ direction

Q233. AIEEE 2006

Open solution

An electric dipole is placed at an angle of 30° to a non-uniform electric field. The dipole will experience

(A) a translation force only in the direction of the field

(B) a translation force only in a direction normal to the direction of the field

(C) a torque as well as a translational force

(D) a torque only

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Q234. AIEEE 2006

Two spherical conductors A and B of radii $1\ mm$ and $2\ mm$ are separated by a distance of $5\ cm$ and are uniformly charged. If the spheres are connected by a conducting wire then in equilibrium condition, the ratio of the magnitude of the electric fields at the surfaces of spheres A and B is

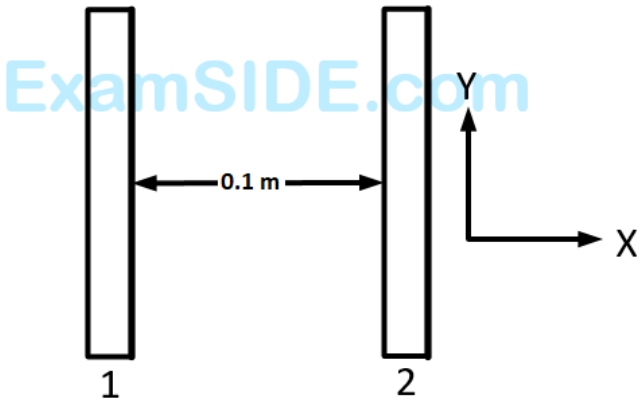
Open solution

(A) $4:1$ (B) $1:2$ (C) $2:1$ (D) $1:4$

Q235. AIEEE 2006

Two insulating plates are both uniformly charged in such a way that the potential difference between them is $V_2 - V_1 = 20\,V$. (i.e., plate 2 is at a higher potential). The plates are separated by $d = 0.1\ m$ and can be treated as infinitely large. An electron is released from rest on the inner surface of plate 1. What is its speed when it hits plate 2?
$(e = 1.6 \times 10^{-19}\,C,\ m_e = 9.11 \times 10^{-31}\,kg)$

Open solution



(A) $2.65 \times 10^{6}\,m/s$ (B) $7.02 \times 10^{12}\,m/s$ (C) $1.87 \times 10^{6}\,m/s$ (D) $32 \times 10^{-19}\,m/s$

Q236. AIEEE 2005

Two point charges $+8q$ and $-2q$ are located at $x = 0$ and $x = L$ respectively. The location of a point on the x axis at which the net electric field due to these two point charges is zero is

Open solution

(A) $\frac{L}{4}$ (B) $2\ L$ (C) $4\ L$ (D) $8\ L$

Q237. AIEEE 2005

Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are $+q$ and $-q$. The potential difference between the centres of the two rings is

Open solution

(A) $\frac{q}{2\pi \in_0}\left[\frac{1}{R} - \frac{1}{\sqrt{R^2+d^2}}\right]$

(B) $\frac{qR}{4\pi \in_0 d^2}$

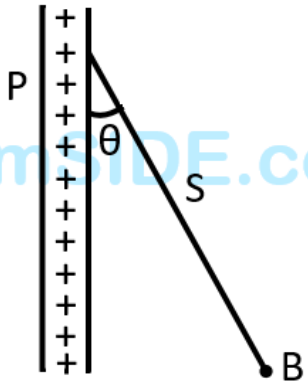
(C) $\frac{q}{4\pi \in_0}\left[\frac{1}{R} - \frac{1}{\sqrt{R^2+d^2}}\right]$

(D) zero

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Q238. AIEEE 2005

A charged ball B hangs from a silk thread S, which makes angle θ with a large charged conducting sheet P, as shown in the figure. The surface charge density σ of the sheet is proportional to



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Open solution

(A) $\cot\theta$ (B) $\cos\theta$ (C) $\tan\theta$ (D) $\sin\theta$

Q239. AIEEE 2004

Two spherical conductors B and C having equal radii and carrying equal charges on them repel each other with a force F when kept apart at some distance. A third spherical conductor having same radius as that of B but uncharged is brought in contact with B, then brought in contact with C and finally removed away from both. The new force of repulsion between B and C is

Open solution

(A) $F/8$ (B) $3F/4$ (C) $F/4$ (D) $3F/8$

Q240. AIEEE 2004

A charged particle 'q' is shot towards another charged particle 'Q' which is fixed, with a speed 'v'. It approaches 'Q' upto a closest distance r and then returns. If q were given a speed of '$2v$' the closest distance of approach would be

Open solution

(A) $r/2$ (B) $2r$ (C) r (D) $r/4$

Q241. AIEEE 2004

Four charges equal to -Q are placed at the four corners of a square and a charge q is at its center. If the system is in equilibrium the value of q is

Open solution

(A) $-\frac{Q}{2}\left(1+2\sqrt{2}\right)$ (B) $\frac{Q}{4}\left(1+2\sqrt{2}\right)$
(C) $-\frac{Q}{4}\left(1+2\sqrt{2}\right)$ (D) $\frac{Q}{2}\left(1+2\sqrt{2}\right)$

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Q242. AIEEE 2004

A charged oil drop is suspended in a uniform field of $3 \times 10^4\ v/m$ so that it neither falls nor rises. The charge on the drop will be (Take the mass of the charge $= 9.9 \times 10^{-15}\ kg$ and $g = 10\, m/s^2$)

(A) $1.6 \times 10^{-18}\, C$ (B) $3.2 \times 10^{-18}\, C$ (C) $3.3 \times 10^{-18}\, C$ (D) $4.8 \times 10^{-18}\, C$

Open solution

Q243. AIEEE 2003

If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2, the electric charge inside the surface will be

(A) $(\phi_2 - \phi_1)\varepsilon_0$ (B) $(\phi_2 + \phi_1)/\varepsilon_0$

(C) $(\phi_1 - \phi_2)/\varepsilon_0$ (D) $(\phi_1 + \phi_2)\varepsilon_0$

Open solution

Q244. AIEEE 2003

A thin spherical conducting shell of radius R has a charge q. Another charge Q is placed at the center of the shell. The electrostatic potential at a point P at a distance $\frac{R}{2}$ from the center of the shell is

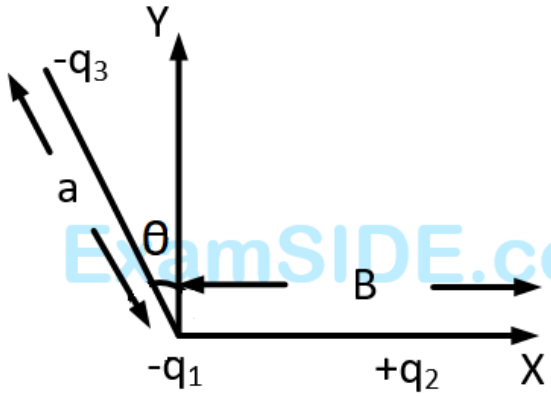
(A) $\frac{2Q}{4\pi\varepsilon_0 R}$ (B) $\frac{2Q}{4\pi\varepsilon_0 R} - \frac{2q}{4\pi\varepsilon_0 R}$

(C) $\frac{2Q}{4\pi\varepsilon_0 R} + \frac{q}{4\pi\varepsilon_0 R}$ (D) $\frac{(q+Q)2}{4\pi\varepsilon_0 R}$

Open solution

Q245. AIEEE 2003

Three charges $-q_1, +q_2$ and $-q_3$ are placed as shown in the figure. The x-component of the force on $-q_1$ is proportional to



(A) $\frac{q_2}{b^2} - \frac{q_3}{a^2}\cos\theta$ (B) $\frac{q_2}{b^2} + \frac{q_3}{a^2}\sin\theta$

(C) $\frac{q_2}{b^2} + \frac{q_3}{a^2}\cos\theta$ (D) $\frac{q_2}{b^2} - \frac{q_3}{a^2}\sin\theta$

Open solution

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Q246. AIEEE 2002

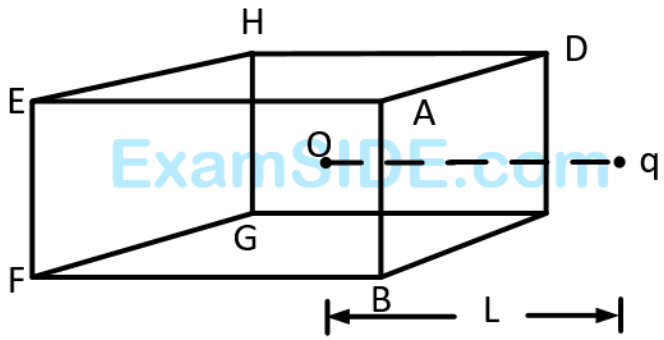
On moving a charge of 20 coulomb by $2\ cm$, $2\ J$ of work is done, then the potential difference between the points is

(A) $0.1\ V$ (B) $8\ V$ (C) $2V$ (D) $0.5\ V$.

Open solution

Q247. AIEEE 2002

A charged particle q is placed at the centre O of cube of length $L(ABCDEFGH)$. Another same charge q is placed at a distance L from O. Then the electric flux through $ABCD$ is



(A) $q/4\pi\in_0 L$

(B) zero

(C) $q/2\pi\in_0 L$

(D) $q/3\pi\in_0 L$

Open solution

Q248. AIEEE 2002

If a charge q is placed at the center of the line joining two equal charges Q such that the system is in equilibrium then the value of q is

(A) $Q/2$ (B) $-Q/2$ (C) $Q/4$ (D) $-Q/4$

Open solution

Numerical 54 questions

Q249. JEE Main 2026 (Online) 6th April Morning Shift

A three coulomb charge moves from the point $(0,-2,-5)$ to the point $(5,1,2)$ in an electric field expressed as $\vec{E}=2x\hat{\mathrm{i}}+3\mathrm{y}^2\hat{\mathrm{j}}+4\hat{\mathrm{k}}\mathrm{N/C}$. The work done in moving the charge is ____ J.

Open solution

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Q250. JEE Main 2026 (Online) 24th January Evening Shift

A point charge $q = 1\mu C$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24\mu C$, distributed uniformly along its length, as shown in figure. Force between q and wire is ____ N.

(Use : $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ N \cdot m^2/C^2$)



Open solution

Q251. JEE Main 2025 (Online) 7th April Evening Shift

The electric field in a region is given by $\vec{E} = (2\hat{i} + 4\hat{j} + 6\hat{k}) \times 10^3\ N/C$. The flux of the field through a rectangular surface parallel to $x - z$ plane is $6.0\ Nm^2C^{-1}$. The area of the surface is _____ cm^2.

Open solution

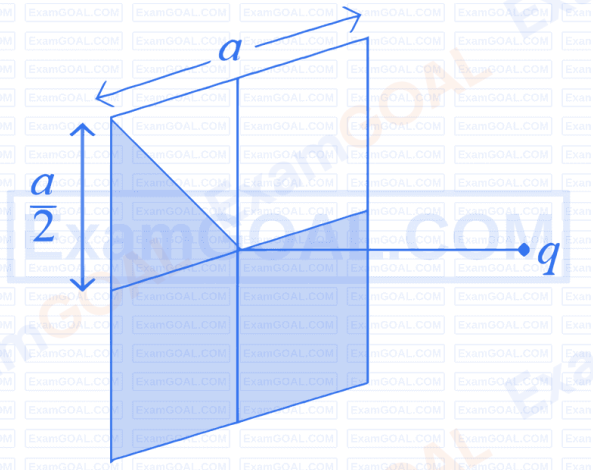
Q252. JEE Main 2025 (Online) 28th January Evening Shift

An electric dipole of dipole moment $6 \times 10^{-6} Cm$ is placed in uniform electric field of magnitude $10^6 V/m$. Initially, the dipole moment is parallel to electric field. The work that needs to be done on the dipole to make its dipole moment opposite to the field, will be _____ J.

Open solution

Q253. JEE Main 2025 (Online) 24th January Morning Shift

A square loop of sides $a = 1\ m$ is held normally in front of a point charge $q = 1C$ at a distance $\frac{a}{2}$. The flux of the electric field through the shaded region is $\frac{5}{p} \times \frac{1}{\varepsilon_0} \frac{Nm^2}{C}$, where the value of p is _____ .



Open solution

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Q254. JEE Main 2025 (Online) 23rd January Morning Shift

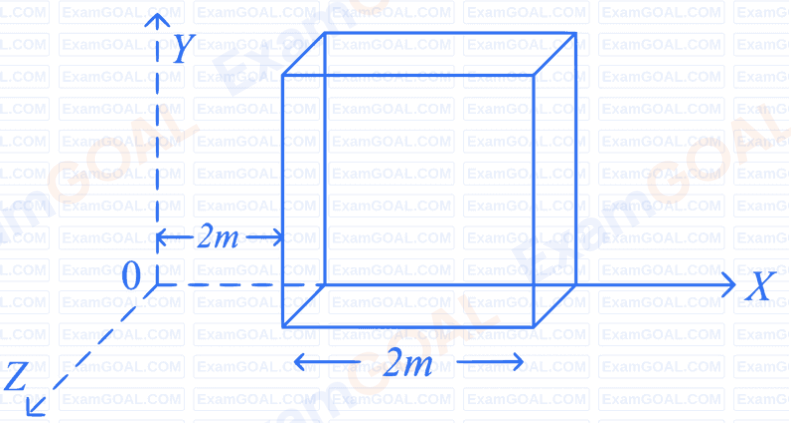
A positive ion A and a negative ion B has charges $6.67 \times 10^{-19}\text{C}$ and $9.6 \times 10^{-10}\text{C}$, and masses 19.2×10^{-27} kg and 9×10^{-27} kg respectively. At an instant, the ions are separated by a certain distance r. At that instant the ratio of the magnitudes of electrostatic force to gravitational force is $\text{P} \times 10^{-13}$, where the value of P is _____.

(Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{Nm}^2\text{C}^{-2}$ and universal gravitational constant as $6.67 \times 10^{-11}\text{Nm}^2\,\text{kg}^{-2}$)

Open solution

Q255. JEE Main 2024 (Online) 9th April Evening Shift

An electric field $\vec{E} = (2x\hat{i})NC^{-1}$ exists in space. A cube of side $2\,\text{m}$ is placed in the space as per figure given below. The electric flux through the cube is _____ Nm^2/C.



Open solution

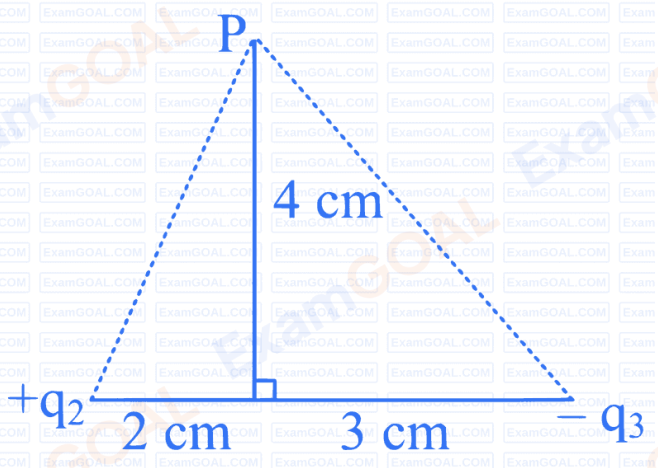
Q256. JEE Main 2024 (Online) 9th April Morning Shift

At the centre of a half ring of radius $\text{R} = 10\,\text{cm}$ and linear charge density $4\,\text{nC}\,\text{m}^{-1}$, the potential is $x\pi\text{V}$. The value of x is _____.

Open solution

Q257. JEE Main 2024 (Online) 8th April Evening Shift

If the net electric field at point P along Y axis is zero, then the ratio of $\left|\frac{q_2}{q_3}\right|$ is $\frac{8}{5\sqrt{x}}$, where $x =$ _____.



Open solution

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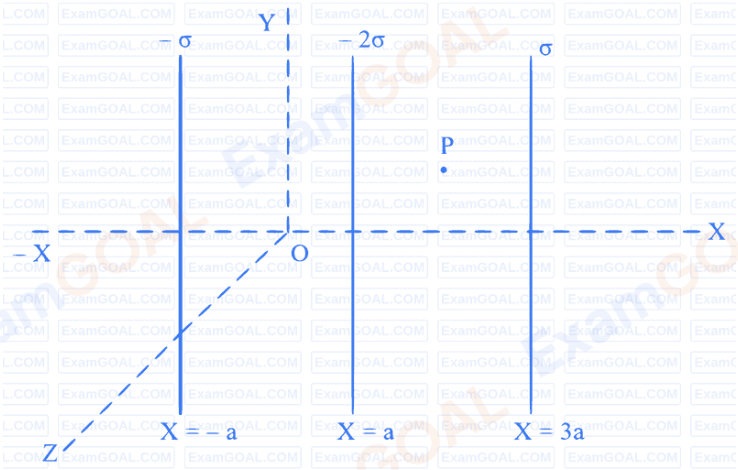
Q258. JEE Main 2024 (Online) 8th April Morning Shift

An electric field, $\overrightarrow{E} = \frac{2\hat{i}+6\hat{j}+8\hat{k}}{\sqrt{6}}$ passes through the surface of $4\ m^2$ area having unit vector $\hat{n} = \left(\frac{2\hat{i}+\hat{j}+\hat{k}}{\sqrt{6}}\right)$. The electric flux for that surface is _____ Vm.

Open solution

Q259. JEE Main 2024 (Online) 6th April Morning Shift

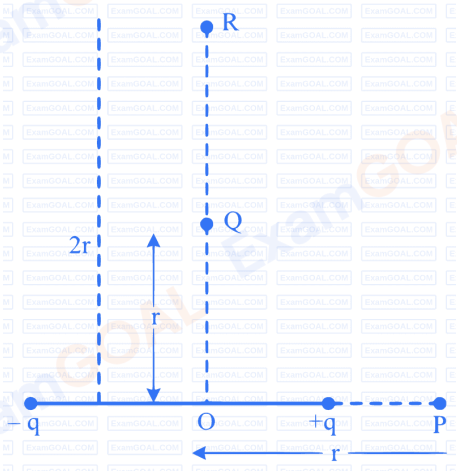
Three infinitely long charged thin sheets are placed as shown in figure. The magnitude of electric field at the point P is $\frac{x\sigma}{\epsilon_0}$. The value of x is _____ (all quantities are measured in SI units).



Open solution

Q260. JEE Main 2024 (Online) 5th April Evening Shift

The electric field at point p due to an electric dipole is E. The electric field at point R on equatorial line will be $\frac{E}{x}$. The value of x :



Open solution

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Q261. JEE Main 2024 (Online) 4th April Morning Shift

An infinite plane sheet of charge having uniform surface charge density $+\sigma_s \mathrm{C/m^2}$ is placed on x-y plane. Another infinitely long line charge having uniform linear charge density $+\lambda_e \mathrm{C/m}$ is placed at $z = 4\,\mathrm{m}$ plane and parallel to y-axis. If the magnitude values $|\sigma_s| = 2\,|\lambda_e|$ then at point $(0, 0, 2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n} : 1$. The value of n is _____.

Open solution

Q262. JEE Main 2024 (Online) 1st February Evening Shift

Suppose a uniformly charged wall provides a uniform electric field of $2 \times 10^4\,\mathrm{N/C}$ normally. A charged particle of mass $2\,\mathrm{g}$ being suspended through a silk thread of length $20\,\mathrm{cm}$ and remain stayed at a distance of $10\,\mathrm{cm}$ from the wall.

Then the charge on the particle will be $\frac{1}{\sqrt{x}}\mu\mathrm{C}$ where $x =$ _____. [use $\mathrm{g} = 10\,\mathrm{m/s^2}$]

Open solution

Q263. JEE Main 2024 (Online) 1st February Morning Shift

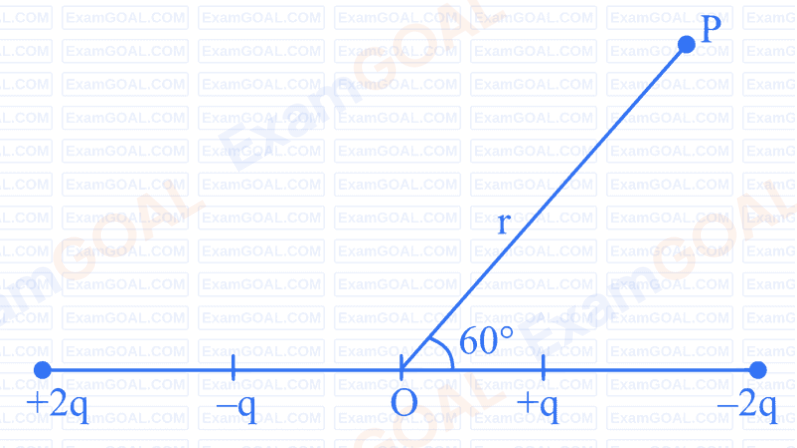
Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is $1.5\,\mathrm{g/cc}$, the dielectric constant of water will be _____.

(Take density of water $= 1\,\mathrm{g/cc}$)

Open solution

Q264. JEE Main 2024 (Online) 31st January Evening Shift

The distance between charges $+q$ and $-q$ is $2l$ and between $+2q$ and $-2q$ is $4l$. The electrostatic potential at point P at a distance r from center O is $-\alpha\left[\frac{ql}{r^2}\right] \times 10^9\,\mathrm{V}$, where the value of α is _____. (Use $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\,\mathrm{Nm^2C^{-2}}$)



Open solution

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Q265. JEE Main 2024 (Online) 30th January Evening Shift

Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 37° with each other. When suspended in a liquid of density $0.7\,\mathrm{g/cm^3}$, the angle remains same. If density of material of the sphere is $1.4\,\mathrm{g/cm^3}$, the dielectric constant of the liquid is _____ $\left(\tan 37^\circ = \frac{3}{4}\right)$

Open solution

Q266. JEE Main 2024 (Online) 29th January Morning Shift

An electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet S having surface charge density $+\sigma$. The electron at $t = 0$ is at a distance of $1\,\mathrm{m}$ from S and has a speed of $1\,\mathrm{m/s}$. The maximum value of σ if the electron strikes S at $t = 1\,\mathrm{s}$ is $\alpha\left[\frac{m\epsilon_0}{e}\right]\frac{\mathrm{C}}{\mathrm{m^2}}$, the value of α is _____.

Open solution

Q267. JEE Main 2024 (Online) 27th January Evening Shift

Two charges of $-4\mu\mathrm{C}$ and $+4\mu\mathrm{C}$ are placed at the points $\mathrm{A}(1,0,4)\mathrm{m}$ and $\mathrm{B}(2,-1,5)\mathrm{m}$ located in an electric field $\overrightarrow{\mathrm{E}} = 0.20\hat{i}\ \mathrm{V/cm}$. The magnitude of the torque acting on the dipole is $8\sqrt{\alpha} \times 10^{-5}\mathrm{Nm}$, where $\alpha =$ _____.

Open solution

Q268. JEE Main 2024 (Online) 27th January Evening Shift

The electric potential at the surface of an atomic nucleus $(z = 50)$ of radius $9 \times 10^{-13}\ \mathrm{cm}$ is _____ $\times 10^6\ \mathrm{V}$.

Open solution

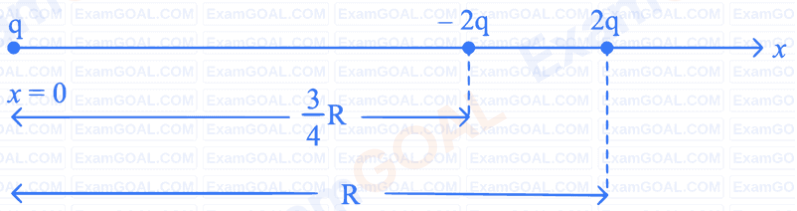
Q269. JEE Main 2024 (Online) 27th January Morning Shift

A thin metallic wire having cross sectional area of $10^{-4}\,\mathrm{m^2}$ is used to make a ring of radius $30\,\mathrm{cm}$. A positive charge of $2\pi\,\mathrm{C}$ is uniformly distributed over the ring, while another positive charge of $30\,\mathrm{pC}$ is kept at the centre of the ring. The tension in the ring is _____ N; provided that the ring does not get deformed (neglect the influence of gravity). (given, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI units)

Open solution

Q270. JEE Main 2023 (Online) 13th April Evening Shift

Three point charges $\mathrm{q}, -2\mathrm{q}$ and $2\mathrm{q}$ are placed on x-axis at a distance $x = 0, x = \frac{3}{4}R$ and $x = R$ respectively from origin as shown. If $\mathrm{q} = 2 \times 10^{-6}\mathrm{C}$ and $\mathrm{R} = 2\,\mathrm{cm}$, the magnitude of net force experienced by the charge $-2q$ is _____ N.



Open solution

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Q271. JEE Main 2023 (Online) 13th April Morning Shift

A thin infinite sheet charge and an infinite line charge of respective charge densities $+\sigma$ and $+\lambda$ are placed parallel at $5\ \mathrm{m}$ distance from each other. Points 'P' and 'Q' are at $\frac{3}{\pi}$ m and $\frac{4}{\pi}$ m perpendicular distances from line charge towards sheet charge, respectively. '$\mathrm{E_P}$' and '$\mathrm{E_Q}$' are the magnitudes of resultant electric field intensities at point 'P' and 'Q', respectively. If $\frac{E_P}{E_Q} = \frac{4}{a}$ for $2|\sigma| = |\lambda|$, then the value of a is _____.

Open solution

Q272. JEE Main 2023 (Online) 12th April Morning Shift

64 identical drops each charged upto potential of $10\ \mathrm{mV}$ are combined to form a bigger drop. The potential of the bigger drop will be _____ mV.

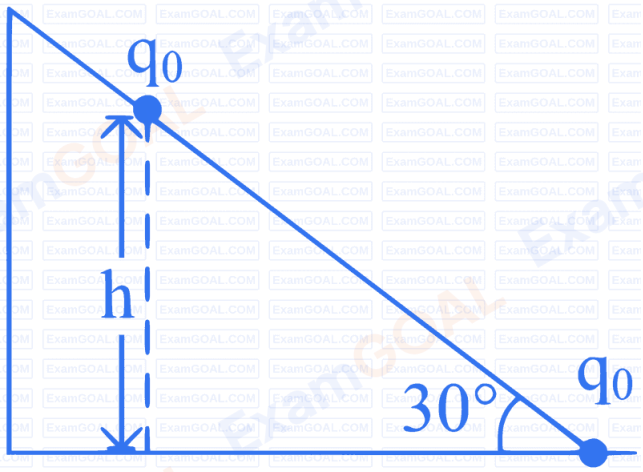
Open solution

Q273. JEE Main 2023 (Online) 11th April Morning Shift

As shown in the figure, a configuration of two equal point charges $(q_0 = +2\mu\mathrm{C})$ is placed on an inclined plane. Mass of each point charge is $20\ \mathrm{g}$. Assume that there is no friction between charge and plane. For the system of two point charges to be in equilibrium (at rest) the height $\mathrm{h} = x \times 10^{-3}\ \mathrm{m}$.

The value of x is _____.

(Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \mathrm{N\,m^2C^{-2}}, g = 10\ \mathrm{m\,s^{-2}}$)

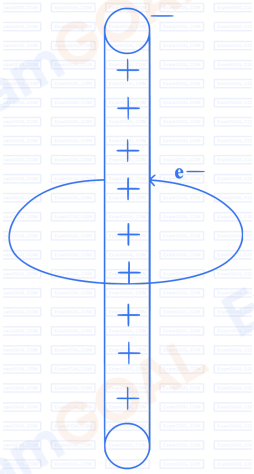


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Q274. JEE Main 2023 (Online) 10th April Evening Shift

An electron revolves around an infinite cylindrical wire having uniform linear charge density $2 \times 10^{-8} \mathrm{Cm}^{-1}$ in circular path under the influence of attractive electrostatic field as shown in the figure. The velocity of electron with which it is revolving is _____ $\times 10^{6}\ \mathrm{m\,s}^{-1}$. Given mass of electron $= 9 \times 10^{-31}\ \mathrm{kg}$



Open solution

Q275. JEE Main 2023 (Online) 10th April Morning Shift

Three concentric spherical metallic shells X, Y and Z of radius a, b and c respectively [a < b < c] have surface charge densities $\sigma, -\sigma$ and σ respectively. The shells X and Z are at same potential. If the radii of X & Y are 2 cm and 3 cm, respectively. The radius of shell Z is _____ cm.

Open solution

Q276. JEE Main 2023 (Online) 8th April Morning Shift

An electric dipole of dipole moment $6.0 \times 10^{-6}\ \mathrm{Cm}$ is placed in a uniform electric field of $1.5 \times 10^{3}\ \mathrm{NC}^{-1}$ in such a way that dipole moment is along electric field. The work done in rotating dipole by 180° in this field will be _____ mJ.

Open solution

Q277. JEE Main 2023 (Online) 1st February Evening Shift

A cubical volume is bounded by the surfaces $\mathrm{x} = 0, x = \mathrm{a}, y = 0, y = \mathrm{a}, \mathrm{z} = 0, z = \mathrm{a}$. The electric field in the region is given by $\overrightarrow{\mathrm{E}} = \mathrm{E}_0 x\hat{i}$. Where $\mathrm{E}_0 = 4 \times 10^{4}\ \mathrm{NC}^{-1}\ \mathrm{m}^{-1}$. If $\mathrm{a} = 2\ \mathrm{cm}$, the charge contained in the cubical volume is $\mathrm{Q} \times 10^{-14}\mathrm{C}$. The value of Q is _____.

(Take $\epsilon_0 = 9 \times 10^{-12}\ \mathrm{C}^2/\mathrm{Nm}^2$)

Open solution

Q278. JEE Main 2023 (Online) 1st February Morning Shift

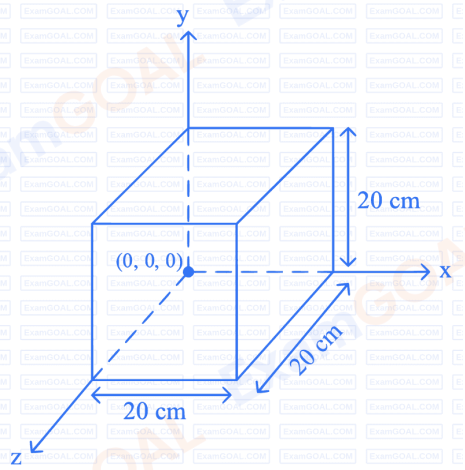
Two equal positive point charges are separated by a distance $2a$. The distance of a point from the centre of the line joining two charges on the equatorial line (perpendicular bisector) at which force experienced by a test charge q_0 becomes maximum is $\frac{a}{\sqrt{x}}$. The value of x is _____.

Open solution

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Q279. JEE Main 2023 (Online) 31st January Morning Shift

Expression for an electric field is given by $\overrightarrow{\mathrm{E}} = 4000x^2\hat{i}\,\frac{\mathrm{V}}{\mathrm{m}}$. The electric flux through the cube of side $20\ \mathrm{cm}$ when placed in electric field (as shown in the figure) is _____ $\mathrm{V\,cm}$.

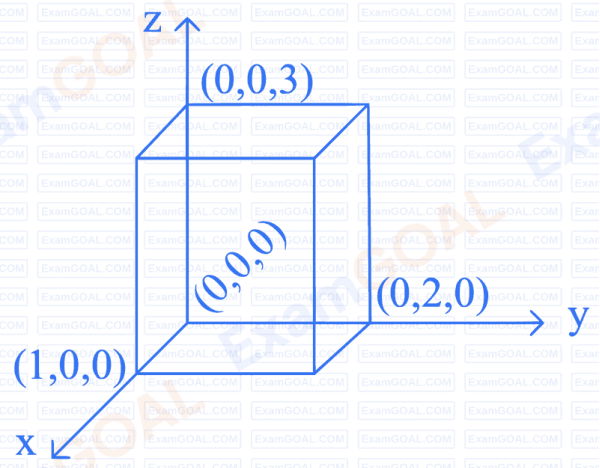


Open solution

Q280. JEE Main 2023 (Online) 30th January Evening Shift

As shown in figure, a cuboid lies in a region with electric field $E = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}\,\mathrm{N/C}$. The magnitude of charge within the cuboid is $n \in_0 C$.

The value of n is _____ (if dimension of cuboid is $1 \times 2 \times 3\ \mathrm{m}^3$)



Open solution

Q281. JEE Main 2023 (Online) 29th January Evening Shift

For a charged spherical ball, electrostatic potential inside the ball varies with r as $\mathrm{V} = 2ar^2 + b$.

Here, a and b are constant and r is the distance from the center. The volume charge density inside the ball is $-\lambda a\varepsilon$. The value of λ is _____.

ε = permittivity of the medium

Open solution

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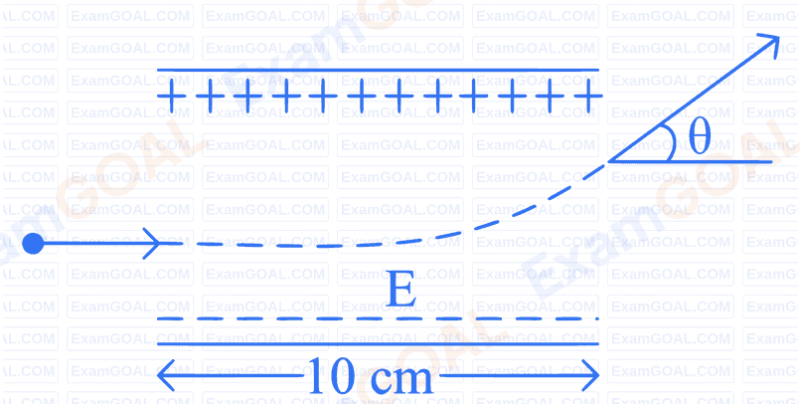
Q282. JEE Main 2023 (Online) 29th January Morning Shift

A point charge $q_1 = 4q_0$ is placed at origin. Another point charge $q_2 = -q_0$ is placed at $x = 12$ cm. Charge of proton is q_0. The proton is placed on x axis so that the electrostatic force on the proton is zero. In this situation, the position of the proton from the origin is _____ cm.

Open solution

Q283. JEE Main 2023 (Online) 25th January Morning Shift

A uniform electric field of 10 N/C is created between two parallel charged plates (as shown in figure). An electron enters the field symmetrically between the plates with a kinetic energy 0.5 eV. The length of each plate is 10 cm. The angle (θ) of deviation of the path of electron as it comes out of the field is _____ (in degree).



Open solution

Q284. JEE Main 2023 (Online) 24th January Morning Shift

A stream of positively charged particles having $\frac{q}{m} = 2 \times 10^{11} \frac{C}{kg}$ and velocity $\overrightarrow{v_0} = 3 \times 10^7 \hat{i}\, m/s$ is deflected by an electric field $1.8\hat{j}$ kV/m. The electric field exists in a region of 10 cm along x direction. Due to the electric field, the deflection of the charged particles in the y direction is _____ mm.

Open solution

Q285. JEE Main 2022 (Online) 28th July Morning Shift

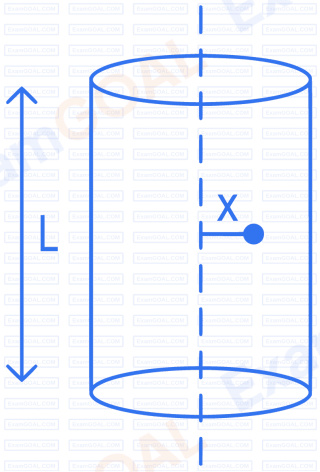
Two electric dipoles of dipole moments 1.2×10^{-30} Cm and 2.4×10^{-30} Cm are placed in two different uniform electric fields of strengths $5 \times 10^4\, NC^{-1}$ and $15 \times 10^4\, NC^{-1}$ respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{x}$. The value of x is _____.

Open solution

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Q286. JEE Main 2022 (Online) 27th July Morning Shift

A long cylindrical volume contains a uniformly distributed charge of density $\rho\,\mathrm{Cm}^{-3}$. The electric field inside the cylindrical volume at a distance $x = \frac{2\varepsilon_0}{\rho}\mathrm{m}$ from its axis is _____ Vm^{-1}.



Open solution

Q287. JEE Main 2022 (Online) 26th July Evening Shift

Three point charges of magnitude $5\mu\mathrm{C}, 0.16\mu\mathrm{C}$ and $0.3\mu\mathrm{C}$ are located at the vertices A, B, C of a right angled triangle whose sides are $AB = 3\ \mathrm{cm}, BC = 3\sqrt{2}\ \mathrm{cm}$ and $CA = 3\ \mathrm{cm}$ and point A is the right angle corner. Charge at point A experiences _____ N of electrostatic force due to the other two charges.

Open solution

Q288. JEE Main 2022 (Online) 25th July Morning Shift

The volume charge density of a sphere of radius $6\ \mathrm{m}$ is $2\ \mu\mathrm{C\,cm}^{-3}$. The number of lines of force per unit surface area coming out from the surface of the sphere is _____ $\times 10^{10}\ \mathrm{NC}^{-1}$.

[Given : Permittivity of vacuum $\epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C}^2\ \mathrm{N}^{-1} - \mathrm{m}^{-2}$]

Open solution

Q289. JEE Main 2022 (Online) 30th June Morning Shift

Eight similar drops of mercury are maintained at 12 V each. All these spherical drops combine into a single big drop. The potential energy of bigger drop will be _____ E. Where E is the potential energy of a single smaller drop.

Open solution

Q290. JEE Main 2022 (Online) 25th June Evening Shift

27 identical drops are charged at 22V each. They combine to form a bigger drop. The potential of the bigger drop will be _____ V.

Open solution

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Q291. JEE Main 2021 (Online) 25th July Morning Shift

A particle of mass 1 mg and charge q is lying at the mid-point of two stationary particles kept at a distance '2 m' when each is carrying same charge 'q'. If the free charged particle is displaced from its equilibrium position through distance 'x' (x << 1 m), the particle executes SHM. Its angular frequency of oscillation will be _____ $\times 10^5$ rad/s if $q^2 = 10\ C^2$.

Open solution

Q292. JEE Main 2021 (Online) 22th July Evening Shift

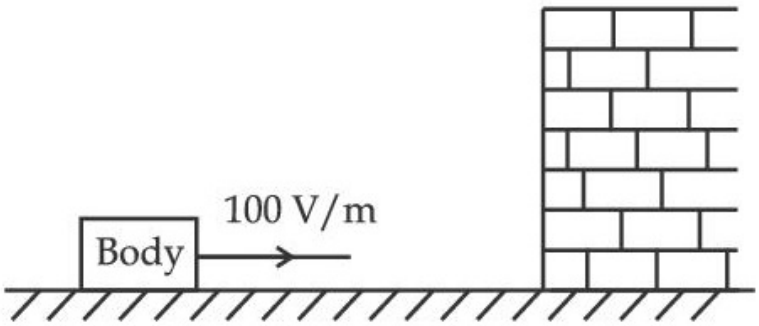
The total charge enclosed in an incremental volume of 2×10^{-9} m^3 located at the origin is _____ nC, if electric flux density of its field is found as

$D = e^{-x} \sin y\, \hat{i} - e^{-x} \cos y\, \hat{j} + 2z\, \hat{k}$ C/m^2

Open solution

Q293. JEE Main 2021 (Online) 20th July Morning Shift

A body having specific charge 8 μC/g is resting on a frictionless plane at a distance 10 cm from the wall (as shown in the figure). It starts moving towards the wall when a uniform electric field of 100 V/m is applied horizontally towards the wall. If the collision of the body with the wall is perfectly elastic, then the time period of the motion will be _____ s.



Open solution

Q294. JEE Main 2021 (Online) 18th March Evening Shift

An infinite number of point charges, each carrying 1 μC charge, are placed along the y-axis at y = 1 m, 2 m, 4 m, 8 m .....

The total force on a 1C point charge, placed at the origin, is x $\times 10^3$ N.

The value of x, to the nearest integer, is _____. [Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ Nm^2/C^2]

Open solution

Q295. JEE Main 2021 (Online) 17th March Evening Shift

The electric field in a region is given by $\vec{E} = \frac{2}{5}E_0\hat{i} + \frac{3}{5}E_0\hat{j}$ with $E_0 = 4.0 \times 10^3 \frac{N}{C}$. The flux of this field through a rectangular surface area 0.4 m^2 parallel to the Y-Z plane is _____ Nm^2C^{-1}.

Open solution

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Q296. JEE Main 2021 (Online) 26th February Evening Shift

27 similar drops of mercury are maintained at 10V each. All these spherical drops combine into a single big drop. The potential energy of the bigger drop is _____ times that of a smaller drop.

Open solution

Q297. JEE Main 2021 (Online) 25th February Evening Shift

Two identical conducting spheres with negligible volume have 2.1 nC and -0.1 nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is _____ $\times 10^{-9}$ N.

[Given : $4\pi\varepsilon_0 = \frac{1}{9\times10^9}$ SI unit]

Open solution

Q298. JEE Main 2021 (Online) 25th February Evening Shift

Two small spheres each of mass 10 mg are suspended from a point by threads 0.5 m long. They are equally charged and repel each other to a distance of 0.20 m. The charge on each of the sphere is $\frac{a}{21} \times 10^{-8}$C. The value of 'a' will be _____. [Given g = 10 ms^{-2}]

Open solution

Q299. JEE Main 2021 (Online) 25th February Morning Shift

The electric field in a region is given by $\overrightarrow{E} = \left(\frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0\hat{j}\right)\frac{N}{C}$. The ratio of flux of reported field through the rectangular surface of area 0.2 m^2 (parallel to y $-$ z plane) to that of the surface of area 0.3 m^2 (parallel to x $-$ z plane) is a : b, where a = _____ [Here $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x, y and z-axes respectively.]

Open solution

Q300. JEE Main 2021 (Online) 25th February Morning Shift

512 identical drops of mercury are charged to a potential of 2V each. The drops are joined to form a single drop. The potential of this drop is _____ V.

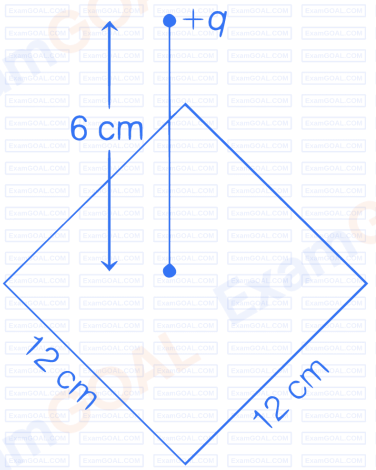
Open solution

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Q301. JEE Main 2021 (Online) 24th February Evening Shift

A point charge of +12μC is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be _____ $\times 10^3$ Nm^2/C.

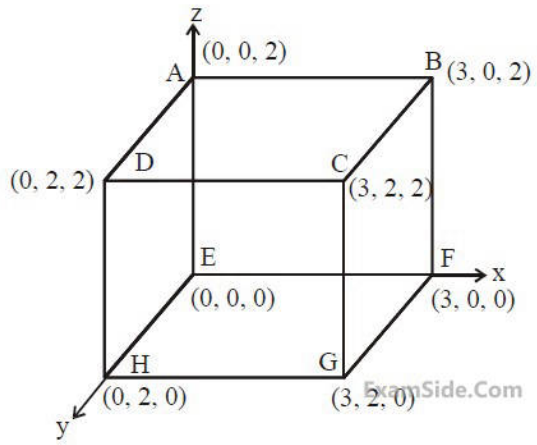
Open solution



Q302. JEE Main 2020 (Online) 9th January Evening Slot

An electric field $\vec{E} = 4x\hat{i} - (y^2+1)\hat{j}$ N/C passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as ϕ_I and ϕ_{II} respectively. The difference between $(\phi_I - \phi_{II})$ is (in Nm^2/C) _____.

Open solution



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MCQ (More than One Correct Answer) 1 questions

Q303. JEE Main 2015 (Offline)

A uniformly charged solid sphere of radius R has potential V_0 (measured with respect to ∞) on its surface. For this sphere the equipotential surfaces with potentials $\frac{3V_0}{2}$, $\frac{5V_0}{4}$, $\frac{3V_0}{4}$ and $\frac{V_0}{4}$ have radius $R_1,\ R_2,\ R_3$ and R_4 respectively. Then

Open solution

(A) $R_1 = 0$ and $R_2 < (R_4 - R_3)$

(B) $2R < R_4$

(C) $R_1 = 0$ and $R_2 > (R_4 - R_3)$

(D) $R_1 \neq 0$ and $(R_2 - R_1) > (R_4 - R_3)$

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **A**	3. **A**	4. **A**	5. **B**
6. **A**	7. **C**	8. **D**	9. **B**	10. **C**
11. **C**	12. **A**	13. **D**	14. **A**	15. **B**
16. **A**	17. **B**	18. **B**	19. **A**	20. **C**
21. **B**	22. **C**	23. **D**	24. **A**	25. **D**
26. **B**	27. **D**	28. **D**	29. **D**	30. **C**
31. **D**	32. **D**	33. **C**	34. **A**	35. **B**
36. **C**	37. **D**	38. **A**	39. **B**	40. **B**
41. **A**	42. **B**	43. **B**	44. **B**	45. **C**
46. **A**	47. **C**	48. **B**	49. **D**	50. **D**
51. **D**	52. **B**	53. **D**	54. **B**	55. **B**
56. **D**	57. **C**	58. **B**	59. **B**	60. **D**
61. **B**	62. **A**	63. **D**	64. **C**	65. **A**
66. **C**	67. **D**	68. **C**	69. **C**	70. **D**
71. **A**	72. **C**	73. **D**	74. **B**	75. **D**
76. **D**	77. **B**	78. **B**	79. **D**	80. **A**
81. **A**	82. **C**	83. **A**	84. **D**	85. **C**
86. **D**	87. **B**	88. **D**	89. **D**	90. **A**
91. **A**	92. **A**	93. **A**	94. **B**	95. **C**
96. **A**	97. **B**	98. **B**	99. **A**	100. **B**
101. **D**	102. **D**	103. **D**	104. **B**	105. **A**
106. **A**	107. **B**	108. **D**	109. **B**	110. **C**
111. **C**	112. **A**	113. **A**	114. **B**	115. **B**
116. **C**	117. **B**	118. **B**	119. **A**	120. **D**
121. **A**	122. **B**	123. **B**	124. **C**	125. **D**
126. **A**	127. **B**	128. **A**	129. **C**	130. **C**
131. **C**	132. **A**	133. **A**	134. **B**	135. **D**
136. **B**	137. **C**	138. **C**	139. **A**	140. **C**
141. **C**	142. **B**	143. **D**	144. **C**	145. **A**
146. **C**	147. **A**	148. **A**	149. **B**	150. **C**

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151. **A**	152. **B**	153. **A**	154. **D**	155. **C**
156. **D**	157. **A**	158. **D**	159. **D**	160. **B**
161. **A**	162. **C**	163. **C**	164. **D**	165. **D**
166. **B**	167. **C**	168. **A**	169. **D**	170. **B**
171. **A**	172. **D**	173. **D**	174. **D**	175. **D**
176. **D**	177. **C**	178. **A**	179. **B**	180. **A**
181. **B**	182. **C**	183. **C**	184. **B**	185. **B**
186. **D**	187. **C**	188. **B**	189. **D**	190. **C**
191. **A**	192. **C**	193. **B**	194. **D**	195. **C**
196. **C**	197. **C**	198. **A**	199. **B**	200. **B**
201. **C**	202. **C**	203. **A**	204. **A**	205. **A**
206. **B**	207. **C**	208. **C**	209. **C**	210. **D**
211. **C**	212. **A**	213. **D**	214. **C**	215. **C**
216. **B**	217. **D**	218. **C**	219. **C**	220. **D**
221. **C**	222. **C**	223. **C**	224. **A**	225. **C**
226. **D**	227. **A**	228. **B**	229. **A**	230. **C**
231. **A**	232. **A**	233. **C**	234. **C**	235. **A**
236. **B**	237. **A**	238. **C**	239. **D**	240. **D**
241. **B**	242. **C**	243. **A**	244. **C**	245. **B**
246. **A**	247. **B**	248. **D**		

Numerical

249. **186**	250. **90**	251. **15**	252. **12**	253. **48**
254. **BONUS**	255. **16**	256. **36**	257. **5**	258. **12**
259. **2**	260. **16**	261. **16**	262. **3**	263. **3**
264. **27**	265. **2**	266. **8**	267. **2**	268. **8**
269. **3**	270. **5440**	271. **6**	272. **160**	273. **300**
274. **8**	275. **5**	276. **18**	277. **288**	278. **2**
279. **640**	280. **12**	281. **12**	282. **24**	283. **45**
284. **2**	285. **6**	286. **1**	287. **17**	288. **45**
289. **32**	290. **198**	291. **6000**	292. **4**	293. **1**
294. **12**	295. **640**	296. **243**	297. **36**	298. **20**
299. **1**	300. **128**	301. **226**	302. **-48**	

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MCQ (More than One Correct Answer)

303. **A, B**

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